

System Startup Guide Cartridge Drive

SW11604

**Implementation
Startup & Reconfiguration - 1**

System Startup Guide Cartridge Drive

**SW11-604
Release 681
9/08**

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About This Publication

This publication provides an example of the startup of a **TotalPlant** Solution (TPS) system for Release 500 and later, using Honeywell-supplied cartridge disks. TPS is the evolution of TDC 3000^X.

The tasks in this publication are intended to show you how a system is initially configured. You should have filled in configuration forms based on the system you are working with, so you know exactly what information needs to be entered. If you are upgrading to a newer release of software, consult the *Customer Release Guide* for procedures. If you are reconfiguring an existing LCN network, refer to the *Network Data Entry* manual (see References).

You should know how to use functions such as the Command Processor, the Data Entity Builder, the Picture Editor, the Text Editor, and so on, before attempting this procedure. This information is explained in the manuals listed in the References section of this manual.

IMPORTANT

- A. A 20 MB cartridge disk should be removed from the drive when not in use to avoid unnecessary shortening of the cartridge or drive life caused by continuous spinning. The 35-150 MB disk drives automatically stop spinning after a period of non use. Disks larger than 20 MB if purchased from Honeywell usually contain a size indication such as "35 MB" stamped on the edge.

- B. The position of the Write-Protect switch on a cartridge disk should not be changed while the disk is in a drive.

- C. Inserting and removing a cartridge

Always insert the cartridge in a smooth fashion and halt immediately if the green LED begins to flash before the cartridge is all the way in. If this happens, wait for the LED to stop flashing before inserting the cartridge the rest of the way.

Upon inserting a cartridge disk into the drive, wait for the following events to occur before proceeding:

1. Green light flashes
2. Yellow light blinks several times
3. Green light remains illuminated

To remove a cartridge disk from the drive

1. Push the Eject button.
2. Wait until the green light stops flashing (is extinguished) and an audible "click" is heard (approximately 5 seconds). If the cartridge resists removal, push the Eject button again.

WARNING

ATTEMPTED REMOVAL OF THE DISK WITHOUT FOLLOWING THE ABOVE PROCEDURE CAN RESULT IN SERIOUS DAMAGE TO BOTH THE DISK AND THE DISK DRIVE.

Change bars are used to indicate paragraphs, tables, or illustrations containing changes that have been made to this manual effective with this update. Pages revised only to correct minor typographical errors contain no change bars.

Support and Other Contacts

Location	Organization	Address	Phone/Facsimile
United States and Canada	Honeywell Solution Support Center	Honeywell TAC, MS P13, 2500 West Union Hills Drive, Phoenix, AZ, 85027	Phone: 1-800 822-7673 602- 313-5558 (In Arizona) Facsimile: (602) 313-3293
Note: Calls are answered by dispatcher between 6:00 A.M. and 4:00 P.M. Mountain Standard Time. Emergency calls outside normal working hours are received by an answering service and returned within one hour.			
Europe	Honeywell TAC-EMEA	TAC-BE02 Hermes Plaza Hermeslaan, 1HB-1831 Diegem, Belgium	Phone: +32-2-728-2732 Facsimile: +32-2-728-2696
Pacific	Honeywell Global TAC – Pacific	Honeywell Limited Australia 5 Kitchener Way Burswood 6100, Western Australia Email: GTAC@honeywell.com	Phone: 1300-300-4822 (toll free within Australia) +61-8-9362-9559 (outside Australia) Facsimile: +61-8-9362-9564
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Elsewhere	Call your nearest Honeywell office.		
World Wide Web	Honeywell Solution Support Online: http://www.honeywell.com/ps		
Training Classes	Honeywell Automation College: http://www.automationcollege.com		

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INTRODUCTION

Section 1

1.1 SCOPE OF COVERAGE AND SUGGESTIONS

This "how-to" document covers the engineering tasks necessary to initially configure a History Module-based **TotalPlant** Solution (TPS) system. Its step-by-step instructions cover all file and cartridge disk manipulations necessary for initial system startup. It does not cover all permutations of data that can be entered. For instance, in developing the guide, we configured one Unit, one Area, a few points, and one Group; otherwise, the step-by-step instructions would have been far too bulky.

During Tasks 6-11 you should configure as much of the network as possible. At the very least you will have to use the On-Line Network Reconfiguration procedures if you later wish to change the Network Configuration. Pay particular attention to accurately and completely configuring your History Modules; otherwise, you will have to perform time-consuming procedures if you have to change the History Module's volume configuration.

Refer to the *Customer Release Guide* to determine exactly what functions are included in the current software.

The procedures described in this publication require at least one Universal Station with dual cartridge drives from which configuration operations are performed.

1.1.1 Universal Station^X, Micro TDC, and Universal Workstation

After the system is configured, any Universal Station^X (U^XS) can easily be loaded as described in section 30.

If the system contains at least one Universal Station^X with an Engineering Keyboard and dual cartridge drives, it should be possible to perform the procedures described in this publication, but they have not been tested for that situation.

U^XS station preparation — The U^XS station must contain software that is compatible with the system software (this software is updated with DAT cartridges). Refer to the *Universal Station^X User's Guide* for the loading procedure.

An Engineer's Workstation or Micro TDC station can also be used, if the slight differences are taken into account.

1.2 HOW THIS STARTUP GUIDE IS ORGANIZED

This guide is organized as follows:

Section (Number and Title)	You Should Use the Information to
1. INTRODUCTION	Learn what this document is all about
2. OVERVIEW	Understand the overall System Startup process
3. INSTRUCTIONS General Instructions System Startup Task Record Procedures One or more Tasks Multiple Instructions	Take action necessary to start up your system Understand the format of standard instructions Check off tasks completed & record information Know major objective for each set of tasks Know subobjective represented by each task Know exact steps to accomplish each task
4. ERROR RECOVERY PROCEDURES	Solve a problem that you have encountered
A. APPENDIX	Load the Operating Personality into a second Universal Station
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C. APPENDIX	Save Application Module checkpoint data in the History Module and on cartridge disks
D. APPENDIX	Build points by exception
E. APPENDIX	Create exception build files from IDFS

1.3 HOW TO USE THE PROCEDURES IN THIS DOCUMENT

The procedures in this manual were developed on a sample system by the writers of this document and you can use them as a set of hard-and-fast rules for starting up a system.

One person can easily perform these procedures, but if two people are available, the most efficient method is to have one person read the step-by-step instructions while another person keys in the data.

1.4 WHAT TO DO BEFORE STARTING UP THE SYSTEM

Format History Modules

The system's hardware must be installed and checked out before performing system startup. Each History Module (HM) must be formatted before system startup. This is normally done at the factory. See your maintenance technician or Honeywell support engineer if there is any question.

Have Reference Material Available

You should have access to the publications listed under References during the entire system startup process.

Understand Various Responses After the Enter Key Is Pressed

In the Command Processor function, when the <ENTER> key is pressed, the command entered is redisplayed with all characters in the command in uppercase. In the Network Configuration functions, when the <ENTER> key is pressed, the data is entered and reappears in upper case and in cyan. Also in the Network Configuration functions, when a target (pick) is selected, it usually fills with full intensity white. After the <ENTER> key is pressed, the target reappears filled with half-intensity white. In Point Build, HG Library, and Area Database Parameter Entry Displays (PEDs), when the <ENTER> key is pressed the data entered reappears in cyan and in upper case.

Understand Responses When Using Cartridge Disks

When a cartridge disk is inserted, the system usually reads the volume name for use in subsequent displays. Wait for the steady green light on the front of the drive before performing the next instruction (see the notice regarding handling of cartridge disks at the front of this manual).

Understand the Function of the HELP key

When you are entering configuration data, help information is available. If you get a red error message or if a data entry region reappears in red after you press the <ENTER> key, press the <HELP> key and any available help information is displayed.

Understand the Concept of Work Files

Work files accumulate configuration data that you enter until you perform an "install" function, which renames the work file to become your permanent site-specific configuration file.

Understand Time Delays

As you perform some of the procedures in this guide, time delays can occur. You may have to wait up to several minutes for a few of the responses listed in the instructions.

Printer Use

It is an advantage to have a printer attached to the Universal Station where the configuration tasks are being performed. This allows you to perform screen-print and file-print operations on that printer.

Understand How to Respond to Messages

The **TotalPlant** Solution (TPS) *Messages Directory* gives you a definition of the message and error recovery procedures, if available.

Know What Files You Are Copying or Deleting

If you add "-D" at the end of command lines such as copy, delete, and so on, the software lets you see exactly which files are being copied, deleted, and so on. Using this feature allows you to know that all file operations are completed successfully.

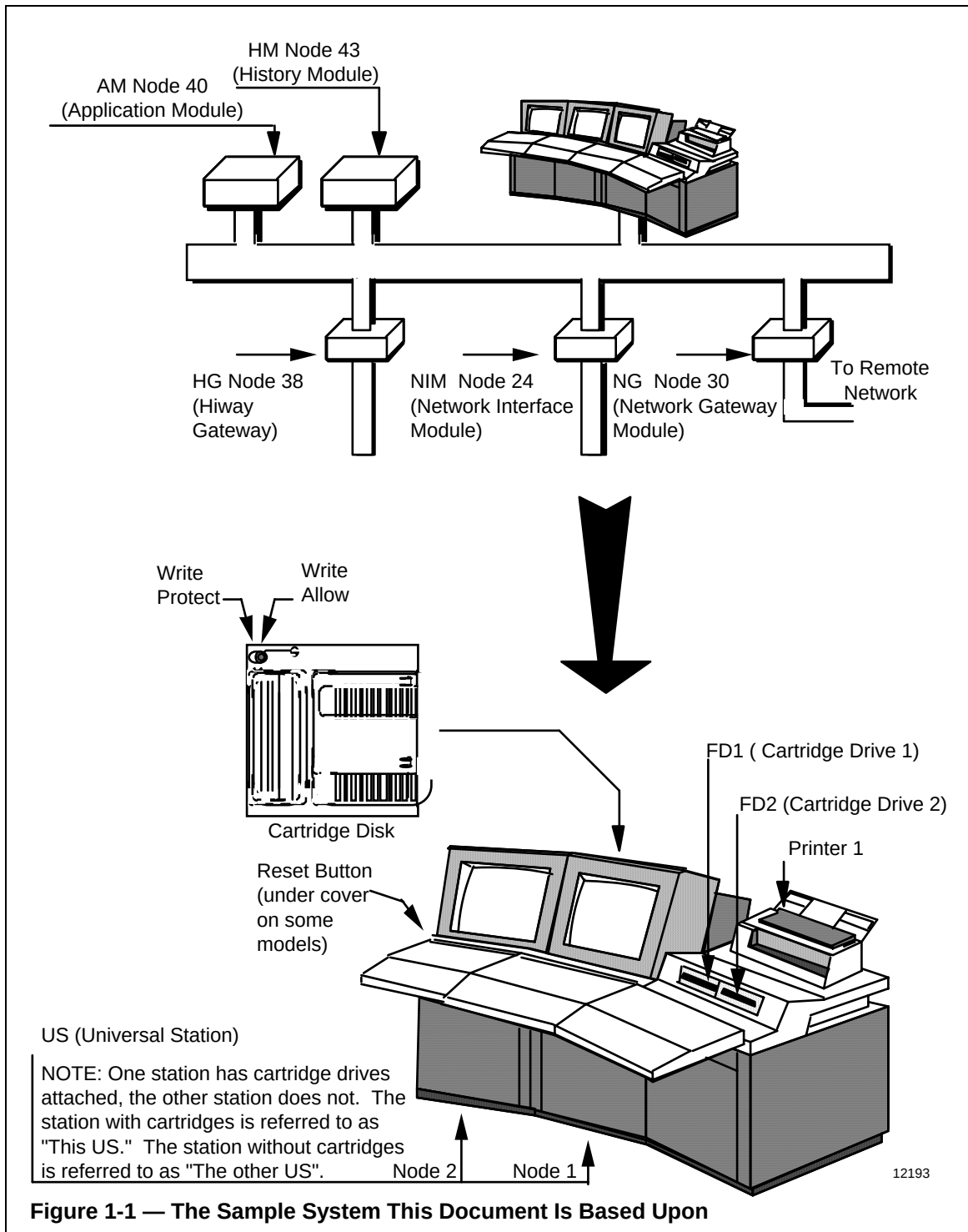
1.5 REFERENCES

Publication Title	Pub Number	Binder
Advanced Process Manager Control Function and Algorithms	AP09-600	Implementation/Advanced Process Manager - 1
Advanced Process Manager Parameter Reference Dictionary	AP09-640	Implementation/Advanced Process Manager - 2
Application Module ^X User Guide	AX09-500	Application Module ^X
Button Configuration Data Entry	SW11-570	Implementation/Engineering Operations - 1
Command Processor Operation	SW11-507	Implementation/Startup & Reconfiguration - 1
Control Language/Application Module Data Entry	AM11-585	Implementation/Application Module - 3
Control Language/MC Reference Manual	PC27-510	Implementation/Hiway Gateway - 2
Control Language/MC Data Entry	PC11-585	Implementation/Hiway Gateway - 2
Configuration Data Collection Guide	SW12-500	Implementation/Startup & Reconfiguration - 2
Data Entity Builder Manual	SW11-511	Implementation/Engineering Operations - 1
Engineer's Reference Manual	SW09-605	Implementation/Startup & Reconfiguration - 2
Free Format Log Data Entry	HM11-660	Implementation/Engineering Operations - 1

High-Performance Process Manager Control Function and Algorithms	HP09-600	Implementation/High-Performance Process Manager - 1
High-Performance Process Manager Parameter Reference Dictionary	HP09-640	Implementation/High-Performance Process Manager - 2
History Module Service	HM13-501	LCN Service - 2
HM History Group Form Instructions	HM12-500	Implementation/Engineering Operations - 1
Logic Manager Control Functions	LM09-500	Implementation/Logic Manager
Logic Manager Parameter Reference Dictionary	LM09-540	Implementation/Logic Manager
Messages Directory	SW09-507	Implementation/Startup & Reconfiguration - 3
Network Form Instructions	SW12-605	Implementation/Startup & Reconfiguration - 1
Network Gateway Implementation and Operation	NG09-510	Implementation/Network Gateway
Picture Editor Data Entry	SW11-650	Implementation/Engineering Operations - 3
PLC Gateway Control Functions	PL09-500	Implementation/PLC Gateway
Process Manager Control Functions and Algorithms	PM09-600	Implementation/Process Manager - 1
Process Manager Parameter Reference Dictionary	PM09-640	Implementation/Process Manager - 2
Control Language/Process Manager Data Entry	PM11-500	Implementation/Process Manager - 2
Process Operations Manual	SW11-601	Operation/Process Operations
Safety Manager Module Control Functions	SM09-500	Implementation/Safety Manager
Safety Manager Module Parameter Reference Dictionary	SM09-550	Implementation/Safety Manager
Text Editor Operation	SW11-506	Implementation/Engineering Operations - 3
Universal Work Station Specification and Technical Data	UW03-500	System Summary - 2

1.6 THE SAMPLE SYSTEM THIS DOCUMENT IS BASED UPON

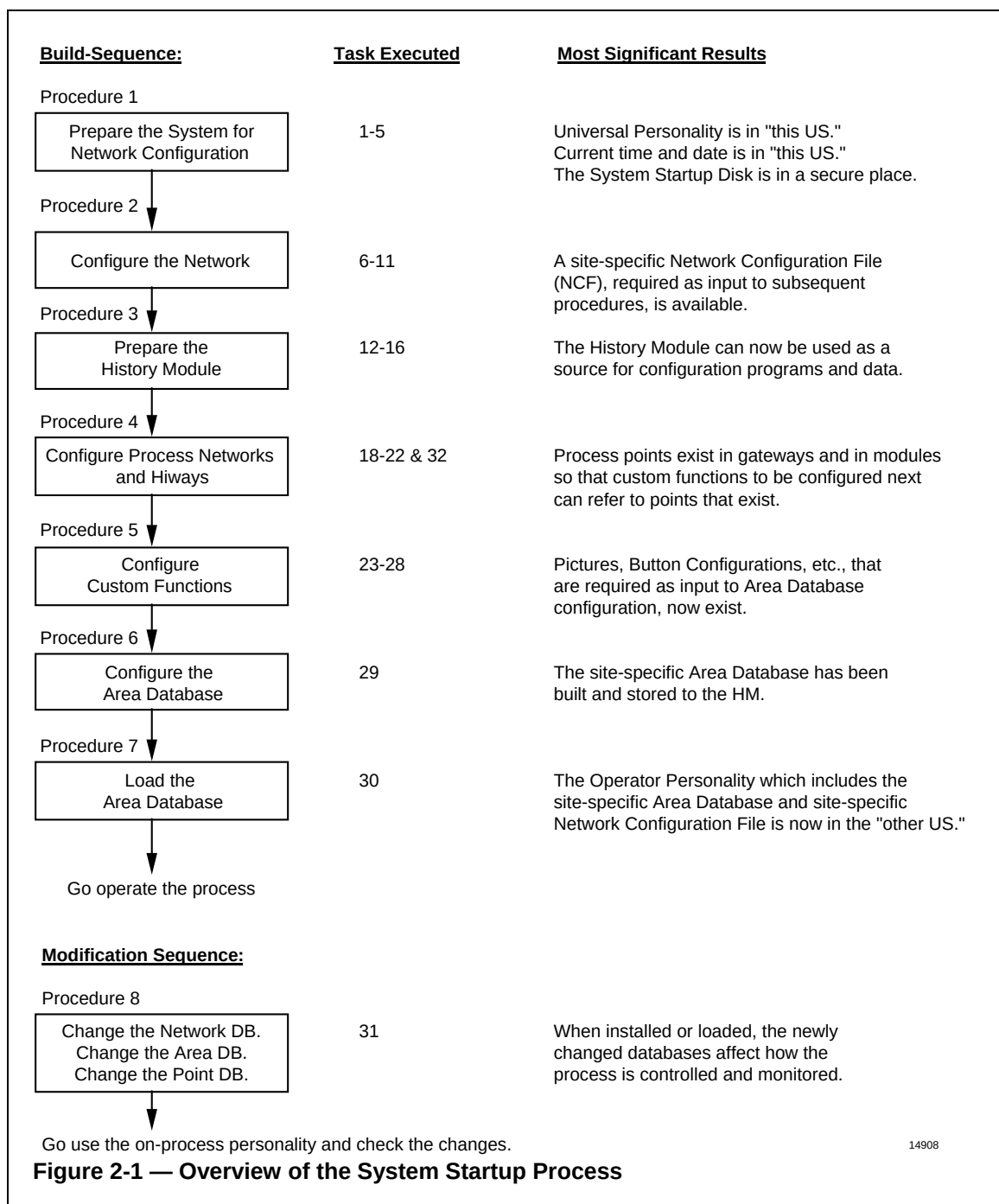
Figure 1-1 shows the system that this guide is based upon. Please memorize the definitions and be aware that your node numbers may be different. A Programmable Logic Controller Gateway (PLCG) and its associated controllers was also temporarily connected.



OVERVIEW

Section 2

Figure 2-1 shows an overview of the build and modification sequences involved in the TotalPlant Solution (TPS) system startup process.



INSTRUCTIONS

Section 3

3.1 GENERAL INSTRUCTIONS

The process of starting up a system is really the process of following step-by-step instructions that tell you how you use configuration programs and configuration data to specify the specifics of your unique site.

Program and Data Sources—During Tasks 2 through 16, the program and data sources are Cartridge Disks. After the History Module (HM) has been loaded in Task 16, if it is necessary to restart the system, either the Universal Personality or the Operator Personality must be loaded from removable media; however, most other data and program sources can be read from the History Module by entering N (for NET) when the source is requested. Some infrequently used software is not loaded onto the HM, but can be accessed from cartridge disks (removable media), if needed.

After an operating personality (Operator's or Universal) is running, cartridge disks are accessed by their logical-device numbers (for example, \$F1 for Drive 1 and \$F2 for Drive 2, and so on).

Special Software—Two types of software are available for most node types. If a node contains an HMPU, K2LCN, or HPK2 printed wire board, you must load 68020 type software into that node. Also, in Task 10 you must specify the software used by these nodes as TYPE II (for example, HG II, USUNP II, etc.). If you do not know which nodes contain these boards, refer to Figure 3-2 for help. If a node contains a K4LCN printed wire board, you must load 68040 type software into that node. In Task 10 you must specify the software used by these nodes as TYPE IV (for example, NIM IV, AM IV).

Organization—The organization of this section is based on the following hierarchy:

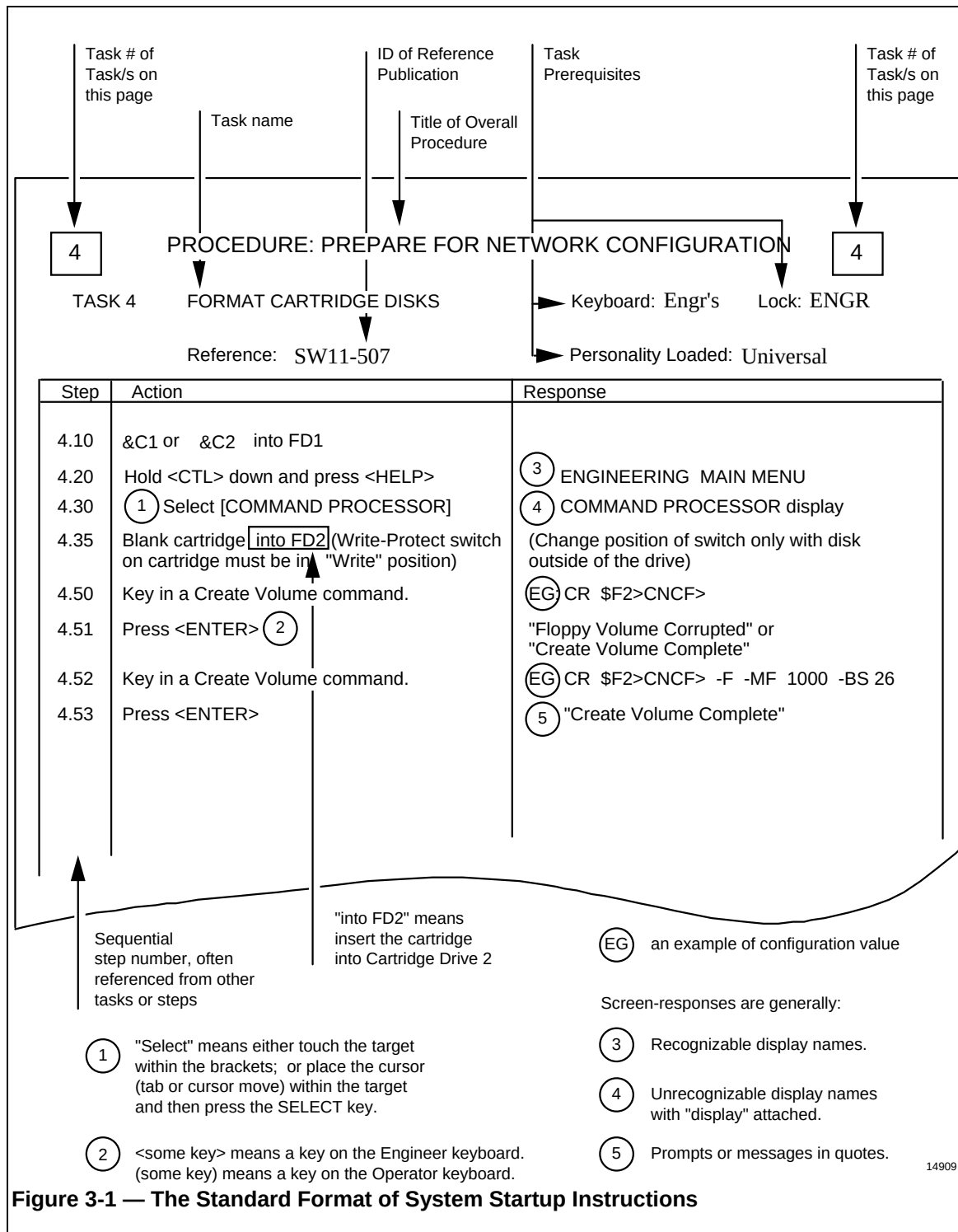
- Procedure
 - One or more Tasks
 - Multiple Instructions

Eight procedures are made up of 32 individual tasks. The first seven procedures (Tasks 1-30 and 32) must be performed in sequence the first time you configure a system. Procedure 8 (Task 31) is a prerequisite for changing the database after your system is up and running on-process. Task 13 was made optional and moved to Appendix A, and Task 32, an addition, is located near the end of this manual. Some tasks instruct you to perform a task previously performed; therefore, the instructions in a given task are generally written as free-standing or re-entrant.

To make it easier for you to find a particular task, a task number has been printed in the upper left-hand and right-hand corner of each page. You can use the Table of Contents to first find Task Numbers and then find instructions.

Instructions—Instructions are presented as Action/Response pairs. Figure 3-1 shows the standard format of a sample task within a procedure. You must understand the parts of the standard format before performing the instructions that follow.

Examples—Examples are used extensively. They are based upon the sample system and example values used for developing this guide. If you decide to use the examples, you must substitute **your site-specific values** for our example values in most cases, but **be consistent**. Use the same Area and Console IDs throughout the procedure.



3.2 SYSTEM STARTUP TASK RECORD

As each task is performed and completed you should record the indicated information and check the associated task box. You will find the recorded information useful to have when performing subsequent tasks.

Task # Information to be recorded or checked off when configured.

As you reset Nodes, check them off; also record Node Nos.

- 1 USs ; AMs ; HMs ; HGs ; NIMs ; CIUs PLCGs NGs ;
 U^XSs A^XMs

 _____; _____; _____; _____; _____; _____; _____; _____
 _____; _____; _____; _____; _____; _____; _____; _____
 _____; _____; _____; _____; _____; _____; _____; _____
 _____; _____; _____; _____; _____; _____; _____; _____
- 2 Physical Device Nos:
 Floppy Drive Nos: _____, _____, _____, _____, _____, _____

 Cartridge Drive Nos: _____, _____, _____, _____

 Printers: _____, _____, _____, _____
- 3 4 5 6
- 7 Unit IDs: _____; _____; _____; _____; _____; _____; _____; _____; _____; _____; _____
 Area Name: _____; Console Name: _____
- 8
- 9 Nodes: Type/Numbers : _____/_____, _____/_____, _____/_____, _____/_____, _____/_____, _____/_____
 _____/_____, _____/_____, _____/_____, _____/_____, _____/_____, _____/_____

 Hiway Numbers: _____, _____, _____, _____,

 UCN Numbers: _____, _____, _____, _____,
- 10 Area Names: _____; _____; _____

 User's Volume IDs: _____, _____, _____

* For configuration tasks in this document, CIU, CG, and CM broadly refer to the same thing.

SYSTEM STARTUP TASK RECORD (continued)

11 12 13 14 15

16 17 18 19

20 Hiway Box Types/Box Nos/Slot Nos:

____/____/____ ____/____/____ ____/____/____ ____/____/____ ____/____/____
 ____/____/____ ____/____/____ ____/____/____ ____/____/____ ____/____/____

21 Point-IDF Names: _____; _____; _____; _____

Point IDs: _____

22 Logic Block Conf.

23 Picture Filenames: _____

24 Button Filenames: _____

Picture _____/Button # _____; Picture _____/Button # _____

Picture _____/Button # _____; Picture _____/Button # _____

Picture _____/Button # _____; Picture _____/Button # _____

25 HG Library Names: _____

26 Free Format Log Filenames

27 HM History Groups

28 CL Configuration

29 Area Database—IDFs: _____

Areas Installed: _____

30 Node Numbers of USs Loaded: _____

32 Node (PMM) No./Primary (P),Redundant (R):

____/____ ____/____ ____/____ ____/____ ____/____

NODE IDENTIFICATION

Figure 3-2 shows how to determine what type of nodes you have, their node numbers, and which ones require 68020 software. Many modules have a tag to identify the module type. You will need this information when loading software, and for Task 10 configuration.

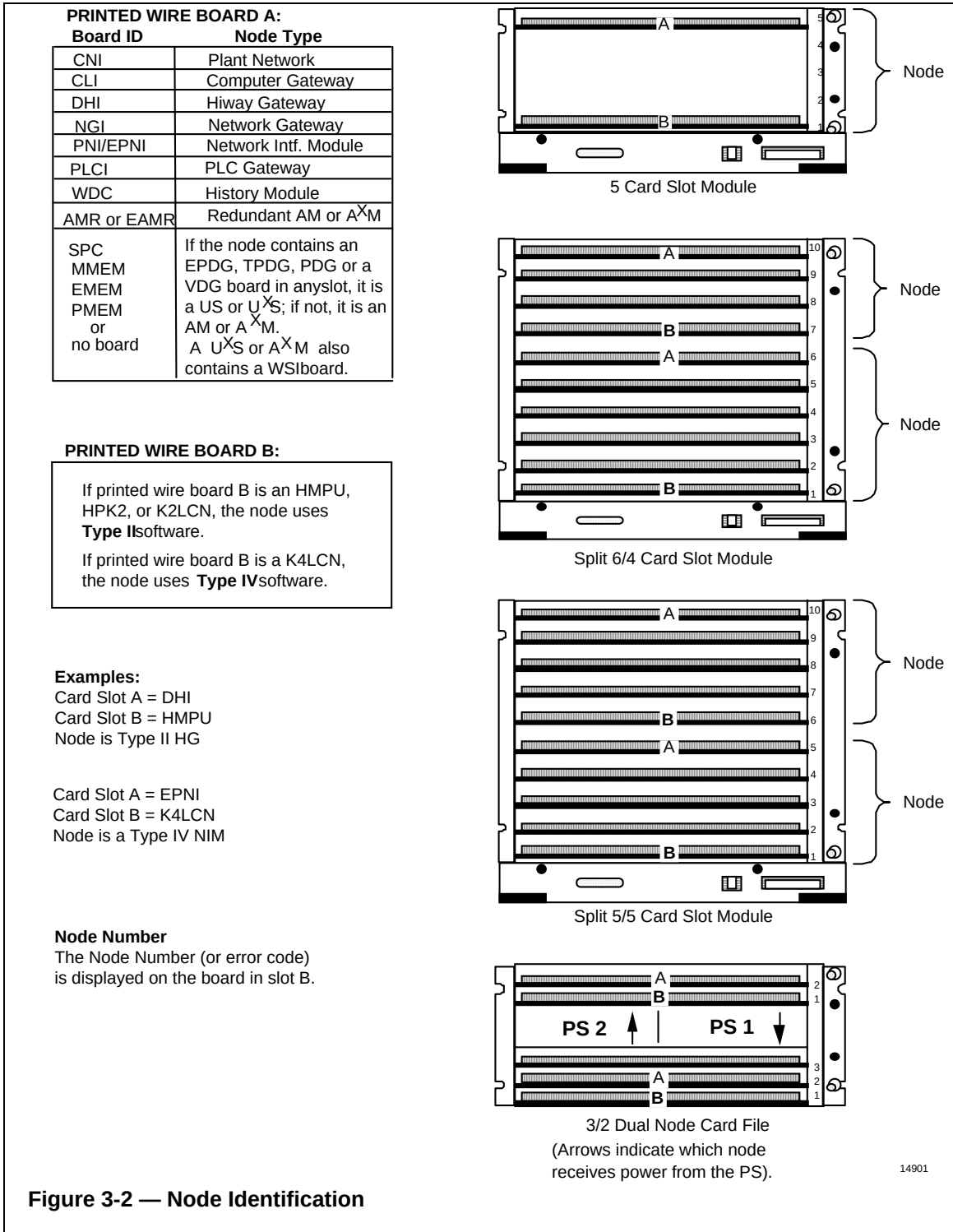


Table 3-1 shows how Honeywell cartridge disks are labeled, their contents and Honeywell Part number.

Table 3-1 — Standard Software and Cartridge Disk Names

Cartridge/ Model No.	Contents	Part Number	
		20 MB	35MB
&C1 M5CD11	Universal Station, HM, HG, and NIM/PM Personalities and supporting files.	51155063	51155066
	The following files for 68040 microprocessors are stored in these directories: &OPR Universal Station Operator Personality &UNP Universal Station Universal Personality &OP1 US Univ. Personality Overlay Files &NMG Startup NIM Global Descriptor Files &NM2 NIM Global Descriptor Files (continued) &LDR Boot Loader &AMG AM, Global Descriptor Files (GDFs) &ARG Area Database GDFs and Initial Area &CUS Cust. Soft./Background CL Prog. (optional) &DSY Universal Station Abstracts (displays) &EC Executable Command Files &HGG Hiway Gateway Global Descriptor Files &OVG Generic Overlay &DS2		
&C2 M5CD21	&C2 contains directories with labels identical to those listed above for &C1; however, the module-dependent directories contain files for 68020 microprocessors.	51155051	51155067
&C3 M5AM31	Application Module Personalities (optional). This cartridge includes software for both 68000 and 68020 microprocessors. &LDR Boot Loader &AMO AM On-Process Personality &CUS Cust. Soft./Background CL Prog. (optional) &CLX Cust. Soft./Background CL Prog. (optional) &EC Executable Command Files	51155052	51155068

(continued)

Table 3-1 — Standard Software and Cartridge Disk Names (continued)

Cartridge/ Model No.	Contents	Part Number	
		20 MB	35 MB
&C4 M5CM31	Computing Module Personalities (optional) This cartridge includes software for both 68020 and 68040 microprocessors &LDR Boot Loader &CIO CM On-Process Personality &EC Executable Command Files &CUS Cust. Soft./Background CL Prog. (optional)	51155053	51155069
&C5 MKHW31	HVTS, CMTS, PM, LCN Test Programs	51155110	51155079
&C6 SWOP31	&KFO Key File (unique for each system &ASY System Startup (Customer Startup)* System Startup (64 Node) &AMC Startup Application Module Checkpoint &CIC Startup CM Checkpoint DIA1 Diagnostics &I01 - Startup checkpoint files for HGs, &I20 PMs/APMs/HPMs, and NIMs TLK1 Toolkit &D01 Null Area Database and Button File	51155054	51155070
&C9 M5NG31	&NGO Network Gateway On-Process Personality &LDR Boot Loader &EC Executable Command Files	51155057	51155073
&C10 M5MG31	&LDR Boot Loader &HGO Hiway Gateway On-Process Personality &HMI History Module Off-Process Personality &HMO History Module On-Process Personality &NMO Network Intf. Mod. On-Process Personality &UCN &EC Executable Command Files	51155058	51155074
*Special file; see the <i>Customer Release Guide</i>			

TASK 1 RESET NODES**Keyboard:** N/A**Lock:** N/A**Reference:** This Document**Personality Loaded:** N/A

Step	Action	Response
1.10	For all LCN nodes: press the <RESET> button on the electronic chassis; Power Off/On Univ. Work Stations	Nodes go to a reset state Note: After reset, the indicators near the reset button display the Node Number
1.15	Record all node numbers	
1.20	Turn power off to all HMs	

CAUTION

In the tasks that follow, you must load software appropriate for the node type. If the node contains an HMPU, HPK2, or K2LCN printed wiring boards, you must use 68020 type software. If the node contains a K4LCN printed wiring board, you must use 68040 type software (see Table 3-1). The first US is started on Area 01.

TASK 2 LOAD THE UNIVERSAL PERSONALITY**Keyboard:** Optr's**Lock:** ENGR

(to a station with cartridge disks and an engineering keyboard).

Reference: This Document**Personality Loaded:** none; reset

Step	Action	Response
2.10	&C1 or &C2 into FD1 Note: Universal Stations with 68040 microprocessors require &C1, 68020 microprocessors require &C2	
2.20	Press (RESET) (see Figure 1-1), or Power Off/On the Universal Work Station (not necessary if coming from Task 1)	Wait for the > sign to appear
2.25	Place keyswitch in ENG position	Engineering Keylock enabled
2.30	Press (LOAD) on the US Operator's Keyboard (or <LOAD> on the Universal Work Station Engineering Keyboard)	"N,1,2,3,4,X?"
2.40	Press (1) and then press (ENTER)	"OPR, UNP?"
2.50	Press the ALPHA SHIFT key until lit, key in UNP and press (ENTER)	Series of messages, ending with "NODE STARTING UP"... "NCF? N,1,2,3,4,X"
2.51	&C6 (CNCF if coming from Task 11) into FD2	
2.52	Press (2) and then press (ENTER)	"ABST N,1,2,3,4,X"
2.56	Press (1) and then press (ENTER)	"BUTT N, 1,2,3,4,X"

TASK 2 LOAD THE UNIVERSALPERSONALITY
(continued)**Keyboard:** Optr's **Lock:** ENGR**Reference:** This Document**Personality Loaded:** None; reset

Step	Action	Response
2.60	Press (2) and then press (ENTER)	SYSTEM STATUS Display appears
2.64	Select " MOUNT STD ABST FOR USnn "	Source targets appear
2.68	Select [ALTERNATE SOUCE]	Additional targets appear
2.72	Select drive that contains &C1 or &C2	<u>EG:</u> [1]
2.74	Select [EXECUTE COMMAND]	Abstract data is read from &C1 or &C2
2.78	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
2.101	Select [SUPPORT UTILITIES]	SUPPORT UTILITY MENU
2.102	Select [MODIFY VOLUME PATHS] display	MODIFY DEFAULT VOL PATHNAMES
2.103	Select [SET DEVICE PATH TO REM. MEDIA]	Paths are changed from NET to \$Fn, \$Fn+1 (EG: \$F1 and \$F2)
2.104	Select [MAIN MENU]	ENGINEERING MAIN MENU
2.110	Select [CONSOLE STATUS]	CONSOLE STATUS AND ASSIGN display
2.120	Record device numbers for the printer and the drives	PRINTER DRIVES ACCESS -----, ----- ENGR
	Skip to step 3.30	Note: If the drive numbers are other than 1 and 2, remember to substitute the correct numbers in place of \$F1 and \$F2 or FD1 and FD2 wherever these are used in the tasks that follow.

TASK 3 SET TIME AND DATE**Keyboard:** Engr's **Lock:** ENGR**Reference:** This Document**Personality Loaded:** Universal

Step	Action	Response
3.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
3.20	Select [SYSTEM STATUS]	CONSOLE STATUS AND ASSIGN display
3.30	Select [TIME/DATE]	"ENTER TIME: HH MM" appears at top left
3.50	Key in hours-space-minutes	<u>EG:</u> 14 25
3.60	Press <ENTER>	<u>EG:</u> 14:25 in upper-right corner, "ENTER DATE: DD MMM YY" in upper left corner
3.80	Key in days-space-month-space-year	<u>EG:</u> 15 JUN 95
3.90	Press <ENTER>	<u>EG:</u> 15: Jun 95 in upper right corner

TASK 4 **FORMAT CARTRIDGE DISKS**

Keyboard: Engr's Lock: ENGR

Reference: SW11-507

Personality Loaded: Universal

Step	Action	Response
4.10	&C1 or &C2 into FD1	
4.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
4.30	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
4.35	Blank cartridge into FD2 (Write-Prot. switch on cartridge must be in "Write" position)	(Change position of switch only with disk outside of the drive)
4.50	Key in a Create Volume command.	<u>EG</u> : CR \$F2>CNCF>
4.51	Press <ENTER>	"VOLUME CORRUPTED" or "CREATE VOLUME COMPLETE"
4.52	Key in a Create Volume command.	<u>EG</u> : CR \$F2>CNCF> -F -MF 1000 -BS 512 -FD
4.53	Press <ENTER>	"CREATE VOLUME COMPLETE"

NOTE

A 20 MB cartridge disk supports up to 9995 files; however, some operations will be slowed as a function of the size of the directory; this example specifies 1000 files. Refer to the Command Processor manual for the maximum number of files supported by 35 MB and larger cartridges.

4.54	Key in a Create Directory command for the &ASY volume.	<u>EG</u> : CD \$F2>CNCF> &ASY
4.55	Press <ENTER>	"CREATE DIRECTORY COMPLETE"
4.58	Repeat steps 4.54 and 4.55 to create &KFO volume.	<u>EG</u> : CD \$F2>CNCF> &KFO
4.65	Remove the formatted cartridge from FD2 and write the date and Volume ID (For example, "CNCF") on the label.	This will be the system-specific NCF cartridge.
4.66	Repeat steps 4.35 – 4.53 (using same commands) and label it (For example, "Backup CNCF")	These volumes are used for backing up customer databases as suggested in the note at the end of Task 29.

TASK 4 **FORMAT CARTRIDGE DISKS**

Keyboard: Engr's Lock: ENGR

Reference: SW11-507

Personality Loaded: Universal

Step	Action	Response
4.70	Repeat steps 4.35 – 4.55 except use Volume/Directory names below to create the two additional volumes for future use. 1. Area Database for area button file for custom files Repeat step 4.70 if you have additional areas. 2. User volumes for data points for custom graphics for CL.	<u>EG:</u> CR \$F2>AR01> CR \$F2>AR01> -F -MF 1000 -FD CD \$F2>AR01> &D01 CD \$F2>AR01> BUTT CD \$F2>AR01> &CUS <u>EG:</u> CD \$F2>AR02> &D02 CD \$F2>AR03> &D03 <u>EG:</u> CR \$F2>USER> CR \$F2>USER> -F -MF 1000 -FD CD \$F2>USER> HMV1 CD \$F2>USER> PICT CD \$F2>USER> CL
4.80	Repeat steps 4.35 – 4.53 to create a checkpoint volume, and label it.	<u>EG:</u> CR \$F2>CKPT> CR \$F2>CKPT> -F -MF 1000 -FD

TASK 5 **COPY FILES**

Keyboard: Engr's Lock: ENGR

Reference: SW11-507

Personality Loaded: Universal

Step	Action	Response
5.10	If the COMMAND PROCESSOR display is still on the US go to step 5.20; if not, hold <CTL> down and press <HELP>. Then, Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
5.20	&C6 into FD1	
5.30	Load the CNCF cartridge disk created in steps 4.50 – 4.65 into FD2. Key in a copy command for the NCF files.	<u>EG:</u> CPV \$F1>&ASY> \$F2>&ASY> -D [where \$F1 and \$F2 are the drive numbers assigned in the &C6 disk (see step 2.120) or in Task 9 if the NCF has been installed (see step 9.95); these may differ]
5.35	Press <ENTER>	Lists files copied, followed by: "COPY VOLUME COMPLETE"
5.36	Key in a copy volume command for the Key File Option files.	<u>EG:</u> CPV \$F1>&KFO> \$F2>&KFO> -D
5.40	Press <ENTER>	"COPY VOLUME COMPLETE"

TASK 5 COPY FILES (continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-507**Personality Loaded:** Universal

Step	Action	Response
5.42	Key in a copy volume for the &D01 File (&D01 contains null Area 1 and Button Files).	<u>EG:</u> CPV \$F1>&D01> \$F2>&D01> -D
5.44	Press <ENTER>	"COPY VOLUME COMPLETE"
5.50	If you wish to copy any other volumes, insert the source into FD1, insert destination into FD2 and perform steps 5.30 – 5.35.	
5.60	If you intend to assign this US to any area 2 - 10, go perform steps 5.64 – 5.78. &C6 should be FD1 and CNCF in FD2.	Note: This US started up on Area 1. You will need the proper area database to reload if you intend to configure this US on another area.
5.64	Create a &Dnn directory for whatever Area you will assign this US to (nn = 02 - 10).	<u>EG:</u> CD \$F2>CNCF> &D03 (where nn = 03 for this example)
5.66	Press <ENTER>	Create Directory Complete
5.70	Copy/Rename the &D01 volume for whatever Area (Dnn) you will assign this US to.	<u>EG:</u> CPV \$F1>&D01> \$F2>&D03> -D (where nn = 03 for this example)
5.74	Press <ENTER>	"COPY VOLUME COMPLETE"
5.76	Rename the added area file within &Dnn	<u>EG:</u> RN \$F2>&D03>AREA01.DA AREA03
5.78	Press <ENTER>	RENAME COMPLETE

TASK 6 ESTABLISH CUSTOMER NCF**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605**Personality Loaded:** Universal

Step	Action	Response
6.10	&C1 or &C2 into FD1, CNCF into FD2	CNCF is the primary copy made in Task 5
6.20	If the COMMAND PROCESSOR display is on screen, go to step 6.50; if not, hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
6.30	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
6.50	Use FD2 device number from step 2.120 and delete the NCF files	<u>EG:</u> \$F2 <u>EG:</u> DL \$F2>&ASY>NCF.* -D
6.60	Press <ENTER>	"DELETE COMPLETE"
6.70	Key in a rename command (Rename customer NCF file)	<u>EG:</u> RN \$F2>&ASY>CSTMRNCF.CF NCF
6.80	Press <ENTER>	"RENAME COMPLETE"
6.90	Go to step 7.20. Be sure to read the warning below.	

WARNING

YOU SHOULD PERFORM TASKS 7 THROUGH 11 SEQUENTIALLY WITHOUT INTERRUPTION. DO NOT INTERRUPT THE SUGGESTED SEQUENCE AND DO NOT REMOVE THE CNCF DISK FROM THE DRIVE UNLESS YOU DO ONE OF THE FOLLOWING FIRST:

1. Return to Engineering Main Menu by holding <CTL> down and pressing <HELP>
2. Perform the Install NCF function (Task 11)

TASK 7 **CONFIGURE UNIT, AREA, CONSOLE NAMES** **Keyboard:** Engr's **Lock:** ENGR

Reference: SW11-605

Personality Loaded: Universal

NOTE

If you are not entering data from filled in configuration forms or if you want a printout of your entries, press <PRT SCN> (hold <CTL> down and press <CLR-ENT>) to get a screen print after each configured page has been entered in Tasks 7-10.

With R681, NCF print option enables you to print NCF to a text file. Initiate a print using F4 function key from any NCF form to print the complete NCF as a text file. It will be stored in NET>&ASY by default with the file name as NCF.XX. If the default filename is not desired, specify a text file name using the extensions ".X?" or ".Y?" or ".Z?".

Where,

? can be any alphabet.

If &ASY has memory shortage, you can print the NCF.XX to a different user volume.

You can use the existing functionality to abort printing – simultaneously press the keys CTL + Break. The printing is aborted and the partially printed text file is retained with a comment PRINT OPERATION ABORTED.

For an NCF parameter with multiple values, the values are printed continuously separated by a space. If the number of characters exceeds 80, the next line begins with &\, indicating it is a continuation of the previous line.

Note that in large networks having more than 64 nodes, the NCF text file takes approximately 30 minutes to print.

TASK 7 CONFIGURE UNIT, AREA, CONSOLE NAMES Keyboard: Engr's Lock: ENGR
(continued)

Reference: SW11-605

Personality Loaded: Universal

Step	Action	Response
7.10	&C1 or &C2 into FD1, CNCF into FD2	
7.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
7.30	Select [UNIT NAMES]	EDIT UNIT NAMES—PAGE 1 display
7.32	If the words OFF LINE appear at the end of the 2nd line on the screen header, skip to step 7.40.	
7.35	Hold <CTL> down and press <3>	Warning Message: "OFF LINE" (Ignore backup reminder for now)
7.38	Press <ENTER>	EDIT UNIT NAMES display
7.40	Key in a UNIT ID and DESCRIPTION for each line item as required.	<u>EG</u> : 01 and UNIT 1 for line 1
7.50	When finished, press <ENTER>	Values and ports reappear in cyan
7.52	Page forward, if necessary, and continue entries. Press <ENTER> after each page.	
7.60	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
7.70	Select [AREA NAMES]	EDIT AREAS NAMES display
7.80	Key in a AREA NAME and DESCRIPTOR for each line item as required. Do not leave blank lines between entries. Note: An Area Name cannot contain spaces.	<u>EG</u> : BOILER1 for line 1 Area Name <u>EG</u> : WEST AREA for Area 1 Description <u>EG</u> : REACTOR for line 2 Area Name <u>EG</u> : AREA 2 for Area 2 Description
7.90	When finished, press <ENTER>	Values and ports appear in cyan
7.100	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
7.110	Select [CONSOLE NAMES]	EDIT CONSOLES NAMES display
7.120	Key in a Console Description for each line item as required.	<u>EG</u> : WEST AREA CONSOLE for line item #1.
7.130	When finished, press <ENTER>	Values and ports appear in cyan

TASK 8A CONFIGURE SYSTEM WIDE VALUES**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605**Personality Loaded:** Universal

Step	Action	Response
8.10	Ensure &C1 or &C2 in FD1, CNCF in FD2	
8.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
8.30	Select [SYSTEM WIDE VALUES]	SYSTEM WIDE VALUES MENU display
8.33	Select [SYSTEM ID]	SYSTEM ID display
8.36	Key into SYSTEM DESCRIPTOR port	<u>EG</u> : PLANT NUMBER 1
8.39	Key into SYSTEM ID port	Note: No entry is required.
8.42	Press <ENTER>	Values and ports reappear in cyan
8.45	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.48	Select [CLOCK SOURCE]	CLOCK SOURCE display
8.51	Key into CLOCK LOCATIONS port (lowest number goes in left port)	<u>EG</u> : 24 and 38
8.54	Press <ENTER>	Values and ports reappear in cyan
8.57	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.60	Select [USER AVERAGE PERIOD]	USER AVERAGE PERIOD display
8.63	Select a User Average Period	<u>EG</u> : 5 MINUTES
8.66	Press <ENTER>	Values and ports reappear in cyan
8.69	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.72	Select [SHIFT DATA]	SHIFT DATA display
8.75	Key in changes as required	
8.78	Press <ENTER>	Values and ports reappear in cyan
8.81	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.84	Select [CONSOLE DATA]	CONSOLE DATA PAGE 1 display
8.87	Key in/select changes as desired	<u>EG</u> : (on page 3) Alarm Summary Freeze Timeout (seconds): 180
	Note: NCF data at the top of the first page is <u>read only</u> (for reference).	

TASK 8A CONFIGURE SYSTEM WIDE VALUES
 (continued)

Keyboard: Engr's **Lock:** ENGR

Reference: SW11-605

Personality Loaded: Universal

Step	Action	Response
8.90	Press <ENTER>	Values and ports reappear in cyan
8.95	Press <PAGE FWD>	CONSOLE DATA—Page n display
8.110	Repeat steps 8.87 – 8.95 until pages 1 through 8 have been viewed and modified as required.	Note: Pages 9 and beyond are reserved. Do not change them without specific instructions.
8.111	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.114	Select [SOFTWARE OPTIONS]	SOFTWARE OPTIONS display
8.118	Ensure CNCF in FD2	"No Software Options" message
8.122	Hold down <CTL> and press <0>	MOUNT port appears
8.124	Select [ALTERNATE]	MOUNT port appears
8.128	Key into MOUNT port	<u>EG:</u> \$F2
8.132	Press <ENTER>	Authorized Options appear in cyan (Do <u>Not</u> press F1 or F2 at this time)
8.136	Press <ENTER>	"BUSY"—SOFTWARE OPTIONS display
8.138	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
	If you do <u>not</u> have a Network Gateway, skip to step 8.172	Note that NG Local System Ids must be configured before NG LCN Nodes
8.140	Select [NG LOCAL SYSTEM ID]	NG LOCAL SYSTEM ID display
8.144	Key into LOCAL SYSTEM NUMBER port	<u>EG:</u> 1
8.148	Key into SYSTEM TITLE port (the first character must be an alpha)	<u>EG:</u> P1
8.152	Key into SYSTEM DESCRIPTION port	<u>EG:</u> Plant Number 1
8.156	Select a PIN ADDRESSING FORM option	<u>EG:</u> LOCAL
8.160	If you selected GLOBAL (above), key into GLOBAL VENDOR NUMBER port	<u>EG:</u> (Honeywell supplied IEEE number)
8.164	Press <ENTER>	Values and ports reappear in cyan
8.168	If you have other NGs repeat steps 8.144 through 8.164 for each. and HM volume configuration	Syst. Wide Value configuration for NGs continues in Task 8B after node configuration

TASK 8A CONFIGURE SYSTEM WIDE VALUES
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605**Personality Loaded:** Universal

Step	Action	Response
8.172	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.176	Select [TAG NAME OPTIONS]	TAG NAMES display
8.180	Select a Tag Name option	<u>EG:</u> LONG
8.184	If you selected LONG, make a Formatting choice	<u>EG:</u> YES
8.188	Press <ENTER>	Values and ports reappear in cyan
8.192	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.196	Select [AM MULTIPLE PRIMMOD ALARMING OPTIONS]	AM MULTIPLE PRIMMOD ALARMING OPTIONS DISPLAY
8.200	Do either Step A. or Step B.: A. Select ENABLED EXCLUSIVE MODE target to cause alarms to be reflected in alarm groups for those \$MPROD parameters that have values. B. Select ENABLED INCLUSIVE MODE target to cause alarms to be reflected in alarm groups for those \$MPROD parameters that have values and also in the alarm group for PRIMMOD parameter.	
8.204	Press <ENTER>	Values reappear in cyan
8.208	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.212	Select [HG SP/OP CLAMP OPTION]	HG SP/OP CLAMP OPTION DISPLAY
8.216	Select ENABLE OPTION target to enable the HG SP/OP Clamp Option functionality.	
8.220	Press <ENTER>	Values reappear in cyan
8.224	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.228	Hold <CTL> down and then press <HELP>	ENGINEERING MAIN MENU

TASK 9 CONFIGURE LCN NODES**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605**Personality Loaded:** Universal**WARNING**

You must configure the Universal Station you are using for system startup; otherwise, you will not be able to successfully start up your system. During Network Configuration, each display where you enter data must say OFF LINE at the end of line 2.

Step	Action	Response
9.10	Ensure &C1 or &C2 in FD1, CNCF in FD2	
9.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU

TASK 9 CONFIGURE LCN NODES
 (continued)

Keyboard: Engr's **Lock:** ENGR

Reference: SW11-605

Personality Loaded: Universal

Step	Action	Response
9.30	Select [LCN NODES] If necessary, <PAGE FWD> until you see the node number for one of your USs.	LCN NODE CONFIG—Page 1 display <u>EG:</u> NODE 1 LCN NODE CONFIG—Page n display
9.50	Select a US Node Number	(<u>EG:</u> 1) SELECT NODE TYPE display
9.60	Select [UNIVERSAL STATION]	UNIVERSAL STATION NODE display
9.70	Key into CONSOLE # port	<u>EG:</u> 01 (for nodes 1 and 2)
9.80	Key into STATION # port	<u>EG:</u> 1 for node 1 (2 for node 2)
9.90	Key into DEFAULT AREA port	<u>EG:</u> BOILER1 (for nodes 1 and 2)
9.95	Key into CARTRIDGE DISK NUMBERS ports	<u>EG:</u> 1 and 2 for node 1 <u>EG:</u> on next pass: 3 and 4 for node 2, and so on.
9.100	Key into FLOPPY DISK NUMBERS ports, if applicable	
9.110	Key into PRINTER NUMBER ports (Printer Assignment function allows screen prints to be made from any station, if at least one printer is present within the console)	<u>EG:</u> 1 for node 1, 2 for node 2, and so on. (configure a printer on any station that has a printer and stations from which you wish to make screen prints.
9.120	Key into TREND PEN NUMBER ports	<u>EG:</u> 1 (Blank if none implemented)
9.130	Select ACCESS LEVEL option	<u>EG:</u> [OPER] for nodes 1 and 2
9.134	Select DEFAULT LOAD PERS. option	<u>EG:</u> [UP] for nodes 1 and 2
9.140	Select OPERATORS KEYBOARD option	<u>EG:</u> [YES] for nodes 1 and 2
9.150	Select TOUCH SCREEN option	<u>EG:</u> [YES] for nodes 1 and 2 (also YES for U ^X S with mouse or trackball)
9.160	Select ENGINEERS KB option	<u>EG:</u> [YES] for nodes 1 and 2
9.175	<PAGE FWD> to enter values on the second page of the Universal Station display (fill in only if extra schematic memory or External Load Module software options are needed). This can be done later.	DO NOT PRESS <ENTER> until all pages of the LCN node are configured. Obtain entry details from the External Load Module documentation. Caution! make sure of adequate Memory for external Load Modules.
9.176	After last page is entered, press <ENTER>	Values and ports reappear in cyan
9.180	If more Universal Stations need to be configured, you can hold <CTL> down, press <PAGE BACK>, and go to step 9.50.	<u>EG:</u> 2 (configure as before, except Station 2, and so on.) Note that U ^X S stations are configured like a US, except as mentioned. DAT (tape) drives do not require configuration entries.

TASK 9 **CONFIGURE LCN NODES**
 (continued)

Keyboard: Engr's **Lock:** ENGR

Reference: SW11-605

Personality Loaded: Universal

Step	Action	Response
9.190	Hold <CTL> down, press <PAGE BACK>	LCN NODE CONFIG—Page 1 display
9.195	<PAGE FWD> look for NIM node number (if no NIM on system, skip to 9.254)	LCN NODE CONFIG—Page n display
9.200	Select a NIM Node Number. If no NIM on system, skip to 9.254.	<u>EG:</u> 24 , SELECT NODE TYPE display
9.210	Select [NETWORK INTERFACE MODULE]	NETWORK INTERFACE MODULE Display
9.220	Key into REDUNDANT MEMBER ID port	<u>EG:</u> 25 or 0 if none
9.230	Key into UNIV. CONT. NETWORK port	<u>EG:</u> 3
9.242	If you have External Load Modules that affect the NIM, press <PAGE FWD>	EXTERNAL LOAD MODULE display This can be done later with the On- Line Reconfiguration feature.
9.243	Key in data, select items as necessary	Obtain entry information from the External Load Module documentation.
9.244	After last data is entered, press <ENTER>	Values and ports reappear in cyan
9.245	Repeat 9.190 – 9.244 for each NIM node pair.	
9.250	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG—Page n display
9.254	<PAGE FWD> (if necessary) to NG node #. If no NG on system, skip to 9.284.	LCN NODE CONFIG—Page n display
9.258	Select a Network Gateway Node Number	<u>EG:</u> 30 , SELECT NODE TYPE display
9.260	Select [NETWORK GATEWAY]	NETWORK GATEWAY MODULE DISPLAY
9.264	Key into PIN LOCAL ADDRESS port	<u>EG:</u> 8
9.266	Key into PIN GLOBAL ADDRESS port (first half Honeywell provided global address, if any)	<u>EG:</u> No entry
9.268	Select CABLES FOR OUTPUT	<u>EG:</u> A
9.272	If you have External Load Modules that affect the NG, press <PAGE FWD>	EXTERNAL LOAD MODULE display This can be done later with the On- Line Reconfiguration feature.
9.274	Key in data, select items as necessary	Obtain entry information from the External Load Module documentation.
9.276	After last data is entered, press <ENTER>	Values and ports reappear in cyan
9.278	Repeat 9.250 – 9.276 for each NG node pair.	

TASK 9 CONFIGURE LCN NODES
 (continued)

Keyboard: Engr's **Lock:** ENGR

Reference: SW11-605

Personality Loaded: Universal

Step	Action	Response
9.280	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG—Page n display
9.284	<PAGE FWD> (if necessary) to HG node # (if the system doesn't have an HG or a PLCG, skip to step 9.318)	LCN NODE CONFIG—Page n display
9.288	Select an HG or PLCG Node Number Note: HG and PLCG LCN nodes are configured exactly the same, but must be on different Data Hiways.	(<u>EG</u> : 38) SELECT NODE TYPE display
9.290	Select [HIWAY GATEWAY]	HIWAY GATEWAY NODE display
9.294	Key into REDUNDANT MEMBER ID port	<u>EG</u> : 39 (using the next highest number for the REDUNDANT ID is recommended or 0 if none.)
9.298	Key into HIWAY NUMBER port	<u>EG</u> : 2
9.304	If you have External Load Modules that affect the HG, press <PAGE FWD>	EXTERNAL LOAD MODULE display. This can be done later with the On-line Reconfiguration feature.
9.308	Key in data, select items as necessary. Press <ENTER> after last page	Obtain entry information from the External Load Module documentation.
9.310	Repeat 9.280 - 9.308 for each HG node pair.	
9.314	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG—Page n display

TASK 9 **CONFIGURE LCN NODES**
 (continued)

Keyboard: Engr's **Lock:** ENGR

Reference: SW11-605, AM12-610, AX11-500 **Personality Loaded:** Universal

Step	Action	Response
9.318	<PAGE FWD>; look for AM node No. (If the system does not have an AM, skip to step 9.370).	LCN NODE CONFIG—Page n display
9.322	Select an AM or A ^X M Node Number	(EG: 40) SELECT NODE TYPE display
9.326	Select [APPLICATION MODULE]	APPLICATION MODULE NODE display
9.328	REDUNDANT MEMBER Node ID port	EG: 0 (Enter node number or enter 0 if no redundant node)
9.340	Select a STARTUP MODE option	EG: [Cold]
9.344	Key into ASSIGNED UNITS port (the units must have been previously configured) <u>Do not press Enter yet.</u>	EG: 01 (multiple units must be separated by commas or spaces, for example, 01, A1,)
9.346	If AM/CL Background functions are used, or if this is an A ^X M, or if Custom AM Software will be used Go to the next step— If not, skip to step 9.360	
9.350	<PAGE FWD> to Page 2. Type in Page 2 information. Obtain from: <i>AM Implementation Manual</i> (Background/CL) <i>A^XM System Administration Manual</i> (A ^X M) or Custom AM Software instructions. These numbers may vary according to system requirements. <u>Do not press Enter yet.</u>	USER MEMORY ALLOCATION display. Note that Page 2 and 3 data can be entered or changed later with the On-Line Recon- figuration feature if you are uncertain. Typical entries for A ^X M/Background/CL are: # BACKGROUND CL TASKS 10 # CONCURRENT DATA ACCESS 4 BACKGROUND TASK SIZE 15000 CVB SIZE (Default) 2000 INCLUDE INTERNETWORK PROC. YES USER MEMORY RESERVED 1

TASK 9 **CONFIGURE LCN NODES**
 (continued)

Keyboard: Engr's **Lock:** ENGR

Reference: SW11-605, AM12-610, AX11-500 **Personality Loaded:** Universal

Step	Action	Response														
9.354	<PAGE FWD> to Page 3. Enter information per same manuals as used in the previous steps. A combination of entries is allowable for multiple functions.	EXTERNAL LOAD MODULE NAMES display. Typical entries for Background/CL are: <table><tr><td>NAME</td><td>PERSONALITY</td></tr><tr><td>FILE</td><td>AMO</td></tr><tr><td>CONV</td><td>AMO</td></tr><tr><td>AMCL03</td><td>AMO</td></tr></table> Typical entries for A^XM are: <table><tr><td>NAME</td><td>PERSONALITY</td></tr><tr><td>AMCL06</td><td>AMO</td></tr><tr><td>XACCES</td><td>AMO</td></tr></table>	NAME	PERSONALITY	FILE	AMO	CONV	AMO	AMCL03	AMO	NAME	PERSONALITY	AMCL06	AMO	XACCES	AMO
NAME	PERSONALITY															
FILE	AMO															
CONV	AMO															
AMCL03	AMO															
NAME	PERSONALITY															
AMCL06	AMO															
XACCES	AMO															
9.360	Press <ENTER>	Page 1 and (if necessary Page 2/3) complete														
9.365	Repeat 9.314- 9.360 for each AM or A ^X M															
9.366	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG—Page n display														
9.370	<PAGE FWD>; look for HM node No. (If the system does not have an HM, skip to step 9.410).	LCN NODE CONFIG—Page n display														
9.372	Select an HM Node Number	<u>EG:</u> 43 SELECT NODE TYPE display														
9.376	Select [HISTORY MODULE]	HISTORY MODULE NODE display														
9.384	Key into REDUNDANT MEMBER NODE port.	<u>EG:</u> 0 (enter redundant node number or enter 0 if no redundant node)														
9.388	Key into NODE PAIR NUMBER port	<u>EG:</u> 1														
9.398	If you have External Load Modules that affect the HM, press <PAGE FWD> (if External Load Modules not used, skip to 9.406)	EXTERNAL LOAD MODULE display This can be done later with the On-Line Reconfiguration feature.														
9.402	Key in data, select items as necessary. Press <ENTER> after last page.	Obtain entry information from the External Load Module documentation.														
9.406	Repeat 9.370 – 9.402 for each HM															

TASK 9 **CONFIGURE LCN NODES**
 (continued)

Keyboard: Engr's **Lock:** ENGR

Reference: SW11-605

Personality Loaded: Universal

Step	Action	Response
9.410	Hold down <CTL>, press<PAGE BACK>	LCN NODE CONFIGURATION display
9.412	<PAGE FWD> look for CM node number (if no CM on system, skip to step 9.454)	LCN NODE CONFIG—Page n display
9.416	Select a CM Node Number	<u>EG:</u> 51 , SELECT NODE TYPE display
9.420	Select [COMPUTING MODULE]	COMPUTING MODULE NODE display
9.422	Key in ASSIGNED UNITS port	<u>EG:</u> 01 ,02,03,04 or 01 02 03 04
9.430	If you have External Load Modules that affect the CM, press <PAGE FWD> (if External Load Modules not used, skip the next step)	EXTERNAL LOAD MODULE display This can be done later with the On- Line Reconfiguration feature.
9.436	Key in data, select items as necessary. Press <ENTER> after last page	Obtain entry information from the External Load Module documentation.
9.440	Repeat 9.410 – 9.436 for each CM	
9.446	Hold <CTL> down, press <PAGE BACK>	LCN NODE CONFIG—Page n display
9.454	To print the LCN Node Config. hold <CTL> down, then press <4>. Select printer target, then press <ENTER>.	(US must be connected to printer) Print instructions appear. LCN configuration is printed.

TASK 10 CONFIGURE HM VOLUMES**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605, HM13-500**Personality Loaded:** Universal

Step	Action	Response
10.10	&C1 or &C2 into FD1, ensure CNCF in FD2	
10.13	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
10.16	Select [VOLUME CONFIGURATION]	HM PAIR SELECTION MENU
10.21	Select a Node Pair number	<u>EG</u> : 1 VOLUME CONFIGURATION display
10.24	Key in the # of Winchester disk drives in the History Module	<u>EG</u> : 1 (each redundant pair counts as 1)
10.27	Key in the type or size of disks in the HM	<u>EG</u> : 3 for Wren 3 <u>EG</u> : 4 or 215040 for 210 MB SCSI drive <u>EG</u> : 5 or 454130 for 445 MB SCSI drive <u>EG</u> : 6 or 896046 for 875 MB SCSI drive

TASK 10 **CONFIGURE HM VOLUMES**
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605, SW09-605**Personality Loaded:** Universal

Step	Action	Response																												
10.180	Select [CHECKPOINT]	P.N. FOR CHECKPOINT CONFIG display																												
10.190	Key into ports for each Node Pair Number line item as required. Note 1: Enter zero for HG or PLCG Vol. Size & No. of Files. Configure an A ^X M the same as an AM. Note 2: Redundant node members— enter sequentially with Size/Files = 0	<table><tr><td><u>EG:</u></td><td><u>NODE NO.</u></td><td><u>VOLUME SIZE</u></td><td><u>NUMBER FILES</u></td></tr><tr><td>NIM</td><td>24</td><td>20000</td><td>100</td></tr><tr><td>NIM</td><td>25</td><td>0</td><td>0</td></tr><tr><td>PLCG</td><td>31</td><td>0</td><td>0</td></tr><tr><td>HG</td><td>38</td><td>0</td><td>0</td></tr><tr><td>HG</td><td>39</td><td>0</td><td>0</td></tr><tr><td>AM</td><td>40</td><td>10000</td><td>200</td></tr></table> Numbers used were for a Wren 3. Increase them for larger HM drives (refer to SW09-505).	<u>EG:</u>	<u>NODE NO.</u>	<u>VOLUME SIZE</u>	<u>NUMBER FILES</u>	NIM	24	20000	100	NIM	25	0	0	PLCG	31	0	0	HG	38	0	0	HG	39	0	0	AM	40	10000	200
<u>EG:</u>	<u>NODE NO.</u>	<u>VOLUME SIZE</u>	<u>NUMBER FILES</u>																											
NIM	24	20000	100																											
NIM	25	0	0																											
PLCG	31	0	0																											
HG	38	0	0																											
HG	39	0	0																											
AM	40	10000	200																											
10.200	Press <ENTER>	Values and ports appear in cyan																												
10.210	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display																												
10.310	Select [CONTINUOUS HISTORY]	CONT. HIST. UNIT AND GROUPS display																												
10.315	Key into UNIT ID port	<u>EG:</u> 01																												
10.320	Key into NBR GROUPS port	<u>EG:</u> 60																												
10.325	Continue entries as required																													
10.330	Press <ENTER>	Values and ports reappear in cyan																												
10.335	Press <PAGE FWD> and repeat steps 10.315 – 10.330 until all required entries are complete.																													
10.350	Select [GROUP OPTIONS] (you can select the [HM/UNIT OPTIONS] display target if desired for a display that allows you to assign options for all).	CONT. HISTORY GROUP OPTIONS																												
10.360	Select a SAVE RATE (you must select a save rate to get <u>any</u> history)	<u>EG:</u> 20																												
10.365	Select either [SNAPSHOTS], [USER AVG] or both (you must select SNAPSHOTS to get trend data).	<u>EG:</u> Select both [SNAPSHOTS] and [USER AVG]																												
10.370	Key into PREARCHIVE HOURS port for SNAPSHOTS	<u>EG:</u> 48																												

TASK 10 **CONFIGURE HM VOLUMES**
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605, SW09-605**Personality Loaded:** Universal

Step	Action	Response
10.373	Key into PREARCHIVE HOURS port for USER AVG	<u>EG:</u> 48
10.376	ARCHIVE [YES] [NO] Function not released	Leave [NO] options selected
10.380	Repeat steps 10.360 – 10.376 for all required entries.	
10.385	Press <ENTER>	Values and ports reappear in cyan
10.390	<PAGE FWD> and repeat steps 10.354 – 10.385 for all required entries.	
10.400	Hold <CTL> down, press <PAGE BACK>	CONT. HISTORY UNIT AND GROUPS display
10.410	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display
10.510	Select [JOURNALS]	PROCESS UNIT JOURNAL CONFIG. display
10.515	Key into UNIT ID port	<u>EG:</u> 01 Note that Unit numbers you specify here are also the units from which SOE events are collected (if you specified an SOE Journal size for this HM node pair).
10.535	Key into the PROCESS ALARM port	<u>EG:</u> 2024
10.540	Key into the PROCESS CHANGE port	<u>EG:</u> 2024
10.545	Key into the OPERATOR MESSG port	<u>EG:</u> 600
10.548	Repeat steps 10.515 – 10.545 for other applicable Units	
10.550	Press <ENTER>	Values and ports reappear in cyan
10.560	Press <PAGE FWD>	Page n of screen forms appears
10.570	Repeat steps 10.515 – 10.560 for all applicable Units	
10.575	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display
10.630	Select [CL STORAGE]	CL STORAGE VOLUME CONFIGURATION display
10.634	Select a HIWAY FILE MGR Description choice	<u>EG:</u> NO
10.638	Select a NETWORK FILE MGR Description choice	<u>EG:</u> NO

TASK 10 **CONFIGURE HM VOLUMES**
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605, SW09-605**Personality Loaded:** Universal

Step	Action	Response																
10.640	Key into HIWAY/NETWORK NBR. port	<u>EG</u> : 02																
10.650	Key into NUMBER OF FILES port	<u>EG</u> : 30																
10.660	Key into VOLUME SIZE port	<u>EG</u> : 100																
10.670	Repeat steps 10.640-10.660 for other applicable hiways																	
10.680	Press <ENTER>	Values and ports reappear in cyan																
10.690	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display																
10.800	Select [USER FILE STORAGE]	USER FILE STORAGE CONFIG. display																
10.850	Key into Node Pair Number n ports for each line item as required (HNV1 is a general HM user volume, CL is for CL programs, CSW1 is for custom programs). Numbers used were for a Wren 3. Increase them for larger drives.	<u>EG</u> : <table><tr><td><u>Vol ID</u></td><td><u># of Files</u></td><td><u>Vol Size</u></td><td><u>File Mgr. Descriptor</u></td></tr><tr><td>HNV1</td><td>1000</td><td>15000</td><td>FAST</td></tr><tr><td>CL</td><td>200</td><td>3000</td><td>NO</td></tr><tr><td>CSW1</td><td>200</td><td>3000</td><td>YES</td></tr></table>	<u>Vol ID</u>	<u># of Files</u>	<u>Vol Size</u>	<u>File Mgr. Descriptor</u>	HNV1	1000	15000	FAST	CL	200	3000	NO	CSW1	200	3000	YES
<u>Vol ID</u>	<u># of Files</u>	<u>Vol Size</u>	<u>File Mgr. Descriptor</u>															
HNV1	1000	15000	FAST															
CL	200	3000	NO															
CSW1	200	3000	YES															
10.860	Press <ENTER>	Values and ports reappear in cyan																
10.870	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display																
	CHECK HM VOLUME CONFIGURATION																	
10.880	Hold <CTL> down and press <1>. This performs the CHECK function	Messages: "NCF FILE IS BEING PACKED INTO MEMORY...NEW NCF.WF IS BEING COPIED....HM CHECKER RUNNING."																
10.881	To configure volumes for another HM, repeat all steps in Task 10.																	
10.885	If errors are indicated, go to step 10.890; otherwise, skip ahead to Task 8B.																	
10.890	Hold <CTL> down and press <8>	NCF.ER file displayed																
10.920	If you want a printed copy, hold <CTL> down and press <4>. Make sure the printer is on.	Print Instructions appear																
10.930	Select printer target, press <ENTER>	NCF.ER file is printed																

TASK 10 CONFIGURE HM VOLUMES
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605, SW09-605**Personality Loaded:** Universal

Step	Action	Response
10.940	If total HM size exceeds 100%, If the number of virtual volumes is exceeded,	Go to step 10.944 and reduce the size of the User Volume. Go to step 10.944 and delete one or more User volumes, or delete the dump volume (10.112). Else call a Honeywell support engineer.
10.944	Hold <CTL> down and press <PAGE BACK>	EDIT VOLUME CONFIG display
10.948	Go repeat steps 10.800 – 10.880 until errors are cleared.	
10.950	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU

TASK 8B CONFIGURE SYSTEM WIDE VALUES
(continued)**Keyboard:** Engr's **Lock:** ENGR

(Network Gateway Completion)

Reference: SW11-505**Personality Loaded:** Universal

If you do **not** have a Network Gateway, skip to Task 11. Note that the following steps could not be completed until the NG nodes were configured (Task 9). The following section completes NG configuration.

Step	Action	Response
8.944	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
8.948	Select [SYSTEM WIDE VALUES]	SYSTEM WIDE VALUES MENU
8.950	Select [NG REMOTE SYSTEMS]	NG REMOTE SYSTEM display
8.952	Key into # port	<u>EG</u> : 2
8.953	Key into TITLE port	<u>EG</u> : P2
8.954	Key into DESCRIPTOR port	<u>EG</u> : Plant NO. 2
8.956	Select a CABLE FOR OUTPUT	<u>EG</u> : A
8.958	Key into RESP NG port	<u>EG</u> : 30
8.960	Key into REMOTE SERVERS RESP LOCAL port	<u>EG</u> : 16
8.962	Key in other ports as desired	
8.964	Press <ENTER>	Values and ports reappear in cyan

TASK 8B CONFIGURE SYSTEM WIDE VALUES**Keyboard:** Engr's **Lock:** ENGR

(continued)

(Network Gateway Completion)

Reference: SW11-605**Personality Loaded:** Universal

Step	Action	Response
8.965	Page forward and continue to make entries if necessary. Press <ENTER> after each page.	
8.966	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.968	Select [NG SECURITY DATABASE]	NG SECURITY DATABASE AUTH. display
8.970	Key in a REMOTE SYSTEM NUMBER or select ALL SYSTEMS	<u>EG:</u> ALL SYSTEMS
8.972	Select [DEFINE ACCESS EXCEPTIONS]	NG SECURITY DATABASE AUTHORIZ. display
8.974	Select a default access for System Vol/Dir.	<u>EG:</u> READ ONLY
8.976	Select a default access for USER Vol/Dir.	<u>EG:</u> READ /WRITE
8.977	Key in a SYSTEM VOL name under EXCEPTIONS if desired and select the access level.	EG: no entry
8.978	Key in a USER VOL name under EXCEPTIONS if desired	<u>EG:</u> CSW1
8.980	Select default access for User Vol/Direc.	<u>EG:</u> READ ONLY
8.982	Page Forward to Parameter Access Defaults	<u>EG:</u> Page 2
8.984	Select a default access for Parameters	<u>EG:</u> READ ONLY
8.985	Select a NODE TYPE for EXCEPTIONS	<u>EG:</u> AM
8.986	Select a HIWAY/UNIT	<u>EG:</u> 01
8.987	Select Exceptions for Parameter Access	<u>EG:</u> READ/WRITE
8.988	Continue entires for other node types. When all entries are complete, press <ENTER>	Values and ports reappear in cyan
8.990	Hold <CTL> down and then press <PAGE BACK>	NG SECURITY DATABASE AUTHORIZ. display
8.994	Hold <CTL> down and then press <PAGE BACK> Unless Honeywell has requested that you change NG Modem parameters, skip the NG MODEM DEFINITION configuration target.	SYSTEM WIDE VALUES MENU

TASK 11 **INSTALL THE NCF**
Reference: SW11-605

Keyboard: Engr's **Lock:** ENGR
Personality Loaded: Universal

Step	Action	Response
11.01	Ensure &C1 or &C2 in FD1, CNCF in FD2	
11.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
11.30	Select [UNIT NAMES]	Units—Page 1 display
11.40	Press <ENTER>	
11.50	Hold <CTL> down and press <1>	"NCF FILE IS BEING PACKED INTO MEMORY...NEW NCF.WF IS BEING COPIED HM CHECKER RUNNING"
11.52	Hold <CTL> down and press <2>	"WARNING:...PRESS ENTER"
11.60	Press <ENTER>	"OFFLINE INSTALL COMPLETE— DATE/TIME)"
11.65	After the "INSTALL COMPLETE" message appears, remove the CNCF cartridge disk from FD2 and write the date on the label.	
11.66	Make a backup copy of your CNCF . Primary CNCF into FD1 and the backup CNCF disk from step 4.66 into FD2.	
11.67	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
11.68	Key in a disk copy command.	<u>EG:</u> FCP \$F1 \$F2
11.70	Press <ENTER>	"Media Copy Complete"
11.80	Remove both cartridges from the drives.	If your primary CNCF disk becomes corrupted, you have a backup copy.

TASK 12 **RELOAD THE UNIVERSAL
PERSONALITY AND SET
TIME AND DATE**

Keyboard: N/A **Lock:** N/A

Reference: This Document

Personality Loaded: N/A

Step	Action	Response
12.05	Go perform Task 1 (Reset all nodes)	Note: Not required for nodes previously reset.
12.10	Go perform Task 2 using the CNCF from Task 11 and return here	You must reload the US using your site-specific CNCF (from Task 11)
12.20	Go perform Task 3 and return here	Current time/date in "this US"
12.30	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU

Figure 3-4 shows a graphic representation of the Tasks (13-16) that must be performed to prepare the History Module.

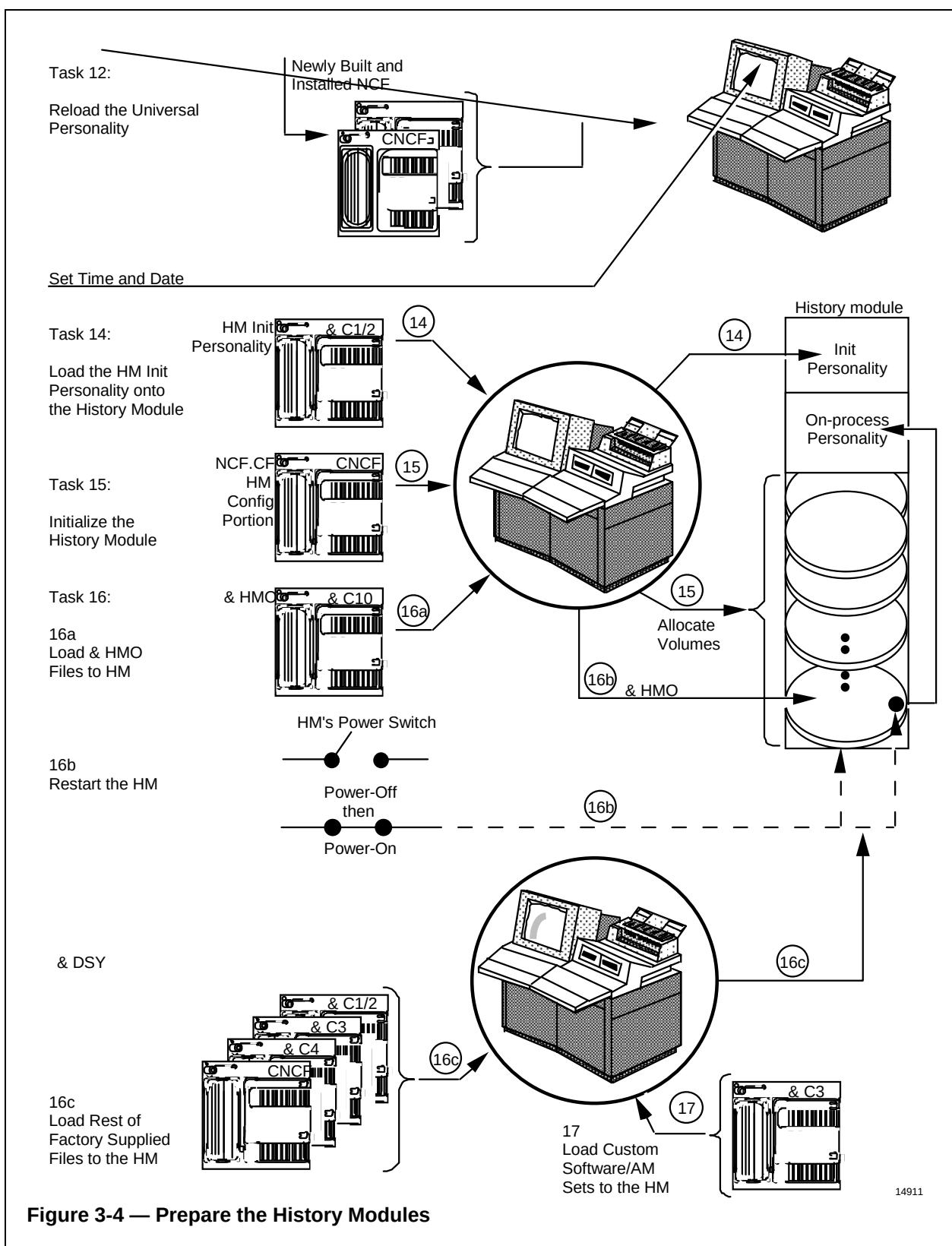


Figure 3-4 — Prepare the History Modules

13, 14 **PROCEDURE 3 – PREPARE THE HISTORY MODULE(S)** 13, 14

TASK 13 **SEE APPENDIX A IF YOU WISH TO LOAD ANOTHER US AT THIS POINT.**

TASK 14 **LOAD THE HISTORY MODULE WITH THE** **Keyboard:** Both **Lock:** ENGR
HM INITIALIZATION PERSONALITY

Reference: This Document

Personality Loaded: Universal

Step	Action	Response
14.05	Turn power <u>ON</u> at all HMs. RECORD THE TIME _____	HM Power was turned off in Task 1
14.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
14.15	Select [SYSTEM STATUS]	SYSTEM STATUS display
14.18	Select [DE-SELECT]	All nodes de-selected
14.20	Select the HISTORY MODULE Node	HM Node box selected
14.24	Select [NODE STATUS]	HISTORY MODULES display (Ignore HM status)

CAUTION

The HMs must have had power applied for at least 5 minutes before you press an HM's Reset button in the next step. Make sure you press it only once.

14.30	Reset each HM by pressing white RESET button on HM module.	Wait until the HM Status changes from FAIL or PWR-OFF to POWER_ON.
14.35	If any HM does not show POWER_ON status, turn its power off for a few seconds; then turn its power on and repeat step 14.30.	
14.40	Insert &C10 in FD1, CNCF in FD2	EG: 43 (Number changes color) LOAD/DUMP targets appear PGM targets appear PGM SOURCE FOR NODE NN display EG: Boxes [1] and [2] appear "VOL ID: &C10" and targets appear DATA SOURCE FOR NODE NN display EG: Boxes [1] and [2] appear "VOL ID: CNCF" and targets appear HISTORY MODULES display
14.50	Select Number under NODE heading on the HISTORY MODULE display	
14.60	Select [LOAD/DUMP]	
14.64	Select [MANUAL LOAD]	
14.70	Select [INIT PROGRAM]	
14.80	Select [ALTERNATE SOURCE]	
14.90	Select [1] (if this is the number for the first cartridge drive entered in step 9.95/9.100)	
14.100	Select [EXECUTE COMMAND]	
14.110	Select [ALTERNATE SOURCE]	
14.120	Select [2]	
14.130	Select [EXECUTE COMMAND]	

TASK 14 **LOAD THE HISTORY MODULE WITH THE HM INITIALIZATION PERSONALITY** **Keyboard:** Both **Lock:** ENGR
(continued)

Reference: This Document

Personality Loaded: Universal

Step	Action	Response
14.140	Press <ENTER>	Messages appear at the top of the screen "SINGLE LOAD; BUSY: HM NN, STATUS: LOAD, ...TYPE: HMOF, . . . OK" This can take several minutes to complete.
14.150	Repeat 14.50 - 14.140 to load the HM Init Personality to other HMs	

NOTE

1. If you get an error message, refer to the TPS system *Messages Directory* for possible error recovery procedures.
2. From this point on, the cartridge drive numbers will be as entered in Step 9.95/9.100 and may differ from the examples.

TASK 15 **INITIALIZE THE HISTORY MODULE**

Keyboard: Engr's **Lock:** ENGR

Reference: This Document

Personality Loaded: Universal

Step	Action	Response
15.10	Insert &C1 or &C2 in FD1, CNCF in FD2	
15.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
15.30	Select [VOLUME CONFIGURATION]	HISTORY MODULE PAIR SEL. Menu
15.40	Select Node Pair Number	<u>EG: 1</u> (for Primary Node 43), VOLUME CONFIGURATION display
15.50	Hold <CTL> down and press <6>	"HM INITIALIZATION RUNNING... DISC SURFACE HAS BEEN INITIALIZED AND A NEW LOCAL-NVCF FILE HAS BEEN WRITTEN"....
	Note: Requires several minutes to complete	

NOTE

If you get an error message, refer to the **TotalPlant** Solution (TPS) system *Messages Directory* publication, for possible error recovery procedures.

15.60	Hold <CTL> down, press <PAGE BACK>	HISTORY MODULE PAIR SELECTION Menu
15.70	Repeat 15.40 – 15.60 for each HM	
15.80	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU

TASK 16 COPY SOFTWARE FROM CARTRIDGES TO THE HISTORY MODULE Keyboard: Engr's Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Step	Action	Response
	HISTORY MODULE PREPARATION	
16.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
16.20	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
16.30	Check that the HM Local Vol. file is on the HM by keying in a Catalog command.	<u>EG:</u> CAT PN:NN>!9np>*.* where NN = HM Node Number and np = node pair (CAT PN:43>!901 for this sample system)
16.40	Press <ENTER>	"CATALOG COMPLETE"
16.50	If the L01_NVCF MM file is listed on the display, continue with step 16.70; else call for help.	Your Honeywell engineer
16.60	Assure printer is powered on. upcoming steps	To obtain a record of the files transferred in the
16.70	Key in " PD ON "	
16.80	Press <ENTER>	"Prompt To Dataout Turned On"
16.90	Key in " DO (Print Device ID) "	<u>EG:</u> DO \$P1
16.100	Press <ENTER>	"DATA OUT COMPLETE" Paper advances in printer
		Note: Printer must be connected to the station
16.110	Insert &C10 in FDI	
16.120	Key in an EC command for the Local Volume	<u>EG:</u> EC \$F1>&EC>LOC_VOL.EC \$F1 43 01
16.130	Press <ENTER>	"EC LOC_VOL.EC THIS EC IS USED TO COPY....."
At this point you have initiated an "Execute Command" program to simplify the task of loading a series of files from cartridge to the HM. You will be asked a series of questions to which you must respond by keying in a "Y" for yes or an "N" for no, followed by pressing <ENTER>. NOTES: 1. The &HMI files can be copied to more than one HM. 2. The &HMO files must be copied to <u>all</u> HMs. 3. &C10 contains both 68020 and 68040 HMO & HMI software. Answers to the EC questions will determine which type is loaded (HMs with K4LCN boards require 68040 software).		
16.140	Shown here are the files that should be transferred during the execution of the above EC file. Check your printout to determine that all required files have been loaded on the HM(s).	
	___ &LDR ___ &HMI (HMOFxxx) files ___ &HMO (HMONxxx) files When the EC program is complete, the message "Write Boot ...EC Complete" appears.	
16.150	If you have other HMs, return to Step 16.120 to copy the &HMO (&HMI if desired) directory to the additional HM(s).	

TASK 16 COPY SOFTWARE FROM CARTRIDGES TO THE HISTORY MODULE (continued) **Keyboard:** Engr's **Lock:** ENGR

Reference: This Document

Personality Loaded: Universal

Step	Action	Response
NOTE The &ASY directory in the following steps must be copied to <u>only one</u> HM, your system HM.		
16.160	Ensure CNCF in FD2	
16.170	Copy the custom NCF files to the HM	<u>EG:</u> CP \$F2>&ASY>*. * PN:43>&ASY>= -D
16.180	Press <ENTER>	"COPY COMPLETE"
RESTART THE HISTORY MODULE		
16.190	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
16.200	Select [SYSTEM STATUS]	SYSTEM STATUS display
16.210	Select the HISTORY MODULE node	Node is selected
16.230	Select [SHUT DOWN]	Target changes color
16.240	Press <ENTER>	Status changes to QUALIF
16.250	At the HM module, turn power OFF , wait at least 5 seconds, and then turn power ON .	Status of the HM changes to: LOC LOAD, then READY, and finally OK. (Redundant disks may have a status of DISKPROB until synchronized.) Type: HMON On-process personality is now running in the HM This process can take several minutes.

NOTE

If you get an error message, check the TPS system *Messages Directory* publication (see References), for possible error recovery procedures.

TASK 16 **COPY SOFTWARE FROM CARTRIDGES TO THE HISTORY MODULE** (continued) **Keyboard:** Engr's **Lock:** ENGR

Reference: This Document

Personality Loaded: Universal

Step	Action	Response
	COPY REST OF SOFTWARE TO THE HM	
16.260	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
16.270	Select [COMMAND PROCESSOR]	Command Processor display
16.280	Assure printer is on	To obtain a record of the files transferred in the upcoming steps
16.290	Key in " PD ON "	
16.300	Press <ENTER>	"Prompt To Dataout Turned On"
16.310	Key in " DO (Printer Device ID)"	<u>EG:</u> DO \$P1
16.320	Press <ENTER>	"DATA OUT COMPLETE" Paper advances in printer
16.330	If you want to enter files for modules with 68020 microprocessors, ensure &C2 in FD1 and key in an EC command for the 68020 personality volume. For 68040 processor files, ensure &C1 in FD1 and key in an EC command.	<u>EG:</u> EC \$F1>&EC>PERS2VOL.EC \$F1
16.340	Press <ENTER>	<u>EG:</u> EC \$F1>&EC>PERS4VOL.EC \$F1 "EC PERS2VOL.EC" or "EC PERS4VOL.EC" "THIS EC IS USED TO COPY....."
You should answer "Y" for each of the following files that is needed by a node that you have on your system. If you have inserted &C2 into FD1, answer "Y" only for nodes that use 68020 microprocessors. Then, for nodes that have 68040 microprocessors, place &C1 in FD1 and repeat the procedure starting with Step 16.330. Answer "Y" for the files applicable to those nodes. In either case, to save HM space, decline the copying of files for nodes that you do not have. As part of this procedure, you may be requested to place other cartridges into the drive. The EC program you invoke will select the correct software from &C3, &C4, &C9, and &C10 for a Yes answer.		

TASK 16 **COPY SOFTWARE FROM CARTRIDGES TO THE HISTORY MODULE** (continued) **Keyboard:** Engr's **Lock:** ENGR

Reference: This Document

Personality Loaded: Universal

Step																																																																																																																								
16.350	This step provides a list of the files that can be transferred during the execution of the above EC file(s), and the module with which each is associated. The "PERS2VOL" EC loads files to nodes with 68020 microprocessors and the "PERS4VOL" EC loads files to nodes with 68040 microprocessors. Most of the files on &C1 are for 68040 microprocessors and most of the files on &C2 are for 68020 microprocessors (in some case there is no distinction). This software is normally copied to <u>only</u> the system HM. Note that file &LDR is loaded along with the other files. <table><tr><th>Module</th><th>&C1</th><th>&C2</th><th>&C3</th><th>&C4</th><th>&C9</th><th>&C10</th></tr><tr><td>US II</td><td>___ &OPR</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>US IV</td><td>___ &OPR^k</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>US II</td><td>___ &UNP</td><td colspan="5">(Also requires overlay files &OP1 on &C1 and &C2).</td></tr><tr><td>US IV</td><td>___ &UNP^k</td><td colspan="5">(Also requires overlay files &OP1 on &C1 and &C2).</td></tr><tr><td>US</td><td>___ &OVG</td><td>___ &OVG</td><td></td><td></td><td></td><td></td></tr><tr><td>HG II</td><td></td><td></td><td></td><td></td><td>___ &HGO</td><td></td></tr><tr><td>HG IV</td><td></td><td></td><td></td><td></td><td>___ &HGO^k</td><td></td></tr><tr><td>NIM II</td><td></td><td></td><td></td><td></td><td>___ &NMO</td><td></td></tr><tr><td>NIM IV</td><td></td><td></td><td></td><td></td><td>___ &NMO^k</td><td></td></tr><tr><td>PM, APM, HPM, LM, SM</td><td></td><td></td><td></td><td></td><td>___ &UCN</td><td></td></tr><tr><td>AM II</td><td></td><td>___ &AMO</td><td></td><td></td><td></td><td></td></tr><tr><td>AM IV</td><td></td><td>___ &AMO^k</td><td></td><td></td><td></td><td></td></tr><tr><td>CG II</td><td></td><td></td><td>___ &CIO</td><td></td><td></td><td></td></tr><tr><td>CG IV</td><td></td><td></td><td>___ &CIO^k</td><td></td><td></td><td></td></tr><tr><td>NG II</td><td></td><td></td><td></td><td>___ &NGO</td><td></td><td></td></tr><tr><td>NG IV</td><td></td><td></td><td></td><td>___ &NGO^k</td><td></td><td></td></tr></table> k means type 68040 software (required when the node contains a K4LCN board). When the EC asks if you want to create/copy &OVG, answer Yes if you want the Equipment List functions. When the EC asks where, specify a user volume such as HMV1 or CSW1. When the EC copies the AM files it asks if you want to create/copy &CUS and CLX. This is optional software, but you should probably copy it in unless you know that it is not needed. If you copy the Computer Gateway files, the EC asks if you want to copy &CUS/&CUS. You probably should. If the EC also asks where you want to create the directories, Key in the name of one of your User volumes such as CSW1 or HMV1. When the EC program is complete, the message "EC COMPLETE" appears. Check the printout to assure that all required files have been loaded.	Module	&C1	&C2	&C3	&C4	&C9	&C10	US II	___ &OPR						US IV	___ &OPR ^k						US II	___ &UNP	(Also requires overlay files &OP1 on &C1 and &C2).					US IV	___ &UNP ^k	(Also requires overlay files &OP1 on &C1 and &C2).					US	___ &OVG	___ &OVG					HG II					___ &HGO		HG IV					___ &HGO ^k		NIM II					___ &NMO		NIM IV					___ &NMO ^k		PM, APM, HPM, LM, SM					___ &UCN		AM II		___ &AMO					AM IV		___ &AMO ^k					CG II			___ &CIO				CG IV			___ &CIO ^k				NG II				___ &NGO			NG IV				___ &NGO ^k		
Module	&C1	&C2	&C3	&C4	&C9	&C10																																																																																																																		
US II	___ &OPR																																																																																																																							
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PM, APM, HPM, LM, SM					___ &UCN																																																																																																																			
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CG II			___ &CIO																																																																																																																					
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NG II				___ &NGO																																																																																																																				
NG IV				___ &NGO ^k																																																																																																																				
16.360	If necessary, return to Step 16.330 and invoke the other EC.																																																																																																																							

TASK 16 COPY SOFTWARE FROM CARTRIDGES TO THE HISTORY MODULE (continued) **Keyboard:** Engr's **Lock:** ENGR
Reference: This Document**Personality Loaded:** Universal

Step	Action	Response
16.370	Ensure &C1 or &C2 in FD1	Since the files on both disks are identical, you can use either one.
16.380	Key in an EC command for the System Volume	<u>EG:</u> EC \$F1>&EC>SYS_VOL.EC \$F1 43 (the last parameter is the system HM node number)
16.390	Press <ENTER>	"EC SYS_VOL.EC....."
16.400	The files that can be transferred using the above EC command are listed here for comparison with your printout to determine that all required files have been loaded on the HM.	
	NOTE To the question "COPY VOLUME &ASY (YES/NO)" answer No ("N"). To the question "DO YOU WISH TO CHANGE TO ANOTHER DRIVE" usually answer No ("N"). To the question "DO YOU WISH TO SKIP &DSY...." usually* answer No ("N"). *This EC loads several small files. If you answer "No," the EC will load all of them. If you answer "Yes," you can choose to load or not load any or all of the following files: ___ &DSY ___ &HGG ___ &AMG ___ &ARG ___ &NMG & NM2 The system does not have to have the node type to load any of these files.	
16.410	Ensure CNCF in FD2	<u>EG:</u> CPV \$F2>&KFO> NET>&KFO> -D <u>EG:</u> CPV \$F2>&D01> NET>&D01> -D "COMMAND COMPLETE"
16.420	Copy the customer key file to the HM	
16.424	Copy the null Area Database file to the HM	
16.430	Press <ENTER>	
16.443	You can copy certain standard displays (NCF Status, etc.) to a user volume on an HM other than that which contains the &DSY volume. The systems hunts for these displays in directory &DS2 on another HM If the main HM fails. To load the displays, ensure cartridge &C1 or &C2 is in \$F1. Key in the EC command EC \$F1>&EC>DS2 BKUP.EC \$F1 . Press <ENTER>. When asked "Do you wish to create &DS2...?", answer Yes. Key in the user volume name when asked.	
16.440	Return to Task 16.330 if any of the above software must be copied to a different HM (as determined earlier during Volume Configuration in Task 10).	
16.450	Key in " PD OFF "	"PROMPT TO DATAOUT TURNED OFF"
16.460	Press <ENTER>	
16.470	Key in " DO "	
		"DATA OUT COMPLETE"

TASK 17 COPY OPTIONAL SOFTWARE TO THE HISTORY MODULE**Keyboard:** Engr's **Lock:** ENGR

Reference: AM27-610, Custom Software Documentation;
 SW09-605 Engineer's Reference Manual
 SW11-507 Command Processor Operation manual

Personality Loaded: Universal

If you purchased optional custom software (load modules) that were not loaded in Task 16 or if you want to load the Tool Kit or Diagnostic software, perform the following steps. If not, skip to Task 18.

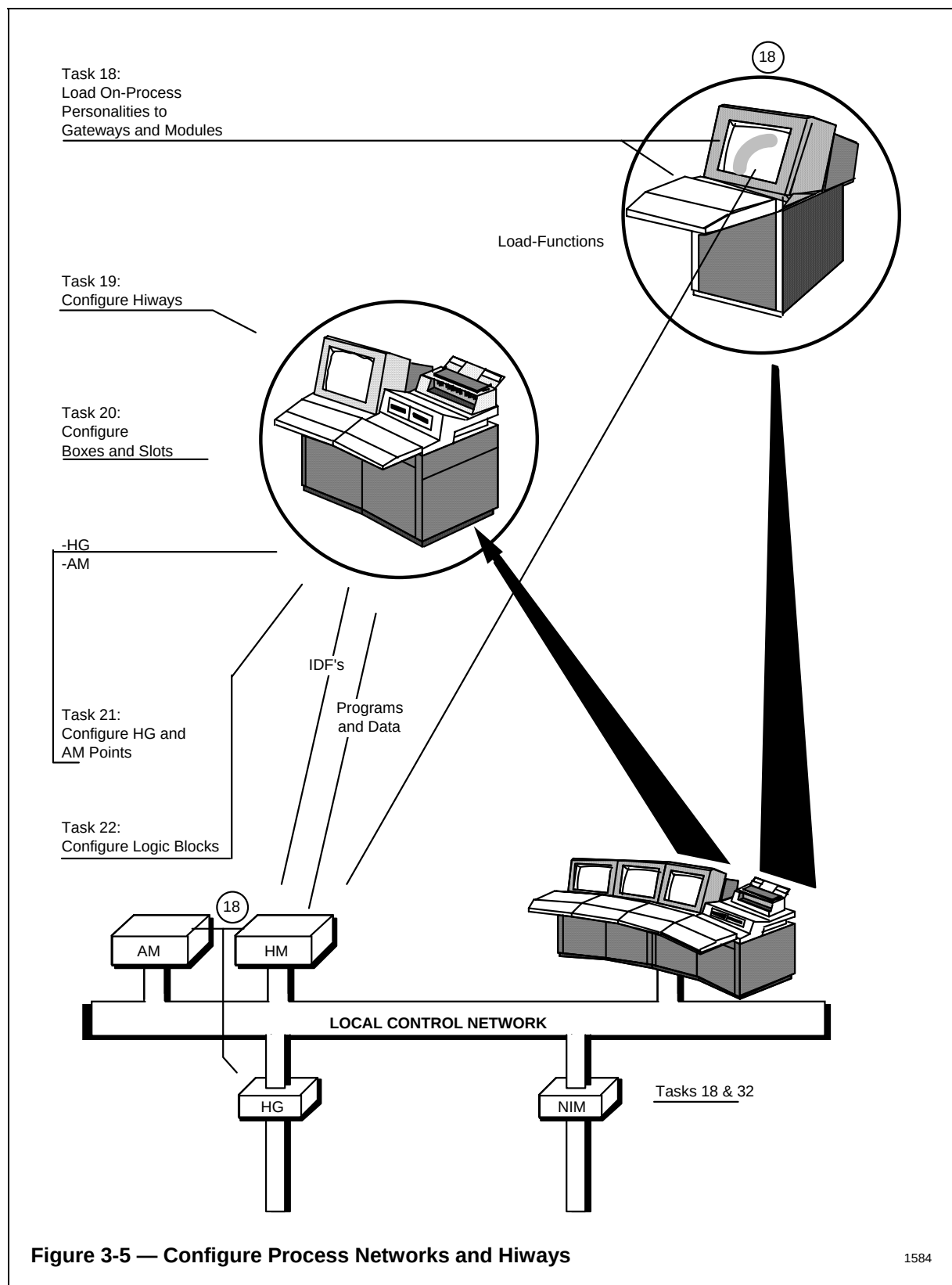
These steps assume the Command Processor is still selected.

Step	Action	Response
17.10	Create directories for the software you intend to load. List the contents of the cartridges, if necessary, to find the directories <u>EG:</u> Key in a Create Directory Command. Press <ENTER> (where &NNN is custom software) <u>EG:</u> Key in other Create Directory Commands. Press <ENTER> <u>EG:</u> Key in other Create Directory Commands. Press <ENTER>	In Task 10 you may have created a user volume for custom software (CSW1 in the example). You can also create a suitable directory under another volume, such as the example user volume HMV1. <u>EG:</u> CD NET>CSW1 &NNN* (custom software) <u>EG:</u> CD NET>HMV1 TLK1 (Tool Kit files) <u>EG:</u> CD NET>HMV1 DIA1 (Diagnostic software)
17.50	Copy the optional software into the History Module—	
	<u>In FD1</u>	<u>Cmd Type</u> <u>Example of command to be keyed</u> <u>Press</u>
17.80	&C__	Copy <u>EG:</u> CP \$F1>&NNN>*** NET>&NNN>= -D <ENTER>
17.70	&C6	Copy <u>EG:</u> CP \$F1>TLK1>*** NET>TLK1>= -D <ENTER>
17.84	&C6	Copy <u>EG:</u> CP \$F1>DIA1>*** NET>DIA1>= -D <ENTER>
17.90	Repeat Task 17 for each HM that you wish custom software copied to.	
*Substitute the volume name of the custom software cartridge(s) for &NNN.		

NOTE

If the system has redundant disk drives, they can be synchronized now (or later). Refer to the *Engineer's Reference Manual*, Section 7.5 and Section 5 in the *Command Processor Operation* manual for the procedure. The sample system did not have this feature.

Figure 3-5 shows a graphic representation of the Tasks (18-22) that must be performed to Configure Process Networks and Hiways.



**TASK 18 LOAD ON-PROCESS PERSONALITIES
TO GATEWAYS AND MODULES****Keyboard:** Both **Lock:** ENGR**Reference:** This Document**Personality Loaded:** Universal

Step	Action	Response
18.01	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
18.03	Select [SUPPORT UTILITIES]	SUPPORT UTILITY MENU
18.04	Select [MODIFY VOLUME PATHS]	MODIFY DEFAULT VOL PATHNAMES display
18.05	Select [SET DEVICE PATH TO "NET"]	LDIDs (Logical Device IDs) are all filled in with NET .
18.06	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
18.10	Select [SYSTEM STATUS]	SYSTEM STATUS display
18.20	Select an AM, NIM, or Gateway Node NIM example: Redundant NIM: NG example: HG/PLCG example, before Tasks 19-20: HG/PLCG example, after Tasks 19-20: AM example: CM example:	Only one node should be selected at a time <u>EG:</u> 24 for Network Interface Mod (NIM) <u>EG:</u> 25 for Redundant NIM <u>EG:</u> 30 for Network Gateway <u>EG:</u> 38 for Primary HG or PLCG Module <u>EG:</u> 39 for Redundant HG or PLCG Module <u>EG:</u> 40 for Application Modules or A ^X M <u>EG:</u> 46 for Computing Modules

WARNING

1. Do not select a redundant (secondary) HG, PLCG, or NG node unless Hiway and Box/Slot Configuration has been done in Tasks 19 and 20.
2. The primary HG or PLCG of a redundant pair must be loaded first, using the &C6 checkpoint disk. After Hiway & Box/Slot configuration, the primary HG or PLCG is checkpointed. The secondary HG or PLCG is loaded later and must be started with a checkpoint volume from the primary node. See *Engineering Reference Manual*, Section 8.
3. Do not attempt to load a Primary HG or PLCG from one station and a Secondary node from a different station at the same time. The nodes will both fail.
4. Always load the primary AM before loading the redundant AM (if any).

18.30	Select [NODE STATUS]	Process Networks Node Status Display or Application Modules Node Status Display
18.40	Select [MANUAL LOAD]	NODE LOAD targets appear.
18.50	Select a LOAD option	<u>EG:</u> [COLD LOAD] for AM or A ^X M <u>EG:</u> [OPERATOR PROGRAM] for HG, NG, PLCG, or NIM PGM SOURCE FOR NODE NN display
18.60	Select [DEFAULT SOURCE]	[EXECUTE COMMAND] target appears.
18.70	Select [EXECUTE COMMAND]	DATA SOURCE FOR NODE NN display

TASK 18 LOAD ON-PROCESS PERSONALITIES TO Keyboard: Both Lock: ENGR
GATEWAYS AND MODULES (continued)**Reference:** This Document**Personality Loaded:** Universal

Step	Action	Response
18.75	If you will be loading the null checkpoint for an HG, move the Write-Protect switch on the &C6 cartridge disk to the Write position.	Otherwise, you won't be able to complete the HG load (write position is away from edge).
18.80	Ensure &C6 in FD1	&C6 contains null checkpoints for all checkpoint nodes (HG, NIM, AM, A ^X M, CG, and PLCG)
18.90	Select [ALTERNATE SOURCE]	Boxes [1], [2], etc. appear.
18.100	Select EG: [1]	&C6 and [EXECUTE CMD]
18.110	Select [EXECUTE COMMAND]	GATEWAYS or AM display reappears.
18.120	Press <ENTER>	"BUSY: AM or GWY nn or CM" Status goes to LOAD.
18.130	Wait a few minutes	Status goes to OK when loaded.
18.134	Repeat 18.06-18.130 including other redundant modules	Note that PMs, APMs, and HPMS are loaded in a later task.
18.136	After loading an A ^X M, Select the Node number under NODE	A ^X M node number color changes to white Status should still be OK
18.138	Select [SAVE DATA]	CHKPT DEST display appears
18.140	Select [DEFAULT SOURCE]	Additional targets appear
18.142	Select [EXECUTE COMMAND]	BUSY AM nn. Node status =SAVE... then OK
18.144	Select [SHUTDOWN]	Node should still be selected
18.145	Press <ENTER>	Node status changes to READY...QUALIF
18.146	Select [LOAD/DUMP]	Additional targets appear
18.147	Select [AUTOLOAD NET]	
18.148	Press <ENTER>	Node status = LOAD...LOC LOAD...OK Node Type changes to AxM
18.149	Repeat steps 18.136 - 18.148 for all A ^X Ms.	
18.150	Remove the cartridge disk from the disk drive and, if necessary, place the Write-Protect switch in the protect position.	

19 **PROCEDURE 4 – CONFIGURE PROCESS NETWORKS AND HIWAYS** 19

TASK 19 **CONFIGURE HIWAYS**

Keyboard: Engr's Lock: ENGR

Reference: SW11-605 & SW11-511

Personality Loaded: Universal

Step	Action	Response
19.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU

NOTE

The primary HG or PLCG must be loaded and running with the startup data base (loaded in Task 18). If the system does not have an HG or PLCG, skip to task 21.

\$HIWAYnn Where nn is hiway number		
19.70	Select [HIWAY GATEWAY]	HG-BUILD TYPE AND CONFIG. MENU
19.75	Select [HIWAY CONFIG]	HG-HIWAY POINT ASSIGN. PED display
19.80	Key into HIWAY NUMBER port	<u>EG</u> : 02
19.90	Select HTD ASSIGNMENT option	<u>EG</u> : [THISHG] (HTD is mandatory if a Basic OP Station is on the Hiway)
19.100	Fill in the remaining ports as required and press <ENTER>	PED reappears with values in cyan
19.110	Press <COMND>	COMMAND DISPLAY

NOTE

1. You should make an IDF with a different name for each HG pair and each PLCG pair.
2. When building a database create the IDFs in this order:
 - a. In this task, create the Hiway Configuration database for the HG pair or PLCG pair.
 - b. In Task 20, create the box/slot database for the HG pair, or PLCG pair and then create the database for all the boxes and devices on the Hiway.

19 PROCEDURE 4 – CONFIGURE PROCESS NETWORKS AND HIWAYS 19**TASK 19 CONFIGURE HIWAYS****Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605 & SW11-511**Personality Loaded:** Universal

Step	Action	Response
19.120	If this is the <u>first</u> write to this IDF, go to step 19.130; If <u>not</u> , go to step 19.160	IDF for this example is HIWAY2
19.130	Select [WRITE TO IDF]	Pathname ports and target appear.
19.140	Key into REFERENCE PATH NAME port	<u>EG:</u> NET>HNV1>
19.150	Key into pathname for IDF port	<u>EG:</u> HIWAY2
19.160	Press <ENTER>	Hiway Config data written to IDF. HG-HIWAY POINT ASSIGNMENT display
19.170	Press <COMND>	COMMAND DISPLAY
19.180	Select [LOAD]	
19.190	Press <ENTER>	Hiway Config data loaded to HG or PLCG. PARAM ENTRY display reappears
19.200	Repeat 19.10 – 19.190 for each other Hiway to be configured.	<u>EG:</u> HIWAY4 for a PLCG

20 PROCEDURE 4 – CONFIGURE PROCESS NETWORKS AND HIWAYS 20

TASK 20 CONFIGURE BOXES AND SLOTS

Keyboard: Engr's Lock: ENGR

Reference: SW11-605 & SW11-511

Personality Loaded: Universal

Step	Action	Response
20.05	Hold <CTL> down and press <9>. If the HG BUILD TYPE MENU appears go to step 20.20; if not go to 20:10	HG-BUILD TYPE AND CONFIG. MENU
20.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
\$HYxxBnn Where xx is the Hiway number and nn is the box number		
20.15	Select [HIWAY GATEWAY]	HG-BUILD TYPE AND CONFIG. MENU
20.20	Select [BOX/SLOT CONFIG]	HG-BOX POINT ASSIGNMENT PED display
20.30	Key into HIWAY NUMBER port	<u>EG</u> : 02
20.40	Key into BOX NUMBER port	<u>EG</u> : 02 for an HG (<u>Configure HG or PLCG before other Hiway boxes</u>) 03 if you have an HG or PLCG pair 04 if you have a Basic Station 05 for a MC, 10 for CB, etc. (configure 1 box/pass)
NOTE If this box is the HG or PLCG, the box number of the first node of a redundant pair must be an even number and the second node must be the next odd number.		
20.50	Select a BOX TYPE option	<u>EG</u> : [HG] [CB] if box is CB [MC] if box is MC
20.55	Select a BOX ASSIGNMENT option	<u>EG</u> : [THISHG]
NOTE If an HG or PLCG pair is not your master HG pair for this Hiway, the BOX ASSIGNMENT for that HG or PLCG pair must be either [REMOTEHG] or [ADDEDHG].		
20.90	Select a LOAD DESTINATION (if appl.)	<u>EG</u> : [HG_HIWAY]
20.95	Press <ENTER>	PED reappears with values in cyan. (If PLCG, perform steps in Note at end of Sec. 20 to clear address error).

TASK 20 CONFIGURE BOXES AND SLOTS
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605 & SW11-511**Personality Loaded:** Universal**NOTE**

If you selected any options outlined in yellow or if you keyed data into any port outlined in yellow, the content of the PED set or PED changes when you press the <ENTER> key. You should press <ENTER> after each PED page.

Step	Action	Response
20.100	Press <PAGE FWD> to next PED page key in data, select options, and press <ENTER> after each page CHECK THE DATA YOU HAVE ENTERED	HG-BOX CONFIGURATION PED
20.110	When you have paged forward and entered all PED pages hold <CTL> down and press <8>.	PARAM ENTRY DISP SET STATUS display
20.120	If "Display Set Entered?" is Yes, and no errors are indicated, go to step 20.150; else go to 20.130	Display Set Entered? Yes Errors? NO
20.130	Select a Display Title option, correct data and press <ENTER>	Applicable PED appears.
20.135	Repeat step 20.120 WRITE TO IDF	Go check that data is all OK.
20.150	Press <COMND>	COMMAND DISPLAY
20.160	If this is the <u>first</u> write to this IDF, go to step 20.165; if not, go to step 20.190	IDF for this example is HY2BXSL .
20.165	Select [WRITE TO IDF]	Pathname ports and target appear.
20.170	Key into REFERENCE PATH NAME port	<u>EG:</u> NET>H MV1>
20.180	Key into pathname for IDF port	<u>EG:</u> HY2BXSL
20.190	Press <ENTER>	Hiway Config data written to IDF. HG-BOX POINT ASSIGNMENT PED display

TASK 20 CONFIGURE BOXES AND SLOTS
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-605 & SW11-511**Personality Loaded:** Universal

Step	Action	Response
20.195	Repeat steps 20.30 – 20.190 for the redundant HG or PLCG.	
20.200	Repeat steps 20.30 – 20.190 for each other box/slot.	All configured boxes/slots on IDF
20.210	Press <COMND>	COMMAND DISPLAY
20.220	Select [LOAD MULTIPLE]	Additional pathname ports appear
20.230	Key into REFERENCE PATH NAME port	<u>EG:</u> Net>HNV1>
20.240	Key into pathname for IDF port	<u>EG:</u> HY2BXSL
20.245	Press <ENTER>	"WAIT" in yellow; "PROCESSING ..." "OPERATION COMPLETE" (See note below if an error results)
20.250	Press <COMND>	COMMAND DISPLAY
20.260	Select [PRINT ENTITIES]	Path name ports/targets appear. REF. PATH NAME = NET>HNV1>
20.270	Select [PRINT IDF ENTITIES]	
20.280	Use default Print Device ID	<u>EG:</u> \$P1
20.290	Press <ENTER>	Box Slot configuration parameters are printed

NOTE

Entity names in the printout identify Hiway and Box numbers as follows:

"\$HIWAYnn" where nn is the Hiway number.

"\$HYnnBnn" where nn after HY is Hiway number and nn after B is Box number.

Address Error—If an address error results when loading Box/Slot data to a Hiway Box, make sure you saved the data in an IDF. Then proceed as follows (doing this on another station is preferable):

Call up the System Status Display.

Select the appropriate HG Node.

Select [NTWK/HWY STATUS] near the bottom of the display.

Select [HIWAY CMND].

Select [INIT ADDR].

Select [EXECUTE COMMAND]. The address errors should be cleared.

Return to the BOX/SLOT configuration display and repeat the LOAD MULTIPLE steps.

This time the result should be "OPERATION COMPLETE."

PLCG Interface—Certain PLCG interface functions have to be programmed with a Local Loader Station, refer to PLCG documentation for details.

TASK 21 CONFIGURE POINTS**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-511**Personality Loaded:** Universal

Step	Action	Response
21.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
21.20	Select [HIWAY GATEWAY] , or [APPLICATION MODULE] , or [COMPUTING MODULE] (if the system has a PLCG, select [HIWAY GATEWAY]. If the system has none of the above, skip to Task 22).	<u>EG:</u> [HIWAY GATEWAY] HG/AM/CM BUILD TYPE MENU
21.21	Select a Point Build option	<u>EG:</u> [REGULATORY]

NOTE

1. For detailed information on point building, refer to the *Data Entity Builder Manual*, (see References). For point building tips and precautions, refer to Section 9 in the *Engineers Reference Manual*.
2. The general scenario of point building is to key into ports, select options on each PED page, press <ENTER> and then <PAGE FWD> to the next PED page. The pages in a particular PED set may change depending upon what type of box, algorithm, or point you are configuring.
3. If you select any **options outlined in yellow** or if you key data into any port outlined in yellow, the content of the PED set or PED changes when you press the <ENTER> key. You should press <ENTER> after each page.
4. Steps 21.30 through 21.420 describe the process of building a typical PID Normal point for a **Basic** Controller. For other types of points see Tables 3-2 through 3-4. When building any of the example points, use default values and options for items not on the list.

	POINT BUILD	
21.30	Key into TAG NAME port	<u>EG:</u> HG1001
21.40	Key into UNIT ID port	<u>EG:</u> 01
21.50	Key into POINT DESCRIPTOR port	<u>EG:</u> CB REG POINT
21.60	Key into POINT KEYWORD port	<u>EG:</u> CBpid
21.64	PRIMARY MODULE POINT ID	-----
21.66	Key into POINT ASSOCIATED DISPLAY	<u>EG:</u> DISP1 (R510 and later systems)
21.70	Key into HIWAY NUMBER port	<u>EG:</u> 02

TASK 21 CONFIGURE POINTS
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-511**Personality Loaded:** Universal

Step	Action	Response
21.75	Select BOX TYPE option	<u>EG</u> : [CB]
21.80	Key into BOX NUMBER port	<u>EG</u> : 10
21.90	Select POINT DISCLOSURE option	<u>EG</u> : [FULL]
21.100	Select LOAD DESTINATION option	<u>EG</u> : [HG_HIWAY]
21.110	Press <ENTER>	PED reappears with values in cyan
21.120	Press <PAGE FWD>	ALGORITHM Selection display
21.125	Select ALGORITHM option	<u>EG</u> : [PIDNORM]
21.130	Press <ENTER>	PED reappears with values in cyan
21.135	Press <PAGE FWD>	HG-PV CONFIGURATION display
21.140	Key into E.U. DESCRIPTOR port	<u>EG</u> : GAL/HOUR
21.150	Select CHARACTERIZATION option	<u>EG</u> : [SQRROOT]
21.160	Select INPUT CONDITIONING option	<u>EG</u> : [SQRROOT]
21.170	Select PV DECIMAL FORMAT option	<u>EG</u> : [D2] (places point at 999.99)
21.180	Key into OVERVIEW VALUE port	<u>EG</u> : 20
21.190	Select PV RANGE OPTION	<u>EG</u> : [CLMPZERO]
21.200	Select CLAMP VALUE OPTION	<u>EG</u> : [NOCLAMP]
21.210	Press <ENTER>	PED reappears with values in cyan
21.220	Press <PAGE FWD>	HG—ALARMING display
21.225	Select ALARM FORMAT option	<u>EG</u> : [ALFMT00]
21.230	Key into PVLOTP port	<u>EG</u> : 10
21.240	Key into PVHITP port (far right)	<u>EG</u> : 100
21.250	Select ALARM PRIORITY option	<u>EG</u> : [HIGH]
21.260	Press <ENTER>	PED reappears with values in cyan

TASK 21 CONFIGURE POINTS
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-511**Personality Loaded:** Universal

Step	Action	Response
21.270	Press <PAGE FWD>	HG-OPERATING CONFIGURATION display
21.290	Select PV SIGNAL Type option	<u>EG</u> : [PV]
21.310	Select INITIALIZATION option	<u>EG</u> : [INIT]
21.320	Key into SETPOINT port	<u>EG</u> : 43
21.330	Press <PAGE FWD>	
21.350	Key into OUTPUT HIGH LIMIT port	<u>EG</u> : 100
21.360	Key into INTEGRAL HIGH LIMIT port	<u>EG</u> : 100
21.370	Key into FILTER CONSTANT port	<u>EG</u> : 1
21.380	Press <ENTER>	PED reappears with values in cyan
21.390	Press <PAGE FWD>	HG-MODE CONFIGURATION display
21.400	Select NORMAL MODE option	<u>EG</u> : [CAS]
21.410	Select NRML MODE ATTRIBUTE option	<u>EG</u> : [OPERATOR]
21.420	Press <ENTER>	PED reappears with values in cyan
	CHECK THE DATA YOU ENTERED	
21.430	When you have paged forward and entered all PED pages, hold <CTL> down and press <8>.	PARAM ENTRY DISPLAY SET STATUS display
21.440	If "Display Set Entered?" is Yes, and no missing data or errors are indicated, go to step 21.460; else 21.450.	
21.450	Select a Display Title option, correct data and press <ENTER>	Applicable PED appears. PED set is corrected.
21.455	Repeat step 21.440.	

TASK 21 CONFIGURE POINTS
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-511**Personality Loaded:** Universal**NOTE**

You should make IDFs with different names for the points that you build in each box on the hiway.
(HG1, CB1, etc.)

Step	Action	Response
	WRITE TO IDF	
21.460	Press <COMND>	COMMAND DISPLAY
21.470	If this is the <u>first</u> write to this IDF, go to step 21.475. If not, go to step 21.510.	In this example, the IDF is HGPNTS for HG, AMPNTS for AM, etc.
21.475	Select [WRITE TO IDF]	Additional port/target appear.
21.480	Key into REFERENCE PATH NAME port	<u>EG:</u> NET>HMOV1>
21.500	Key into pathname for IDF port	<u>EG:</u> HGPNTS for HG, AMPNTS for AM
21.510	Press <ENTER>	Point Config data written to IDF PARAMETER ENTRY DISPLAY
21.520	Repeat steps 21.30 – 21.510 for each other point.	All configured points are on IDF for HG, or IDF for AM, etc.
21.530	Press <COMND>	COMMAND DISPLAY
21.540	Select [LOAD MULTIPLE]	Additional ports/target appear
21.550	Key into Reference Pathname port	<u>EG:</u> NET>HMOV1>
21.555	Select [WITH OVERWRITE]	
21.560	Key into Pathname For IDF port	<u>EG:</u> HGPNTS for HG; AMPNTS for AM
21.570	Press <ENTER>	"WAIT" in yellow; "ENTITY NAMES" Point Config data loaded to HG. "OPERATION COMPLETE"
21.572	If errors occur, select the point on the error display. Make corrections; then repeat steps 21.530 – 21.570.	

TASK 21 CONFIGURE POINTS
(continued)

Keyboard: Engr's Lock: ENGR

Reference: SW11-605 & SW11-511

Personality Loaded: Universal

Step	Action	Response
21.580	If the Command Display is not on the screen, press <COMND>	COMMAND DISPLAY
21.590	Select [LIST ENTITIES IN IDF]	Additional ports/target appear.
21.600	Key into REFERENCE PATH NAME	<u>EG</u> : NET>HMV1>
21.605	Key into pathname for IDF port	<u>EG</u> : HGPNTS for HG, AMPNTS for AM
21.610	Key into pathname For SEL LIST	<u>EG</u> : BXSL.EL
21.614	Select BUILD TYPES of entities to list	<u>EG</u> : HG or AM or CIU as applies. Additional targets appear.
21.618	Select types of entities	<u>EG</u> : ALL
21.620	Press <ENTER>	"BUSY", DISPLAY OF LIST FILE display
21.630	For a printout, press <COMND>	COMMAND DISPLAY
21.640	Select [PRINT FILE]	Ports/std. suffixes appear
21.650	Key into pathname for FILE port	<u>EG</u> : BXSL.EL
21.655	Key into Printer Print Device ID port	<u>EG</u> : \$P1 (default used)
21.660	Press <ENTER>	Point IDs are printed. (For more details, select PRINT ENTITIES)
21.670	Repeat Task 21 to build "master" points for other Data Hiway boxes.	<u>EG</u> : MC point (see Table 3-3)
21.680	Go to Appendix D of this document to EXCEPTION BUILD other Data Hiway box points as required.	
21.690	Repeat Task 21 to build other "master" points for AM and CL.	<u>EG</u> : AM points (vratio is used in task 28 for CL). See Table 3-2.
21.680	Go to Appendix D of this document to EXCEPTION BUILD other AM or CL points as required.	

NOTE

You should save HG and AM checkpoints to cartridge disk at this time. To do so, go perform instructions in Appendix B (HG) or C (AM) of this document.

Table 3-2 — Example AM Points**a. AM/CL Point VLRATIO (Regulatory)**

TAG NAME	VLRATIO
UNIT ID	01
POINT DESCRIPTOR	CL Point
ENGR UNITS DESCRIP.	Fur/Ftnt
POINT KEYWORD	CLPOINT
POINT ASSOCIATED DISPLAY	(blank) (R510 and later systems)
PRIMARY MODULE POINT ID	-----
POINT DISCLOSURE	[FULL]
PV ALGORITHM	[CL]
CONTROL ALGORITHM	[NULL]
OVERVIEW VALUE	25
I/O DISPLAY SUPPR.	[NOSUPPR]
POINT EXEC. PERIOD	[NOPERIOD]
PV DECIMAL FORMAT	[D1]
PV RANGE HIGH	100.0
PV RANGE LOW	0.0
PV RANGE EXT. HIGH	102.9
PV RANGE EXT. LOW	-2.9
NUMBER ASSC. CL BLKS	1

All others—default values.

b. AM Point (Regulatory)

TAG NAME	AM101
UNIT ID	01
POINT DESCRIPTOR	TEMPERATURE
ENGR UNITS DESCRIP.	KBTU/MCF
POINT KEYWORD	TEMPA
POINT ASSOCIATED DISPLAY	(blank) (R510 and later systems)
PRIMARY MODULE POINT ID	-----
POINT DISCLOSURE	[FULL]
PV ALGORITHM	[SUMMER]
CONTROL ALGORITHM	[NULL]
POINT EXEC. PERIOD	5SEC
PV DECIMAL FORMAT	[D2]
PV RANGE HIGH	1.0
PV RANGE LOW	0.0
PV RANGE EX. HI	1.0
PV RANGE EX. LOW	-1.0
ALGORITHM EQN TYPE	[EQA]
OVERALL GAIN	0.01
NUMBER INPUT CONN.	1
SOURCE POINT. PARAM.	HG1001.OP
ALM ENB STATE	DISABLE

All others—default values.

Table 3-3 — Example Parameters for (Regulatory) MC Point HG0501

TAG NAME	HG0501
UNIT ID	01
POINT DESCRIPTOR	MC REGULATORY POINT NO 1
POINT KEYWORD	MCPID
PRIMARY MODULE POINT ID	-----
POINT ASSOCIATED DISPLAY	(blank) (R510 and later systems)
POINT DISCLOSURE	[FULL]
HIWAY NUMBER	02
BOX TYPE	[MC]
BOX NUMBER	05
POINT DISCLOSURE	[FULL]
SLOT NUMBER	01
LOAD DESTINATION	[HG]
ALGORITHM	[PIDNORM]
E. U. DESCRIPTOR	GAL/HR
OVERVIEW VALUE	20
PV RANGE OPTION	[CLMPZER0]
ALARM FORMAT	[ALFMT00]
ALARM PRIORITY	[LOW]
DISPLAY SUPPRESSION	[NOSUPPR]
OPERATOR MODE CHANGE	[PERMIT]
All others—default values.	

Table 3-4 — Example Parameters for PLCG Digital Input Point PLC0801

TAG NAME	PLC0801
UNIT ID	01
POINT DESCRIPTOR	PL DIGITAL POINT NO 1
POINT KEYWORD	PL DIGIN1
PRIMARY MODULE POINT ID	-----
POINT ASSOCIATED DISPLAY	(blank) (R510 and later systems)
HIWAY NUMBER	04
BOX TYPE	[DHP]
BOX NUMBER	8
POINT DISCLOSURE	[BRIEF]
LOAD DESTINATION	[HG_HIWAY]
INPUT SLOT NUMBER	1
INPUT SUBSLOT NUMBER	1
ALARM FORMAT	[STATE1]
ALARM PRIORITY	[HIGH]
UPPER BOX TEXT	ON
LOWER BOX TEXT	OFF
PC TYPE	[MODICON]
INPUT 1 SPECIFIER CODE	0
All others—default values.	

22 PROCEDURE 4 – CONFIGURE HIWAYS, BOXES, SLOTS AND POINTS 22

TASK 22 CONFIGURE LOGIC BLOCKS

Keyboard: Engr's Lock: ENGR

Reference: MC11-500

Personality Loaded: Universal

NOTE

If the system has no Multifunction Controllers, skip to Task 32 to configure NIMs, PMs, APMs, and HPMs. If neither Multifunction Controllers nor NIMs, go to Task 23.

Step	Action	Response
22.01	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
22.03	Select [SYSTEM STATUS] display	
22.03	Select the Hiway Gateway Node	Node box is selected
22.05	Select [NTWK/HWYSTATUS]	Multifunction Controller Status display
22.07	Select [BOX STATUS]	
22.09	Enter Box Number	<u>EG</u> : 5
22.11	Press <ENTER>	BOX STATUS DISPLAY
22.14	Select [IDLE]	Target changes color
22.16	Select [EXECUTE COMMAND]	MF CTLR STATUS: IDLE
22.18	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
CONFIGURE LOGIC BLOCKS		
22.20	Select [LOGIC BLOCKS]	LBC-BOX SELECTION display appears
22.22	Key into PATHNAME port	<u>EG</u> : NET>H MV1>LB020520 (02 = Hiway 2; 05 = Box 5, 20 = revision level 20)
22.24	Press <ENTER>	Message: "SELECT FUNCTION"
22.25	Hold <CTL> down and press <2> (function key = F2)	Message: "TEXT BUILDER IS SELECTED"
22.26	Select [CREATE]	LBC-TEXT BUILDER display
22.27	Key into BLOCK NUMBER	<u>EG</u> : 1
22.28	Press <ENTER>	Additional ports displayed
22.29	Key into ENTER ALGORITHM port	<u>EG</u> : 1
22.30	Press <ENTER>	Algorithm number changes to AND; additional ports appear
22.31	Key into INPUT PARAMETER #1	<u>EG</u> : LP(01).SOA
22.32	Key into INPUT PARAMETER #2	<u>EG</u> : LP(11).SOA

22 PROCEDURE 4 – CONFIGURE HIWAYS, BOXES, SLOTS AND POINTS 22

TASK 22 CONFIGURE LOGIC BLOCKS (continued)

Keyboard: Engr's Lock: ENGR

Reference: MC11-500

Personality Loaded: Universal

Step	Action	Response
22.33	Key into INPUT PARAMETER #3	<u>EG</u> : LP(12).SOA
22.34	Tab to NEXT BLOCK, [YES] , press <SELECT>	Block 1 information appears in lower half of display. At top left, Block Number changes to 2
22.35	Tab to ENTER ALGORITHM; key into port	<u>EG</u> : 3
22.36	Press <ENTER>	Algorithm number changes to XOR; Additional ports appear
22.37	Key into INPUT PARAMETER #1	<u>EG</u> : LB(102)
22.38	Key into INPUT PARAMETER #2	<u>EG</u> : LB(103)
22.39	Key into INPUT PARAMETER #3	<u>EG</u> : LB(104)
22.40	Tab to NEXT BLOCK, [YES] , press <SELECT>	Block 2 information is added to lower half of display. At top left, Block Number changes to 3
22.41	Tab to ENTER ALGORITHM; key in	<u>EG</u> : 7
22.42	Press <ENTER>	Algorithm number changes to FLIP- FLOP; additional ports appear
22.43	Key into INPUT PARAMETER #1	<u>EG</u> : LB(102)
22.44	Key into INPUT PARAMETER #2	<u>EG</u> : LB(103)
22.45	Key into OUTPUT PARAMETER	<u>EG</u> : FL(001)
22.46	Tab to NEXT BLOCK [YES] , press <SELECT>	Block 3 information is added to lower half of display
22.47	Hold <CTL> down and press <^> (function key = F12)	Message: "WRITING FILE NET> HMF1>LB020520.JS"
22.48	Hold <CTL> down and press <4> (function key = F4)	LBC Compile Display appears; Message: "LBC COMPILE STARTED" Message: "WRITING FILE NET> HMF1>LB020520.JL/.JO"
	Wait	Message: "COMPILE COMPLETED" when compile is finished.
22.49	Hold <CTL> down and press <5> (function key = F5)	Message: "LBC DATA LOAD STARTED"

22 PROCEDURE 4 – CONFIGURE HIWAYS, BOXES, SLOTS AND POINTS 22

TASK 22 CONFIGURE LOGIC BLOCKS (continued)

Keyboard: Engr's Lock: ENGR

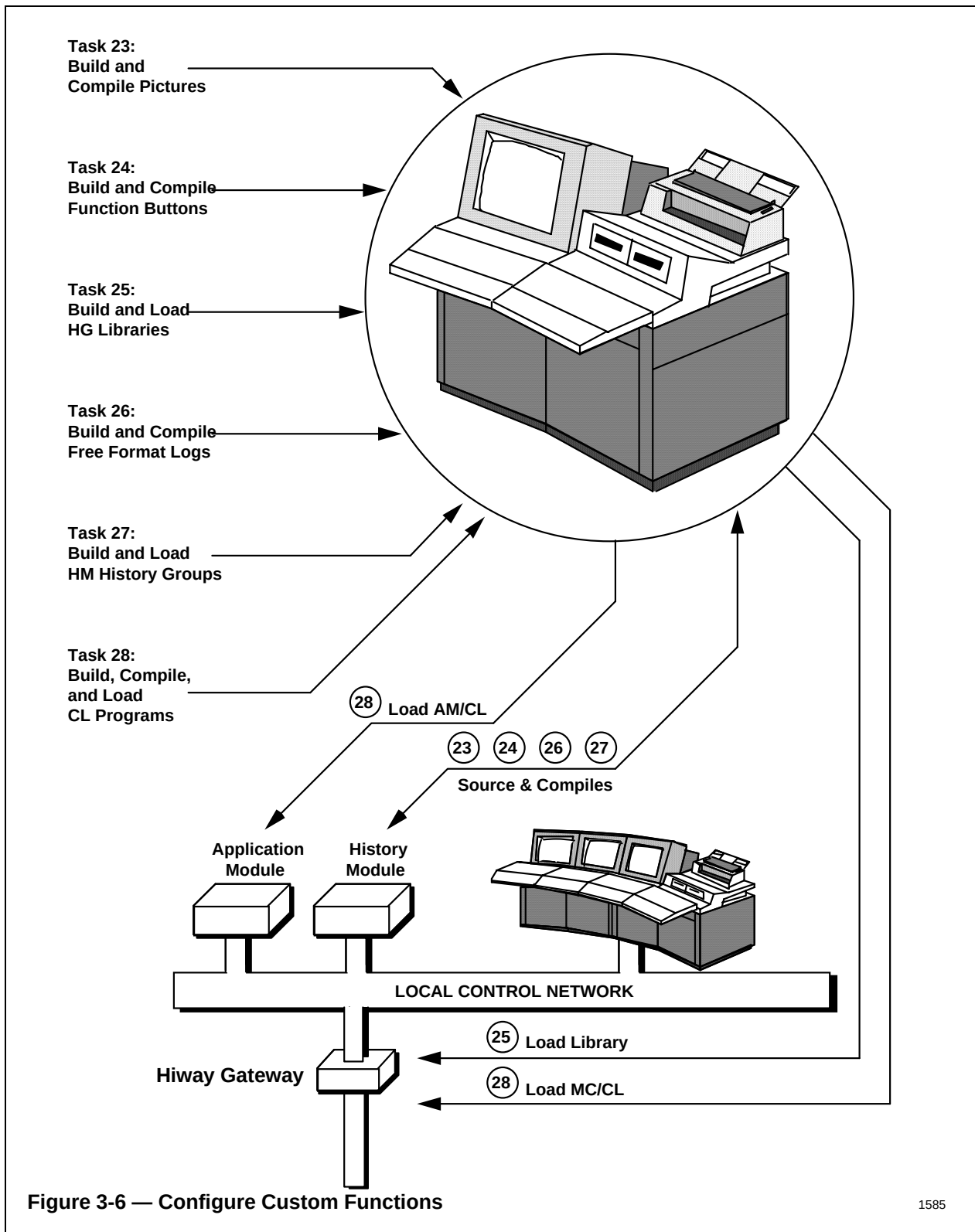
Reference: MC11-500

Personality Loaded: Universal

Step	Action	Response
22.50	Hold <CTL> down and press <6> (function key = F6)	Message: "READING DATA BASE"- then LBC Compare Display appears with last line "NO DIFFERENCES FOUND"
22.51	For other Multifunction Controllers, hold <CTL> down and press <1>. Then, repeat 22.22-22.50	
	The following steps add comments to the file. If you do not want to add comments, skip to 22.68	Comments are used to document the logic block configuration.
22.52	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
22.53	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
22.54	Type in EDIT command	<u>EG</u> : EDIT NET>H MV1>LB020520.JS (.JS = Source File)
22.55	Press <ENTER>	Logic Block data from previous steps appears
22.56	Move the cursor to the end-of-file: Hold <CTL> down and press <3>, Hold <CTL> down and press <2>	Jump Menu appears Cursor moves to end-of-file following existing text
22.58	Press <RETURN>	
22.60	Type in the comments, for example: — <RETURN> — LOGIC BLOCKS 001 THROUGH 003 ARE OVERTEMP/CONTROL ALARMS.<RETURN> — THE OUTPUT OF THIS CIRCUIT CONTROLS A SECONDARY CIRCUIT<RETURN> — CONSISTING OF LOGIC BLOCKS 010 THROUGH 017.<RETURN>	
22.62	Move the cursor to the beginning-of-file: Hold <CTL> down and press <3>, Hold <CTL> down and press <1>	Jump Menu appears Cursor moves to beginning-of-file
22.64	Update and exit from the file: Hold <CTL> down, press <1>, Hold <CTL> down, press <2>	Quit Menu appears Message: "FILE WAS UPDATED- X LINES WRITTEN"
22.66	Repeat 22.54-22.64 for other Logic Block Files.	
22.68	Hold down <CTL> and press <HELP>	ENGINEERING MAIN MENU

If the system has a **Network Interface Module/ PM, APM, or HPM**, go to Task 32 near the back of this manual. The sample system did not have an LM or SM, but this would be the time to configure them also. When the NIM and UCN nodes have been configured, continue on to Task 23.

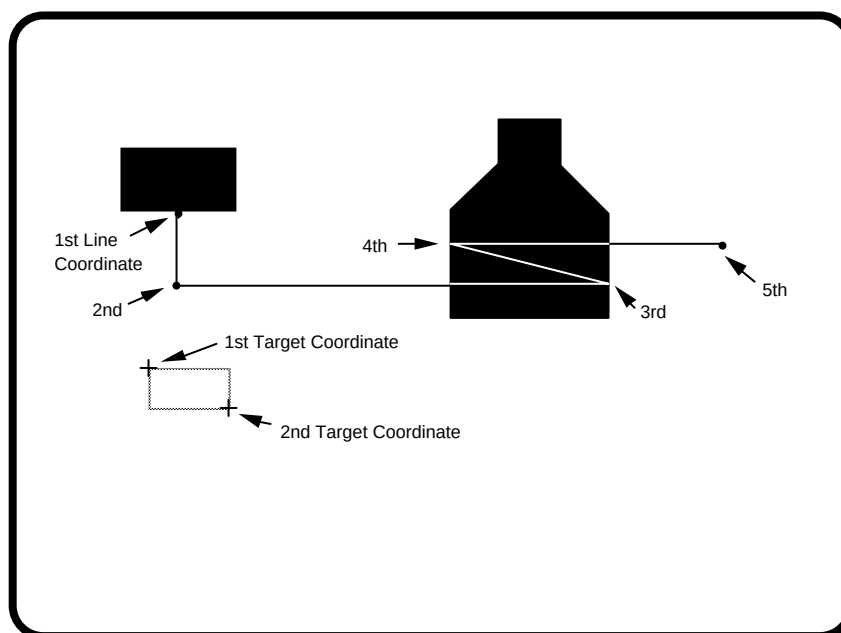
Figure 3-6 shows a graphic representation of the Tasks (23-28) that must be performed to Configure Custom Functions.



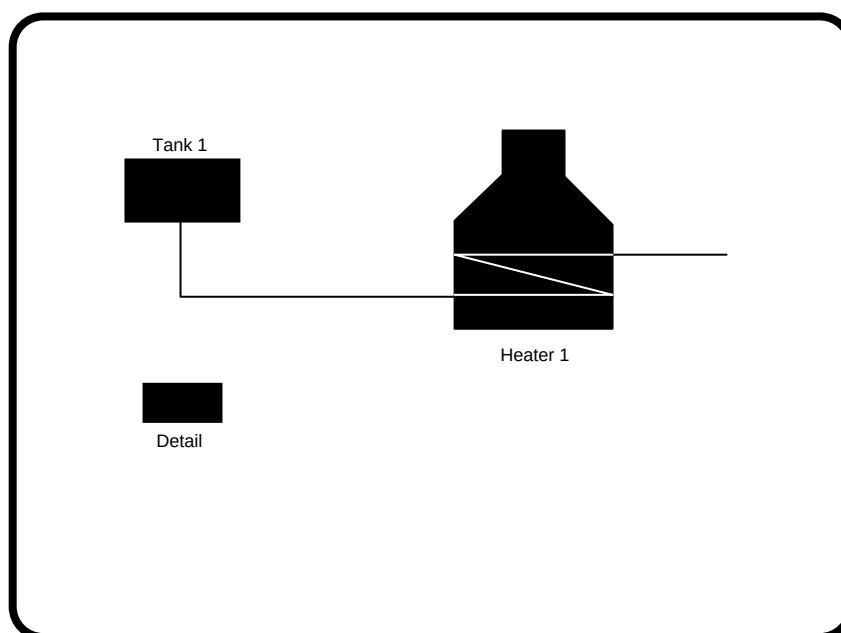
1585

TASK 23 BUILD AND COMPILE PICTURES**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW09-650 & SW11-650**Personality Loaded:** Universal

Step	Action	Response
23.10	Make sure points referenced by pictures have been built and loaded.	
23.14	Hold <CTL> down and press <HELP>	ING MAIN MENU
23.18	Select [PICTURE EDITOR]	PICTURE EDITOR display
23.19	The following steps and Figure 3-7 show how to build a simple Custom Graphic Display.	If pictures were prebuilt, go to step 23.160 to read them in and then compile them on the HM.
23.22	Key in a Picture Editor command	<u>EG:</u> ADD SOLID
23.28	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
23.32	Move cursor to desired location	Hold <CTL> down, press <TAB> keys
23.40	Press <SELECT>	Cross marks the coordinate (see Figure 3-7a)
23.44	Repeat steps 23.32 – 23.40 until all of the coordinates are selected.	Note that deletes the last coordinate selected if you need to make a correction.
23.48	Press <ENTER>	Command Completed Note: The NEW command clears the screen if you want to redraw.
23.56	Repeat steps 23.22 – 23.48 to build additional objects.	<u>EG:</u> Another Solid. (Figure 3-7b)
23.60	Hold <CTL> down and press a color key (color keys are on the keypad).	<u>EG:</u> <CTL> and <4> Status indicator at top center of screen changes to white.
23.65	Key in a Picture Editor command	<u>EG:</u> ADD LINE
23.70	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
23.75	Move cursor to desired location	<u>EG:</u> Hold <CTL> down; press TAB
23.80	Press <SELECT>	Cross marks the Coordinate
23.85	Repeat steps 23.75 – 23.80 until all of the coordinates are selected.	<u>EG:</u> Lines connect objects (See Figure 3-7c).
23.90	Press <ENTER>	Command Completed



(c)



(d)

Figure 3-7 — Picture Building (Continued)

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TASK 23 BUILD AND COMPILE PICTURES
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW09-650 & SW11-650**Personality Loaded:** Universal

Step	Action	Response
23.94	Key in Picture Editor commands to add conditional behavior, and so on.	<u>EG:</u> SEL SOL;ADD COND
23.98	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
23.102	Move cursor to desired location	<u>EG:</u> Place cursor on solid rectangular object
23.106	Press <SELECT>, then press <ENTER>	Object turns white and blinks; Screen Form appears.
23.110	Key in screen form information (points you refer to must have been built and loaded in Task 21)	<u>EG:</u> (in condition block): IF HG1001.PV > 75 THEN SET RED ELSE SET CYAN
23.114	Press <ENTER>	Picture returns (or point Type/Format requested if point not found)
23.118	Key in commands to add targets	<u>EG:</u> ADD TARG
23.124	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
23.128	Move cursor to desired location	<u>EG:</u> Place cursor under first solid rectangular object (see Figure 3-7c)
23.132	Press <SELECT>	Cross marks 1st target coordinate
23.136	Move cursor; press <SELECT>	Rectangle drawn around coordinates
23.140	Press <ENTER>	Target screen form appears
23.145	Key into Action area	<u>EG:</u> DETAIL("HG1001")
23.150	Press <ENTER>	Picture returns (or point Type/Format requested if point not found)
23.154	Enter any other commands to complete desired custom display.	<u>EG:</u> ADD TEXT Label objects (see Figure 3-7d)
23.156	Press <ENTER> to complete command	
23.159	Go to step 23.210 to compile the display	
Follow steps 23.160 – 23.190 to read in prebuilt pictures from floppies or cartridge disks.		
23.160	Insert cartridge with picture files in FD1; key in the SET PATHNAME command	<u>EG:</u> S P \$F1>VOL1>DISP1 (where VOL1 is a cartridge volume name and DISP1 is a picture file on the cartridge)
23.170	Press <ENTER>	Pathname appears at top of display

TASK 23 BUILD AND COMPILE PICTURES
(continued)**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW09-650 & SW11-650**Personality Loaded:** Universal

Step	Action	Response
23.180	Key the READ command onto the command line.	<u>EG:</u> READ DISP1 (Assumes DISP1 is filename)
23.190	Press <ENTER>	Picture appears on US screen
23.210	Key in a COMPILE command	<u>EG:</u> COMPILE NET>H MV1>DISP1
23.220	Press <ENTER>. Picture on the screen is compiled onto the HM To have a redundant schematic, you should compile it to other HMs, cartridge disks, or floppies now.	Message: "WRITING SOURCE FILE" Message: "WRITING OBJECT FILE" No redundant schematics were configured on the example system.
23.230	Repeat steps 23.19 – 23.220 until each of your pictures is compiled.	
23.240	If you want to save picture source files insert cartridge in FD1.	<u>EG:</u> USER (volume created in task 4)
23.241	Press <ESC>	"ESCAPED FROM PICTURE EDITOR"
23.244	Key in a copy command	<u>EG:</u> CP NET>H MV1>DISP1.DS \$F1>PICT>DISP1 -D
23.248	Press <ENTER> Source file is copied	Message: "COPY COMPLETE"
23.249	Hold <CTL> down and press <HELP>	Return to Picture Editor

TASK 24 BUILD AND COMPILE BUTTONS

Keyboard: Engr's Lock: ENGR

Reference: SW11-570

Personality Loaded: Universal

Step	Action	Response
Make sure all the points and pictures you need were created in previous tasks		
24.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
24.30	Select [BUTTON CONFIGURATION]	BUTTON CONFIGURATION MENU
24.35	Select a button (<u>EG</u> : 7)	ACTION & LAMP DATA regions appear
24.40	Key into the Action region	<u>EG</u> : DETAIL("HG1001")
24.45	Press <ENTER>	BUTTON CONFIGURATION MENU appears
24.50	Select another button (<u>EG</u> : 8)	ACTION & LAMP DATA regions appear
24.55	Key into the Action region	<u>EG</u> : SCHEM("DISP1")
24.60	Press <ENTER>	BUTTON CONFIGURATION Menu
24.65	Select another button (<u>EG</u> : 9)	Action & Lamp Data regions appear
24.70	Key into the Action region	<u>EG</u> : SCHEM(R_STR)
24.75	Press <ENTER>	PARAMETER ENTRY display
24.80	Key into ports exposed by Actor.	<u>EG</u> : 10 into "Tip Length" <u>EG</u> : "ENTER SCHEMATIC NAME" into "Prompt 1–30 Char"
24.85	Press <ENTER> CONFIGURE A DISPLAY SET	BUTTON CONFIGURATION MENU
24.90	Select another button	<u>EG</u> : 10 ACTION & LAMP DATA regions appear
24.100	Key into the Action region	<u>EG</u> : CROSSCRN(1);SCHEM("DISP1"); CROSSCRN(2);GROUP(1,0)
24.140	Press <ENTER>	BUTTON CONFIGURATION MENU
24.150	Repeat steps 24.35 – 24.140 to configure rest of buttons in set	Buttons for all stations in an Area are configured by this task.

NOTE

If you want a button on the Operator Keyboard to call up either the Overview Display or the Area Trend Display, configure them here. For the Universal Work Station's Engineering Keyboard, <PAGE FWD> from the main Button Menu to the PF Button Menu and configure as desired.

24.160	Key COMPILE xxx into the cmd line	<u>EG</u> : COMPILE NET>&D01>BUTTON
24.170	Press <ENTER>	"WRITING SOURCE/OBJECT FILE"
24.180	Key in a PRINT command	<u>EG</u> : PRINT 7..10 \$P1
24.190	Press <ENTER>	Button Config. record is printed

TASK 25 BUILD AND LOAD HG LIBRARIES**Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-511**Personality Loaded:** Universal

Step	Action	Response
NOTE: If you do not have a Hiway Gateway, skip to Task 26.		
HG LIBRARY \$HnnLIBx where nn = Hiway No. and x = Library No.		
25.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
25.15	Select [HIWAY GATEWAY]	HG-BUILD TYPE AND CONFIG. MENU
25.20	Select [HG LIBRARY]	HG LIBRARY CONFIGURATION display
25.30	Key into HIWAY NUMBER port	<u>EG</u> : 02 (configured in Task 9)
25.40	Key into LIBRARY NUMBER port	<u>EG</u> : 1
25.50	Key into INDEX ports	<u>EG</u> : ON (into "Index 001" port) <u>EG</u> : OFF (into "Index 002" port) <u>EG</u> : TRUE (into "Index 003" port) <u>EG</u> : FALSE (into "Index 004" port)
25.60	Key into other Index ports as required. Press <ENTER> after page is done	PED reappears with values in cyan
25.70	Press <PAGE FWD> and key into Index ports as required.	New PED page appears.
25.80	When a given library is complete, press <COMND>	COMMAND DISPLAY
25.82	Select [WRITE TO IDF]	Pathname Ports and target appear.
25.83	Key in a REFERENCE PATHNAME	<u>EG</u> : NET>H MV1>
25.84	Key in an IDF Pathname	<u>EG</u> : HGLIB
	Press <ENTER>	Library is saved in IDF. HG-LIB. CONFIG display
25.85	Press <COMND>	COMMAND DISPLAY
25.90	Select [LOAD]	
25.100	Press <ENTER>	"BUSY" HG-LIBR. CONFIGURATION display

NOTE

You should save the HG checkpoint(s) on HM and perhaps cartridge disk at this time. See Appendix B of this document for instructions.

25.110	Repeat Task 25 for each other HG Library to be configured.	<u>EG</u> : Library numbers 2, 3, and 4
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TASK 26 BUILD AND COMPILE FREE FORMAT LOGS Keyboard: Engr's Lock: ENGR

Reference: HM11-660

Personality Loaded: Universal

Step	Action	Response
26.10	Make sure points referenced in the logs have been built and loaded in Task 21.	Prebuilt logs can be read in from removable media and compiled on the HM per step 26.90
26.14	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
26.18	Select [FREE FORMAT LOGS]	FFL BUILDER display. At top of screen, R- 0, 42 indicates the Edit region is now at top left of drawing area.
26.22	On Cmd line, type: SET GRID ON	
26.26	Press <ENTER>	Grid appears
26.34	Enter Command: ADD TEXT , press <ENTER>	"ENTER TEXT" System ready for text input
26.38	Move cursor to locations where you want text and type it in. Figure 3-8 shows an example of text in a Free Format Log.	<u>EG</u> : WEST AREA LOG (title) FURNACE INLET TEMPERATURE = COIL OUTLET TEMPERATURE = and so on. See Figure 3-8a.
26.42	Press <ENTER>	Text is added to display
26.46	Enter Command: ADD VALUE Press <ENTER>	"ENTER VALUE COORDINATES" System waits for you to specify Value coordinates
26.50	Move cursor to each point where you want a value; press <SELECT>. Note cursor location (For example, C-224, 944) on 2nd line at that time.	+ Marks Value locations. (see Figure 3-8b)
26.54	Press <ENTER>	Screen form appears requesting EXPRESSION for Value at XXX, YYY. (For example, VALUE AT 224, 944)

TASK 26 BUILD AND COMPILE FREE FORMAT LOGS Keyboard: Engr's Lock: ENGR
(continued)

Reference: HM11-660

Personality Loaded: Universal

Step	Action	Response
26.58	Enter EXPRESSION information for each value (one per pass)	EG: SYS_TIME (1st Value) SYS_TIME (2nd Value) HG1001.PV HG1002.PV and so on for others (points must exist)
26.62	Press <ENTER>	Screen form requests FORMAT
26.74	Enter Format information (one per pass)	EG: DATEMM-DD-YYENDDATE (1st Value) TIMEHH:MM:SSENDTIME (2nd Value) R-ZZZZ9.99 (other Values)
26.78	Press <ENTER>	String of letters replaces cross at value location (For example, DTDTDT, R-ZZZZ9.99, and so on). EXPRESSION for the next value appears.
26.79	Repeat steps 26.58 – 26.78 until all Value information is entered	
26.80	If TYPE is requested, Enter TYPE information for that Value.	EG: DATE, TYPE, REAL, and so on. (If type is known to the system, this screen form is omitted.)
26.82	Enter command: SET GRID OFF , press <ENTER>	Grid disappears
26.90	If FFL was prebuilt you can read it in from cartridge disk at this time	EG: READ \$F1>VOL2>FFL001 (reads file FFL001 from cartridge named VOL2)
26.94	Enter Command to compile log	EG: COMPILE NET>HNV1>FFL001
26.96	Press <ENTER>	"WRITING SOURCE/OBJECT FILE" Log named FFL001 is compiled on HM
26.100	Repeat steps 26.22 – 26.96 for each FFL.	
26.102	If you want to save a copy of the FFL, you can copy it to a cartridge disk at this time. Insert cartridge in FD1	EG: USER (volume created in task 4)
26.106	Press <ESC>	"ESCAPED FROM PICTURE EDITOR"
26.108	Key in a copy command	EG: CP NET>HNV1>FFL001.DS \$F1>PICT>FFL001 -D
26.110	Press <ENTER> Source file is copied	Message: "COPY COMPLETE"
26.114	Hold <CTL> down and press <HELP>	Return to FFL Builder display

TASK 27 **BUILD AND LOAD HM HISTORY GROUPS** Keyboard: Engr's Lock: ENGR

Reference: HM12-500 & SW11-511

Personality Loaded: Universal

Step	Action	Response
	HM HISTORY \$CHuu(n)	uu = Unit No., n = Hist. Group
	Note: The HM On-Line Personality must be running for Task 27.	Check the System Status display for "HISTORY MODULES-STATUS OK"
27.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
27.20	Select [HM HISTORY GROUPS]	CH HEADER DISPLAY
27.30	Key into UNIT port	<u>EG</u> : 01
27.40	Key into HISTORIZATION GROUP	<u>EG</u> : 1
27.50	Key into ENTITY.PARAMETER table Suggestion: Move cursor to the port; then, press <CLR-ENT>.	<u>EG</u> : HG1001.PV
27.60	When all applicable entries to the ENTITY.PARAM table are complete, press <ENTER>	Values and ports turn cyan
27.70	Press <COMND>	COMMAND DISPLAY
27.80	Select [WRITE TO IDF]	Pathname ports and target appear.
27.90	Key in a REFERENCE PATHNAME	<u>EG</u> : NET>H MV1>
27.100	Key in an IDF Pathname	<u>EG</u> : HM HIS
	Press <ENTER>	Library is saved in IDF. PED display.
27.105	Press <COMND>	COMMAND DISPLAY
27.110	Select [LOAD]	
27.115	Press <ENTER>	PARAMETER ENTRY DISPLAY

TASK 28 BUILD, COMPILE, AND LOAD CL PROGRAMS
Reference: AM11-575 & SW27-500

Keyboard: Engr's **Lock:** ENGR

Personality Loaded: Universal

Step	Action	Response
	CREATE CL SOURCE FILE	If you are not using CL, skip to Section 29.
28.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
28.20	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
28.30	Key in an EDIT command	<u>EG:</u> EDIT NET>CL>VLR_CALC.CL
28.40	Press <ENTER>	Message: "OPEN FILE - FILE NOT FOUND"
28.50	Key in CL source file statements	<u>EG:</u> See Figure 3-9. This example is similar to Fig. 3-4 in the <i>CL/AM Overview Manual</i> . The point named in the BLOCK header must be a point built in task 21 (For example, VLRATIO). All points referenced in the statement must exist (For example, HG1001, HG1002). CLSLOTS must equal 1 or be greater than 1. In this example, PVALGID = CL.
28.60	Move cursor to 1st letter on line 1: Hold down <CTL>, then press <3>, hold down <CTL>, then press <1>	JUMP MENU Cursor moves to start of file.
28.70	Update and exit: Hold down <CTL>, then press <1>, hold down <CTL>, then press <2>	QUIT MENU Source file is transferred from temporary file to user volume. Message: "FILE WAS UPDATED xx Lines...."
	COMPILE THE CL/AM BLOCK	
28.90	Key in a Compile command	<u>EG:</u> CL VLR_CALC
28.100	Press <ENTER>	Elapsed time for Pass 1, Pass 2, Pass 3, time for compilation = nn.n sec. "Errors detected:None"
	If errors were detected, examine the .LE file. A caret (^) marks the error(s). Make corrections, then recompile the file.	<u>EG:</u> PRINT NET>CL>VLR_CALC.LE
	LINK (AND LOAD) THE CL BLOCK	
28.110	Key in a Link Command	<u>EG:</u> LK VLR_CALC VLRATIO
28.120	Press <ENTER>	Elapsed time for linker = nn.n seconds. "Errors Detected None"

```

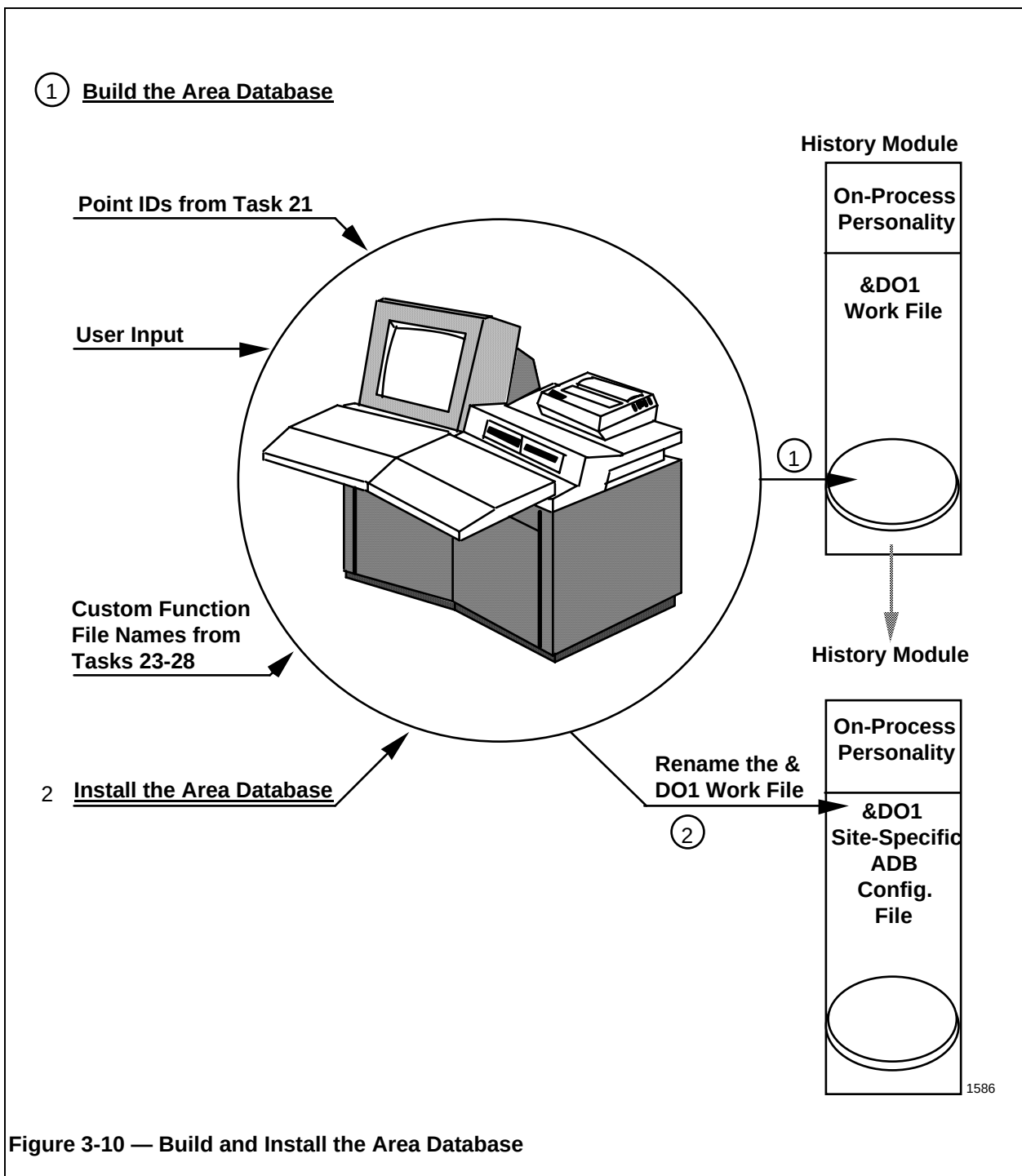
BLOCK VLR_CALC  (POINT VLRATIO; AT PV_ALG)
--
-- CALCULATE INTERNAL VAPOR/LIQUID RATIO FOR COLUMN NO. 2.
--
EXTERNAL  HG1001,  HG1002
LOCAL WL, WV, RATIO
LOCAL CP = 0.806    --LIQUID HEAT CAPACITY
LOCAL LAMBDA = 653.0    --HEAT OF VAPORIZATION
--
-- CALCULATE INTERNAL LIQUID FLOW RATE FROM HEAT AND MATERIAL BALANCE
SET WL = ((HG1001.PV - HG1002.PV)
&        * CP/LAMBDA)
--
-- CALCULATE INTERNAL VAPOR FLOW RATE FROM MATERIAL BALANCE
SET WV = HG1001.PV + WL
--
-- CALCULATE VAPOR/LIQUID RATIO AND STORE AS PV
SET RATIO = WV/WL
--

IF BADVAL (RATIO) THEN (CALL SET_BAD (PVCALC);
&                      SET PVAUTOST = BAD)
&                      ELSE (SET PVCALC = RATIO;
&                            SET PVAUTOST = NORMAL)
--
END VLR_CALC

```

Figure 3-9 — CL Source File Example

Figure 3-10 shows a graphic representation of the Tasks (23-28) that must be performed to Build and Install the Area Database.



**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES****Keyboard:** Engr's **Lock:** ENGR**Reference:** SW11-511**Personality Loaded:** Universal**NOTE**

1. Points to be referenced in the Area Database must already be configured in Task 21. Custom functions to be referenced in the Area Database must already be configured in Tasks 23, 24, 25, 26, 27, and 28.
2. In the steps that follow, the Area Database entity IDs required for reconstituting, loading, or reading a given part of the Area Database is listed just to the right of the bold heading for each part.

Step	Action	Response
COPY BLANK AREA DATA BASE		
29.01	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
29.03	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
29.04	Key in a Copy File command to copy the blank Area Data Base file to an Area nn Volume on &D01. (nn = 01 for this example) Press <ENTER> after each Copy command above.	<u>EG:</u> CP NET>&ARG>AREA00.DA NET>&D01>AREA01 "COPY COMPLETE"
29.06	Key in an Unprotect command to the Area Data Base; then press <ENTER>	<u>EG:</u> UNPT NET>&D01>AREA01.DA "FILE UNPROTECT COMPLETE"
UNIT ASSIGN \$RUNTSTS		
29.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
29.20	Select [AREA DATA BASE]	AREA CONFIGURATION Menu
29.30	Select [UNIT ASSIGNMENT]	UNIT ASSIGNMENT DISPLAY CONFIG.
29.40	Select Unit ports that you want to be in the area.	<u>EG:</u> 01 Selected port(s) turn green
29.45	Press <ENTER>	UNIT ID targets appear

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**
Keyboard: Engr's **Lock:** ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
29.47	Enter Unit ID(s)	<u>EG</u> : 01
29.50	Select INITIALLY ASSIGNED? option	Select [YES] for each unit initially assigned to this area.
29.70	Press <ENTER>	PED reappears with values in cyan
	SET A PATHNAME AND WRITE TO IDF	
29.80	Press <COMND>	COMMAND DISPLAY
29.90	Select [WRITE TO IDF]	Pathname ports and target appear
29.100	Key into Reference Path Name port	<u>EG</u> : NET>HVM1>
29.110	Key into pathname for IDF port	<u>EG</u> : BOILER1
29.120	Press <ENTER>	Unit Assignment data written to IDF
GROUP DISPLAY \$OGROUP(n) where n can be 1 to 400		
29.130	If UNIT ASSIGN display is on screen, hold <CTL> down and press <9>	AREA CONFIGURATION Menu
	If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu
29.140	Select [GROUP]	GROUP DISP. CONFIGURATION display
29.150	Key into GROUP NUMBER port	<u>EG</u> : 1
29.160	Key into GROUP TITLE port	<u>EG</u> : WEST GROUP
29.170	Key into the POINT IDS port	<u>EG</u> : HG1001
29.180	Select IN THE TREND SET? option	<u>EG</u> : [YES]
29.190	Repeat 29.170 – 29.180 for each other point in this group.	
29.200	Select a TREND TIMEBASE option	<u>EG</u> : The default option
29.201	Press <ENTER>	PED reappears with values in cyan
29.203	<PAGE FWD> to the Associated and Help display definitions. Key into ports as desired.	
29.210	Press <ENTER>	PED reappears with values in cyan

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**
Keyboard: Engr's **Lock:** ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
WRITE TO IDF WHEN PATHNAME IS ALREADY SET		
29.220	Press <COMND>	COMMAND DISPLAY with [WRITE TO IDF] selected
29.230	Press <ENTER>	GROUP DISPLAY CONFIG data to IDF
29.240	To configure more groups, hold <CTL> down and press <PAGE BACK>. Go repeat 29.140 – 29.230	
UNIT TREND DISPLAY \$OUNTRND(n) where n can be 1 to 36		
29.250	If GROUP DISP CONFIG is on screen hold <CTL> down and press <9> If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.255	Select [UNIT TREND]	UNIT TREND DISPLAY CONFIG display
29.260	Key into TREND DISP NUMBER port	<u>EG:</u> 1
29.265	Key into TREND TITLE port	<u>EG:</u> UNIT 1 TRENDS
29.270	Select TREND TIMEBASE option	<u>EG:</u> [2 HOURS]
29.275	Key in names of point(s) to be trended in cyan and/or yellow	<u>EG:</u> HG1001 (CYAN)
29.280	Press <ENTER>	PED reappears with values in cyan
29.285	<PAGE FWD>	RELATED DISPLAY 001 DEF display
29.290	Key into SCREEN NUMBER port(s) and DISPLAY TYPE port(s)	<u>EG:</u> 1 <u>EG:</u> ALARMSUM
29.295	Press <ENTER>	PED reappears with values in cyan or SCHEMATIC NAME port appears if DISPLAY TYPE entered was SCHEM

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**
Keyboard: Engr's **Lock:** ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
29.300	Key into SCHEM. NAME port (if any)	<u>EG:</u> DISP1
29.305	Press <ENTER>	PED reappears with values in cyan
29.310	<PAGE FWD>	Next page appears
29.315	Repeat 29.290 – 29.310 to configure remaining Related Displays, and on Page 6, Assoc. Display and Help Display	Related, Associated, and Help displays configured
WRITE TO IDF WHEN PATHNAME IS ALREADY SET		
29.320	Press <COMND>	COMMAND DISPLAY with correct pathnames entered.
29.325	Press <ENTER>	UNIT TREND CONFIG. data to IDF
29.330	Repeat 29.260 – 29.330 to configure other Unit Trend displays. Use <PAGE FWD> and <PAGE BACK> as necessary.	All Unit Trend displays configured and stored in IDF
SYSTEM STATUS DISPLAY \$OSYSTAT		
29.350	If PED display is still on the screen Hold <CTL> down and press <9>	AREA CONFIGURATION Menu
	If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu
29.360	Select [SYSTEM STATUS]	SYSTEM STATUS DISPLAY CONFIG display
29.364	If the screen is blank, press <PAGE FWD>	Next page appears
29.370	Key into Associated Display ports: SCREEN NO. port DISPLAY TYPE port Press <ENTER> CONSOLE NUMBER	<u>EG:</u> 0 <u>EG:</u> CONSTAT CONSOLE # port appears <u>EG:</u> 1

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**
Keyboard: Engr's **Lock:** ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
	SYSTEM STATUS DISPLAY (continued)	
29.380	Key into HELP DISPLAY DEF ports: SCREEN NO. port DISPLAY TYPE port	<u>EG:</u> 1 <u>EG:</u> SYSSTAT
29.390	Press <ENTER>	PED reappears with values in cyan
	WRITE TO IDF	
29.400	If not first write to IDF, press <COMND> and go to 29.410. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY with [WRITE TO IDF] selected. Example IDF is BOILER1.
29.410	Press <ENTER>	SYS STAT DISP CONFIG-data to IDF
FREE FORMAT LOGS \$OFFLOG(n) where n can be 1 to 100		
29.420	If PED display is on the screen, hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION MENU AREA CONFIGURATION MENU
29.430	Select [FREE FORMAT LOG]	FREE FORMAT LOG CONFIG. display
29.440	Key into LOG NUMBER	<u>EG:</u> 1
29.450	Key into LOG NAME	<u>EG:</u> FFL001
29.470	Key into LOG DESCRIPTION	<u>EG:</u> WEST AREA LOG
29.480	Accept default for INVOC. TYPE	<u>EG:</u> DEMAND
29.490	Accept default for PRINTER ID	<u>EG:</u> 1
29.500	Press <ENTER>	PED reappears with values in cyan

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**
Keyboard: Engr's **Lock:** ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
WRITE TO IDF		
29.510	If not first write to IDF, press <COMND> and go to 29.515 If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.515	Press <ENTER>	FFL CONFIG DATA to IDF
29.520	Repeat 29.440 – 29.515 for other Free Format Logs	All FFL configuration data in IDF
HIWAY BOX ANNUN POLICY \$DMONBLK		
29.610	If PED display is on the screen, hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.620	Select [PROC NTWK ANNUN POLICY]	PROCESS NETWORK DEV. ANNC. CONFIG display
29.630	Key into PROCESS NTWK NBR port	<u>EG:</u> 2
29.640	Select ADDRESS ANNUNC. options	<u>EG:</u> [YES] for all addresses
29.650	Press <ENTER>	PED reappears with values in cyan
29.660	Check PED set status (<CTL> <8>)	Display Set Entered?-Yes If errors, see: 29.1060 – 29.1080
29.670	If not first write to IDF, press <COMND> and go to 29.680. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
29.680	Press <ENTER>	HIWAY ANN PLCY CONFIG-data to IDF
PATHNAME CATALOG \$OABSTRT		
29.690	If PED display is on the screen, hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATABASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**
Keyboard: Engr's **Lock:** ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
29.700	Select [PATHNAME CATALOG]	PATHNAME CATALOG CONFIG. display
29.710	Key into BUTN CONF VOLUME ID port	<u>EG:</u> &D01
29.720	Key into BUTN CONF FILE NAME port	<u>EG:</u> BUTTON
29.724	Key into SCHEMATIC or FFL PATHNAMES—most critical pathname in top left-hand port, then left to right, then second line left to right, and so on.	<u>EG:</u> NET>HNV1 (your user volume), NET>&DSY , NET>&D01 , NET>DIA1 , and so on.
29.750	Key into MEM. RES SCHEMATIC OR FFL FILE NAME port and select [FFL] if you entered a FFL filename	<u>EG:</u> FFL001 [FFL] , DISP1 , DISP2 , DISP3 , and so on.
29.760	Press <ENTER>	PED reappears with values in cyan
29.764	Use <PAGE FWD> to add more schematic/FFL names	All schematic/FFL file names entered
	WRITE TO IDF	
29.770	If not first write to IDF, press <COMND> and go to 29.780. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathnames set)
29.780	Press <ENTER>	PATHNAME CAT CONFIG-DATA to IDF

PROCESS MODULE GROUP DISPLAY \$OMODGRP(n) where n can be 1-50
NOTE

If an MC process module point wasn't configured in Task 21, go to 29.890.

29.790	If PED display is on the screen, hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATABASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.795	Select [PROCESS MODULE GROUP]	MODULE GROUP DISP CONFIG display
29.800	Key into GROUP NUMBER port	<u>EG:</u> 1
29.810	Key into GROUP TITLE	<u>EG:</u> MIX AND SOAK GROUP

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**
Keyboard: Engr's **Lock:** ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
29.815	Key into PROCESS MODULE ID ports	<u>EG:</u> SCALE1
29.820	Press <ENTER>	PED reappears with values in cyan
29.825	<PAGE FWD>	RELATED DISPLAY 001 DEF display
29.830	Key into Screen Number port(s) and Display Type port(s)	<u>EG:</u> 1 <u>EG:</u> SCHEM
29.835	Press <ENTER>	PED reappears with values in cyan and additional port(s) displayed.
29.840	Key into added port(s)	<u>EG:</u> DISP1
29.845	Press <ENTER>	PED reappears with values in cyan
29.850	<PAGE FWD>	Next page appears
29.855	Repeat 29.830 – 29.850 to configure ASSOCIATED DISPLAY and HELP DISPLAY	RELATED, ASSOCIATED, and HELP displays configured
WRITE TO IDF		
29.865	If not first write to IDF, press <COMND> and go to 29.870. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] (with path name BOILER1 set)
29.870	Press <ENTER>	PED display
29.880	Repeat 29.800 – 29.870 to configure other Process Module Groups. Use <PAGE FWD> and <PAGE BACK> as necessary.	All Process Module Groups configured and stored in IDF

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**
Keyboard: Engr's **Lock:** ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
MODULE SUMMARY DISPLAY \$OUNITSM(n) where n can be 1 to 36		
29.890	If PED display is on the screen, hold <CTL> down and press <9>.	AREA CONFIGURATION MENU
	If Engineering Main Menu is on the screen, select [AREA DATABASE]	AREA CONFIGURATION MENU
29.900	Select [MODULE SUMMARY]	UNIT SUMMARY DISP CONFIG display
29.910	Select an option for each Line	<u>EG</u> : [HEADER] for Line 01, page 1
29.920	Press <ENTER>	<u>EG</u> : TITLE port appears for Line 1
29.930	Key into each Title port	<u>EG</u> : WEST UNIT 1
29.940	Press <ENTER>	PED reappears with values in cyan
29.950	Use <PAGE FWD> to get to more pages to configure and repeat steps 29.910 – 29.940 for each.	UNIT SUMMARY DISP CONFIG display – n
29.960	<PAGE FWD> to RELATED DISP DEF	RELATED DISP DEFINITION display – 1
29.970	Key into each pair of SCREEN NUMBER and DISPLAY TYPE ports	<u>EG</u> : 0 <u>EG</u> : ALARMSUM
29.980	Press <ENTER>	PED reappears with values in cyan
29.990	If desired, <PAGE FWD> to next page and repeat steps 29.970 – 29.980.	RELATED DISP DEFINITION display – n
29.1000	<PAGE FWD> to ASSOCIATED DISPLAY DEFINITION	ASSOCIATED & HELP DISPLAY DEFINITION
29.1010	Key into each pair of SCREEN NUMBER and DISPLAY TYPE ports	<u>EG</u> : 0 <u>EG</u> : UNITALRM
29.1020	Press <ENTER>	Unit ID port appears.
29.1030	Key into Unit ID port	<u>EG</u> : 01
29.1040	Press <ENTER>	PED reappears with values in cyan

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**
Keyboard: Engr's **Lock:** ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
	CHECK PED SET STATUS	
29.1050	Hold <CTL> down and press <8>	PED Set Stat display
29.1060	If "Display Set Entered?" is Yes, and no missing data or errors are indicated, go to 29.1090; else 29.1070.	If "No," check "Missing Data?" and "Errors?"
29.1070	Select a Display Title option, correct data and press <ENTER>	Applicable PED appears.
29.1080	Repeat step 29.1060	
	WRITE TO IDF	
29.1090	If not first write to IDF, press <COMND> and go to 29.1100. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set).
29.1100	Press <ENTER>	MOD SUM DISP CONFIG data to IDF
ANNUNCIATOR DISPLAY \$ANNDATA		
29.1110	If PED display is on the screen, hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.1115	Select <ANNUNCIATOR>	ANNUNCIATOR DISPLAY CONFIG. display
29.1120	Key into BOX NUMBER port	<u>EG:</u> 1
29.1125	Key into BOX TITLE port	<u>EG:</u> AIR FLOW Note: Annunciator display boxes have two lines of four characters each.
29.1128	Select a Point Usage	<u>EG:</u> [ENTITY ID]
29.1130	Key into NUMBER OF POINTS port	<u>EG:</u> 1
29.1131	Press <ENTER>	
29.1132	Key into POINT NAMES ports	<u>EG:</u> HG1001
29.1135	Key into SCREEN NUMBER port	<u>EG:</u> 1
29.1140	Key into DISPLAY TYPE	<u>EG:</u> SCHEM

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**
Keyboard: Engr's **Lock:** ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
29.1145	Press <ENTER>	Additional port(s) appear
29.1155	Key into added port(s)	<u>EG:</u> DISP1
29.1160	Press <ENTER>	PED reappears with values in cyan
	WRITE TO IDF	
29.1165	If not first write to IDF, press <COMND> and go to 29.1170 If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.1170	Press <ENTER>	ANNUNC. CONFIG DATA to IDF
29.1175	Repeat 29.1110 – 29.1170 for other boxes on Annunciator	All Annunciator Box configuration data in IDF
PROCESS JOURNALS \$OPRCJOR(n)		
29.1200	If PED display is on the screen, hold <CTL> down and press <9>. If Engineer Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.1202	Select [PROCESS JOURNAL]	PROCESS JOURNAL CONFIG. display
29.1204	Key into JOURNAL NUMBER port	<u>EG:</u> 1
29.1206	Key into JOURNAL NAME port	<u>EG:</u> PRJNL001
29.1208	Key into JOURNAL TITLE port	<u>EG:</u> WEST AREA PROCESS ALARMS
29.1212	Select JOURNAL TYPE	<u>EG:</u> [ALARM]
29.1214	Select INVOCATION TYPE	<u>EG:</u> [PERIODIC]
29.1216	Select INITIALLY ACTIVE option	<u>EG:</u> [NO]
29.1220	Select DAYS OF WEEK option	<u>EG:</u> ALL (default)
29.1222	Key into START TIME port	<u>EG:</u> 0:10
29.1224	Key into PERIOD port	<u>EG:</u> 12:0
29.1226	Key into PRINTER ID port	<u>EG:</u> 1
29.1228	Press <ENTER>	PED reappears with values in cyan
29.1230	Press <PAGE FWD>	Next page of PED appears
29.1232	Key into TIME SPAN port	<u>EG:</u> 12

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
29.1234	Key into NUMBER OF UNITS port	<u>EG</u> : 1
29.1236	Press <ENTER>	PED reappears with Unit ID Box(s)
29.1240	Key into (Unit ID) box	<u>EG</u> : 01
29.1243	Press <ENTER>	PED reappears with values in cyan
29.1246	If not first write to IDF, press <COMND> and go to 29.1248. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.1248	Press <ENTER>	PROC. JOURNAL CONFIG data to IDF
29.1250	Repeat 29.1204 – 29.1248 to configure remaining journals	
SYSTEM JOURNALS \$OSYSJOR		
	If PED display is on the screen, hold <CTL> down and press <9>.	AREA CONFIGURATION Menu
	If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu
29.1300	Select [SYSTEM JOURNAL]	SYSTEM JOURNAL CONFIG. display
29.1310	Key into JOURNAL NUMBER port	<u>EG</u> : 1
29.1320	Key into JOURNAL NAME port	<u>EG</u> : SYJNL001
29.1330	Key into JOURNAL TITLE port	<u>EG</u> : SYSTEM JOURNAL NO. 1
29.1340	Select JOURNAL TYPE	<u>EG</u> : [SYSTAT]
29.1350	Select INVOCATION TYPE	<u>EG</u> : [DEMAND]
29.1360	Key into PRINTER ID port	<u>EG</u> : 1
29.1370	Key into TIME SPAN port	<u>EG</u> : 24
29.1390	Press <ENTER>	PED reappears with values in cyan

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
29.1400	If not first write to IDF, press <COMND> and go to 29.1410. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.1410	Press <ENTER>	SYS. JOURNAL CONFIG-data to IDF
29.1420	Repeat 29.1310 – 29.1410 to configure remaining journals	
REAL-TIME JOURNALS \$RTJNL		
29.1430	If Command display is on the screen, hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATABASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.1440	Select [R-T JOURNAL ASSIGNMENT]	R-T JOURNAL ASSIGNMENTS DISPLAY CONFIGURATION
29.1450	Key into PRINTER ID ports	<u>EG</u> : 1 (into all ports)
29.1460	Select JRN TYPE-INIT ACT? options	<u>EG</u> : [No] for SYS STATUS CHANGES
29.1470	Press <ENTER>	PED reappears with values in cyan
29.1490	If not first write to IDF, press <COMND> and go to 29.1500. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.1500	Press <ENTER>	REAL-TIME JRNL config-data to IDF

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

OVERVIEW DISPLAY \$OOVERVW		
Step	Action	Response
29.1510	If PED display is on the screen, hold <CTL> down and press <9>.	AREA CONFIGURATION Menu
	If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu
29.1520	Select [OVERVIEW]	AREA OVERVIEW DISP CONFIG display
29.1530	Key into GROUP NUMBERS ports	<u>EG</u> : 1 (into Index 01)
29.1540	Key into NUMBER OF FIELDS port	<u>EG</u> : 1
29.1550	Press <ENTER>	FIELD DEFINITION ports appear
29.1560	Key into FIELD DEF ports: FLD TITLE TOP LEFT ELEMENT WIDTH HEIGHT	<u>EG</u> : WEST GROUP <u>EG</u> : 1 <u>EG</u> : 1 <u>EG</u> : 1
29.1570	Press <ENTER>	PED reappear with values in cyan
29.1573	<PAGE FWD> to the Related/Assoc. and Help Display definitions. Key into ports as desired.	
29.1578	Press <ENTER>	PED reappears with values in cyan
29.1580	Check PED-set status: Hold <CTL> down and press <8>	"Display Set Entered?"-Yes If errors, see: 29.1060 – 29.1080
29.1590	If not first write to IDF, press <COMND> and go to 29.1600. If first write to IDF, go perform 29.80 - 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.1600	Press <ENTER>	OVERVIEW DISP CONFIG data to IDF

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

AREA TREND DISPLAY \$OTREND		
Step	Action	Response
29.1610	If PED display is on the screen, hold <CTL> down and press <9>.	AREA CONFIGURATION Menu
	If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu
29.1615	Select [AREA TREND]	AREA TREND OV. DISP CONFIG display
29.1620	Key into TREND TITLE port	<u>EG</u> : WEST AREA TRENDS
29.1625	Select TREND TIMEBASE option	<u>EG</u> : [8 HOURS]
29.1630	Key in names of points to be trended in cyan and yellow	<u>EG</u> : HG1001 (cyan)
29.1635	Press <ENTER>	PED reappears with values in cyan
29.1640	<PAGE FWD>	RELATED DISP 001 DEF display
29.1645	Key into each SCREEN NUMBER port and each Display Type port	<u>EG</u> : 1 <u>EG</u> : SCHEM
29.1650	Press <ENTER>	PED reappears with values in cyan and SCHEMATIC NAME appears (or other port depending on the DISPLAY TYPE entered).
29.1655	Key into each additional name port	<u>EG</u> : DISP1
29.1660	Press <ENTER>	PED reappears with values in cyan
29.1665	<PAGE FWD>	Next page appears
29.1670	Repeat 29.1650 – 29.1665 to configure remaining Related Displays, and on page 7, Associated and Help display	RELATED, ASSOCIATED, and HELP displays configured
WRITE TO IDF WHEN PATHNAME IS ALREADY SET		
29.1675	Press <COMND>	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathnames set)
29.1680	Press <ENTER>	AREA TREND CONFIG. data to IDF

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

STANDARD LOGS \$OLOG(n) Where n can be 1 to 100		
Step	Action	Response
29.1900	If PED display is on the screen, hold <CTL> down and press <9>.	AREA CONFIGURATION Menu
	If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu
29.1905	Select [STANDARD LOG]	STANDARD LOG CONFIGURATION display
29.1910	Key into LOG NUMBER port	<u>EG</u> : 1
29.1920	Key into LOG NAME port	<u>EG</u> : BLR1RPT
29.1930	Key into LOG DESCRIPTION port	<u>EG</u> : DAILY BOILER LOG
29.1940	Enter other data as required.	<u>EG</u> : Use default values.
29.1950	Key into NUM VALUES PER POINT port	<u>EG</u> : 1
29.1960	Key into NUMBER OF POINTS port	<u>EG</u> : 1
29.1965	Press <ENTER>; press <PAGE FWD>	INDEX ports appear.
29.1970	Key into INDEX ports until all Point IDs have been keyed in.	<u>EG</u> : HG1001 (for Index 001) (from step 21.30)
29.1980	Press <ENTER>	PED reappears with values in cyan

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
	WRITE TO IDF	
29.2000	If not first write to this IDF, press <COMND> and go to 29.2005. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; with correct pathnames.
29.2005	Press <ENTER>	Standard Log config data to IDF
PRINTED TREND \$OPRTRND(n) where n can be 1 to 100		
29.2010	If PED display is on the screen, hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.2012	Select [PRINTED TREND]	PRINTED TREND DISP CONFIG display
29.2014	Key into TREND NUMBER port	<u>EG</u> : 1
29.2016	Key into TREND NAME port	<u>EG</u> : BOILER 1
29.2018	Key into TREND TITLE port	<u>EG</u> : BOILER 1 TREND RECORD
29.2020	Select INVOCATION Type	<u>EG</u> : [PERIODIC]
29.2024	Select INITIALLY ACTIVE option	<u>EG</u> : [NO]
29.2028	Select START TIME port	<u>EG</u> : [0]:[10]
29.2032	Key into PERIOD port	<u>EG</u> : 1:30
29.2036	Key into PRINTER ID port	<u>EG</u> : 1
29.2040	Key into TIME SPAN port	<u>EG</u> : 2
29.2044	Key into GROUP NUMBER port	<u>EG</u> : 1
29.2050	Press <ENTER>	PED reappears with values in cyan

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
29.2070	WRITE TO IDF If not first write to this IDF press <COMND> and go to 29.2080. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (pathname is set)
29.2080	Press <ENTER>	PRINTED TREND CONFIG. data to IDF
29.2085	Repeat 29.2014 – 29.2080 for other Printed Trends.	All printed trends configured and stored in IDF
LCN NODE ANNUNCIATION POLICY \$ANPOLC		
29.2090	If PED display is on the screen, hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
29.2094	Select [LCN NODE ANNUNC. POLICY]	LCN NODE ANNUNC. POLICY display
29.2102	Use <PAGE FWD> to get to all node numbers you wish annunciated. Select [ADDRESS ANNUNCIATED] option	<u>EG</u> : 01, 02, 24, 25, 30, 38, 39, 40, 43, and 46 <u>EG</u> : [YES] for each of above nodes
29.2106	Press <ENTER>	PED reappears with values in cyan
29.2108	If not first write to IDF, press <COMND> and go to 29.2112. If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (pathname is set)
29.2112	Press <ENTER>	LCN NODE ANNUN config data to IDF

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

SEQUENCE OF EVENTS JOURNAL \$SOEJOR(n)		
Step	Action	Response
29.2120	If PED display is on the screen, hold <CTL> down and press <9>.	AREA CONFIGURATION Menu
	If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu
29.2124	Select [SOE JOURNAL ASSIGNMENT]	SEQUENCE OF EVENTS JOURNAL display
29.2128	Key into JOURNAL NUMBER	<u>EG</u> : 1
29.2130	Key into JOURNAL NAME	<u>EG</u> : SOEJNL01
29.2134	Key into JOURNAL TITLE	<u>EG</u> : BOILER1 SEQ OF EVENTS
29.2138	Key into HM NODE NUMBER	<u>EG</u> : 43
29.2140	Press <ENTER>	PED reappears with values in cyan
29.2144	If not first write to IDF, press <COMND> and go to 29.2148.	COMMAND DISPLAY, [WRITE TO IDF] selected; (pathname is set)
	If first write to IDF, go perform 29.80 – 29.110 and return here.	
29.2148	Press <ENTER>	SOE JOURNAL config data to IDF

REPORT DISPLAY \$SOREPORT(n) where n can be 1 to 100		
Step	Action	Response
29.2150	If PED display is on the screen, hold <CTL> down and press <9>.	AREA CONFIGURATION MENU
	If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION MENU
29.2154	Select [REPORT]	REPORT DISPLAY CONFIGURATION display
29.2156	Key into REPORT NUMBER port	<u>EG</u> : 1
29.2160	Key into REPORT NAME port	<u>EG</u> : WESTRPT
29.2164	Key into REPORT TITLE port	<u>EG</u> : WEST AREA REPORT
29.2170	Select INVOCATION TYPE option	<u>EG</u> : [DEMAND]
29.2175	Key into NO. ITEMS IN REPORT port	<u>EG</u> : 1

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

Step	Action	Response
29.2180	Press <ENTER>	ITEMS IN REPORT port appears.
29.2185	Key into Items In Report port Note: Items must be previously built as trends or logs	<u>EG:</u> BLR1RPT (configured in 29.1920)
29.2190	Press <ENTER> WRITE TO IDF	PED reappears with values in cyan
29.2194	If not first write to IDF, press <COMND> and go to 29.2198 If first write to IDF, go perform 29.80 – 29.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
29.2198	Press <ENTER>	REPORT CONFIGURATION data to IDF

NOTE

Nodes associated with the area being multiple loaded must have a status of OK before the load multiple can be successfully performed.

LOAD MULTIPLE ENTITIES FROM IDF		
Step	Action	Response
	SELECT AN AREA	
29.2205	Press <COMND>	COMMAND DISPLAY
29.2208	Select [SELECT AREA]	Area Ports/Target appear
29.2212	Key into REFERENCE PATH NAME port	<u>EG:</u> NET>H MV1>
29.2216	Select an area Current AREA DATA BASE pathname port is filled in automatically	<u>EG:</u> [1] <u>EG:</u> NET>&D01>AREA01
29.2218	Press <ENTER>	"OPERATION COMPLETE"
29.2220	Select [LOAD MULTIPLE]	Additional Ports/Target appear

**TASK 29 BUILD, SELECT, AND INSTALL
AREA DATABASES (continued)**

Keyboard: Engr's

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

INSTALL THE AREA DATABASE
NOTE

The Area Database is installed to a volume and file with the volume name of &Dnn and with the file name of AREAnn. For &Dnn, the directory names of &D01 through &D10 were automatically created during the HM Volume Configuration task, Task 10. For AREAnn, nn is the index number from the Area Names configuration list that you configured in Task 7. For example, the first area name you entered into the list is automatically accessed by AREA01, the second name by AREA02, etc.

Step	Action	Response
29.2230	Key into REFERENCE PATH NAME port	<u>EG</u> : NET>HNV1>
29.2240	Key into pathname for IDF port	<u>EG</u> : BOILER1
29.2250	Key into current AREA DATA BASE pathname port	<u>EG</u> : NET>&D01>AREA01
29.2260	Press <ENTER> Wait—Entities are listed twice	"PROCESSING ENTITIES..." "OPERATION COMPLETE"
29.2270	Press <COMND>	COMMAND DISPLAY
29.2280	Select [INSTALL AREA]	Additional targets appear.
29.2290	Key a (destination) pathname into Current AREA DATA BASE p/nm port	<u>EG</u> : NET>&Dnn>AREAnn (where nn is equal to 01–10 in both cases)
29.2295	Key a (source pathname) into the Pathname for WORKFILE port.	<u>EG</u> : NET>&D01>AREA01
29.2300	Press <ENTER>	"OPERATION COMPLETE"

NOTE

1. Most of the Command Display functions deal with IDFs; however, Exception Build files are more portable. If you want to create Exception Build files from your IDFs, follow the instructions in Appendix E.
2. After the Area Database is completely built, the IDFs and Area Database files can be copied to cartridge disk and then deleted from the User Volume NET>HNV1>Boiler1.
3. For security, you should copy the following onto disk: Area Database (&D01), Pictures, Free Format Logs, Button source files (.DS and .KS files), and custom files (CUS/CLX, etc.).
4. Go perform the EC in Task 31, steps 31A.10 - 31A.50 to save all the &ASY files. Use the resulting disk as the new backup if you have to reload from disk or to perform On-Line Network Reconfiguration.

TASK 30 **LOAD THE OPERATOR OR UNIVERSAL PERSONALITY TO UNIVERSAL STATIONS** **Keyboard:** Both **Lock:** ENGR

Reference: This Document

Personality Loaded: Universal

Step	Action	Response
30.01	If the other US is a U ^X S, skip to 30.150 to load the U ^X S personality	See 1.1.1 for U ^X S preparation. (Or load U ^X S with a standard personality by following the US procedure starting at 30.10.)
30.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
30.20	Select [CONSOLE STATUS]	CONSOLE STATUS AND ASSG. display
30.30	Under the STN header, select the number of the US you wish to load	<u>EG:</u> 2
30.60	Select [LOAD/DUMP]	LOAD/DUMP NODE Targets appear
30.64	Select [MANUAL LOAD]	NODE LOAD targets appear
30.70	Select [OPERATOR PROGRAM] or [UNIVERSAL PROGRAM]	PGM SOURCE FOR NODE nn display (US must have enough memory to load UP)
30.80	Select [DEFAULT SOURCE]	EXECUTE COMMAND target appears
30.90	Select [EXECUTE COMMAND]	DATA SOURCE FOR NODE nn
30.100	Select [DEFAULT SOURCE]	EXECUTE COMMAND target appears
30.110	Select [EXECUTE COMMAND]	CONSOLE STATUS AND ASSG. display "PRESS ENTER TO EXECUTE"
30.120	Press <ENTER>	STN NN STATUS goes to LOAD
30.130	Wait for several minutes. Watch the Console Status display at the Station being used.	"BUSY: US nn" Console Status Display appears with a status of READY on Station NN. STN nn STATUS eventually goes to OK . Type goes to OPR for Operator Personality or UNVL for Universal Personality.
30.140	Repeat Task 30 to load other Universal Stations.	If no other US or U ^X S stations, this completes Task 30.
30.150	To load a U ^X S, station with the U ^X S personality, press (LOAD) on the U ^X S Operator's Keyboard	"W, N, 1, 2, 3, 4, X?"
30.160	Press (W) and then press (ENTER)	U ^X S loads from internal Winchester disk, then accesses LCN for NCF and area database. U ^X S Station Status goes to OK; Type = U ^X S.

TASK 31 CHANGE NETWORK CONFIGURATION**Keyboard:** Engr's**Lock:** ENGR**Reference:** SW11-605**Personality Loaded:** Universal

Step	Action	Response
	MAKE A BACKUP COPY OF &ASY	
31.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
31.30	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
31.35	&C1 or &C2 into FD1	
31.36	Place a blank cartridge disk in FD2 (Cartridge Write-Protect switch must be in the write position)	
31.40	Key in an EC command	<u>EG:</u> EC \$F1>&EC>ASY_BKUP.EC \$F2 \$P1 (printer \$P1 is optional)
31.45	Press <ENTER>	Message: "THIS EC WILL...." "....." "..... CONTINUE (Yes/No)?"
31.50	Key in YES , then press <ENTER>	Message: "THIS EC... IS THIS CORRECT?"
31.55	Key in YES , then press <ENTER>	Message: "MOUNT BACKUP MEDIA...."
31.60	Press <ENTER> (the EC prepares to update the Backup Log).	"UNPT NET>&ASY>BKUP_LOG.XX...." "DO YOU WANT TO RE-INITIALIZE...."
31.65	Key in YES , then press <ENTER> (the EC formats the cartridge in FD2, and begins copying files to it).	"CR \$F2>&ASY> -F -MF 500" "COPYING" "EC COMPLETE"
31.68	Remove the cartridge from FD2. Label it BACKUP &ASY and write the date on it.	
31.70	Make backup copies of other files that you created (for example insert the AREA cartridge in FD2 and copy the area files, button files, and custom software files.	<u>EG:</u> CP NET>&D01>*. * \$F2>&D01>= -D CP NET>&CUS>*. * \$F2>&CUS>= -D CP NET>&CLX>*. * \$F2>&CLX>= -D

NOTERefer to the *Network Data Entry* manual (see References) for information on network reconfiguration.

CAUTION

The NIM(s) must be loaded with its operating personality and have a status of OK on the System Status Display before starting this procedure. The NIM was loaded in Task 18.

If you are configuring a PM or HPM, the procedure is nearly identical to the **APM** configuration described below except where noted. The sample system did not have an SM or LM. Refer to the manuals in Section 1, References, if necessary, to configure these.

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS **Keyboard:** Engr's **Lock:** ENGR

Reference: PM09-640, AP09-640, & HP09-640 **Personality Loaded:** Universal

Step	Action	Response
32.110	Hold <CTL> down and press <HELP> If you don't have a NIM, skip to Task 23	ENGINEERING MAIN MENU
\$NMhhNnn Where hh is the UCN number and nn is the NIM Node number		
	CONFIGURE THE NIM UCN ADDRESSES	
32.115	Select [NETWORK INTF. MODULE]	NIM BUILD TYPE SELECT MENU
32.120	Select [UCN NODE CONFIGURATION]	UCN-NODE CONFIGURATION display
32.130	Key into NETWORK NUMBER port	<u>EG:</u> 03
32.140	Key into UCN NODE NUMBER port (NIM node number)	<u>EG:</u> 01 (primary NIM node # on 1st pass) <u>EG:</u> 02 (redundant NIM node number, if any, on 2nd pass)
32.150	Select an NODE TYPE option	<u>EG:</u> [NIM]
32.160	Select a NODE ASSIGNMENT option	<u>EG:</u> [THISNIM]
32.165	Press <ENTER>	PED reappears
32.169	Select a LOAD SCOPE option	<u>EG:</u> [NIMANDPM]
32.173	Select a NUMBER OF MESSAGE TEXTS	<u>EG:</u> [3]
32.180	Press <ENTER>	Message Text ports appear
32.184	Key in message text	<u>EG:</u> [OUT/SERV] [RED TAG] [IN USE]
32.188	Select an SOE SYNC option	<u>EG:</u> [ENABLE] (if NIM has EPNI board)
32.190	Press <ENTER>	PED reappears with values in cyan

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS Keyboard: Engr's
(continued)

Lock: ENGR

Ref: PM09-640/AP09-640/HP09-640, & SW09-605

Pers. Loaded: Universal

NOTE

If you select any options outlined in yellow or if you key data into any port outlined in yellow, the content of the PED-set or PED changes when you press the <ENTER> key. You should press <ENTER> after each PED page.

Step	Action	Response
32.210	Press <PAGE FWD> to next PED page (if any) key in data, select options, and press <ENTER> after each page WRITE TO IDF	UCN NODE CONFIGURATION Ped
32.320	Press <COMND>	COMMAND DISPLAY
32.400	If this is the <u>first</u> write to this IDF, go to step 32.460; if not, go to step 32.485	IDF for this example is NIMIDF
32.460	Select [WRITE TO IDF]	Pathname ports and target appear.
32.470	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>H MV1>
32.480	Key into pathname for IDF port	<u>EG</u> : NIMIDF
32.485	Press <ENTER>	NIM Config data written to IDF. UCN NODE CONFIGURATION display
32.488	Repeat steps 32.130 – 32.485 for redundant NIM (if any)	
32.490	Repeat steps 32.130 – 32.488 for each other NIM pair.	All configured NIMs on IDF
32.500	Press <COMND>	COMMAND DISPLAY
32.510	Select [LOAD MULTIPLE]	Additional pathname ports appear
32.515	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>H MV1>
32.520	Key into pathname for IDF port	<u>EG</u> : NIMIDF
32.525	Press <ENTER>	"WAIT" in yellow; "PROCESSING ..." "OPERATION COMPLETE" COMMAND DISPLAY
32.530	Hold <CTL> down and press <1>	UCN NODE CONFIGURATION display

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS Keyboard: Engr's Lock: ENGR
(continued)

Ref: PM09-640/AP09-640/HP09-640, & SW09-605 Pers. Loaded: Universal

Step	Action	Response
\$NMhhNnn Where hh is the UCN number and nn is the APMM Node number		
	PM/APM/HPM NODE CONFIGURATION	The UCN NODE CONFIGURATION display should be on the screen.
32.550	Key into NETWORK NUMBER port	<u>EG</u> : 03
32.560	Key into UCN NODE NUMBER port (APMM node number)	<u>EG</u> : 03 (Primary node # on first pass, redundant node #, if any, on second pass) <u>EG</u> : 04
32.570	Select [UCN NODE TYPE]	<u>EG</u> : [APM]
32.580	Select a NODE ASSIGNMENT option	<u>EG</u> : [THISNIM]
32.590	Press <ENTER>	PED reappears with values in cyan
32.595	Key in any other data/select options press <ENTER>	
	WRITE TO IDF	
32.610	Press <COMND>	COMMAND DISPLAY
32.620	If this is the <u>first</u> write to this IDF, go to step 32.630; if not, go to step 32.660	IDF for this example is APMMIDF
32.630	Select [WRITE TO IDF]	Pathname ports and target appear.
32.640	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
32.650	Key into pathname for IDF port	<u>EG</u> : APMMIDF
32.660	Press <ENTER>	APM Config data written to IDF. UCN NODE CONFIG. PED display
32.670	Repeat steps 32.550 – 32.660 for the redundant APMM (if any).	
32.680	Repeat steps 32.550 – 32.670 for each other APMM	All configured APMMs in IDF
32.700	Press <COMND>	COMMAND DISPLAY
32.710	Select [LOAD MULTIPLE]	Additional pathname ports appear
32.720	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>
32.730	Key into pathname for IDF port	<u>EG</u> : APMMIDF
32.740	Press <ENTER>	"WAIT" in yellow;"...OPERATION COMPLETE"
	Repeat steps 32.550 - 32.740 for other PMs, or HPMs (if any).	Use different IDF names for PMs and HPMs

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS Keyboard: Engr's **Lock:** ENGR
(continued)

Ref: PM09-640/AP09-640/HP09-640, & SW09-605 **Pers. Loaded:** Universal

Step	Action	Response
	LOAD THE APM	
32.742	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
32.744	&C6 copy into FD1	
32.746	Select [SYSTEM STATUS]	SYSTEM STATUS display
32.748	Select the NIM Node	Node Box is selected
32.750	Select [NTWK/HWY STATUS]	UCN nn STATUS display
32.754	Select [nn APM nn] The status must be <i>ALIVE</i> (Select target 2nd time if loading the redundant node)	Target fills with color; selected node is highlighted
32.758	Select [LOAD/SAVE RESTORE]	Additional targets appear
32.762	Select [PROGRAM LOAD]	Target turns white
32.766	Select [ENTER]	PGM SOURCE FOR NODE nn display
32.770	Select [DEFAULT SOURCE]	EXECUTE target appears
32.774	Select [EXECUTE]	DATA SOURCE FOR NODE nn display
32.778	Select [ALTERNATE SOURCE]	Targets [1], [2], etc. appear
32.782	Select <u>EG</u> : [1]	&C6 and [EXECUTE COMMAND]
32.786	Select [EXECUTE COMMAND]	" BUSY UCN nn " [nn APM nn LOADING] " * " "MOUNTAPM nn DATA BASE"
32.790	Select [MOUNT APMnn DATA BASE] (Primary node only)	Additional targets appear
32.794	Select [ABORT REQUEST] (Primary node only)	<u>EG</u> : [nn APM nn] [IDLE] [ALIVE/IDLE] [LOADED]
32.796	Repeat Tasks 32.754 – 32.794 for redundant APM (if any)	<u>EG</u> : [nn APM nn] [IDLE] [BACKUP] [LOADED]
32.798	Repeat Tasks 32.754 – 32.796 for other PMs, APMs, or HPMs	

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS Keyboard: Engr's **Lock:** ENGR
(continued)

Ref: PM09-640/AP09-640/HP09-640, & SW09-605 **Pers. Loaded:** Universal

Step	Action	Response
\$NMhhBuu Where hh is the UCN number and uu is the APMM Node number		
	PM/APM/HPM NODE SPECIFIC CONFIGURATION	
32.800	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
32.802	Select [NETWORK INTF. MODULE]	NIM BUILD TYPE SELECT MENU
32.806	Select [NODE SPECIFIC CONFIG.]	NODE SPECIFIC CONFIGURATION MENU
32.808	Key into NETWORK NUMBER	<u>EG:</u> 03
32.814	Key into NODE NUMBER	<u>EG:</u> 03 (Primary APMM)
32.816	Key into NODE TYPE	<u>EG:</u> APM
32.818	Press <ENTER>	
32.820	Press <PAGE FWD> to next PED page, key in data, select options, and press <ENTER> after each page or gold port	NODE SPECIFIC CONFIG. PED
32.822	Select a CONTROL PROC. OPTION (Not Applicable for HPMS)	<u>EG:</u> [ON]
32.826	Key into NUM OF REG CONTROL SLOTS	<u>EG:</u> 25
32.830	Key into NUMBER OF FAST REG	<u>EG:</u> 5
32.834	Key into NUMBER OF REG PV SLOTS	<u>EG:</u> 20
32.840	Key into NUM OF FAST REG PV SLOTS	<u>EG:</u> 5
32.844	Key into NUMBER OF LOGIC SLOTS	<u>EG:</u> 10
32.846	Key into NUM OF FAST LOGIC SLOTS	<u>EG:</u> 3
32.850	Key into NUM OF DIG. COMP. SLOTS	<u>EG:</u> 15
32.852	Key into NMBER OF FAST DIG. COMP.	<u>EG:</u> 2
32.854	Key into NUM OF DEV. CTL SLOTS	<u>EG:</u> 4
32.856	Key into NMBER OF FAST DEV. CTL	<u>EG:</u> 2
32.858	Key into SEQUENCE PROCESSING	<u>EG:</u> 1_PU
32.860	Key into NUM OF PROC MODULE SLOTS	<u>EG:</u> 10

32 PROCEDURE 4 – CONFIGURE PROCESS NETWORKS AND HIWAYS 32

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS **Keyboard:** Engr's **Lock:** ENGR
(continued)

Ref: PM09-640/AP09-640/HP09-640, & SW09-605 **Pers. Loaded:** Universal

Step	Action	Response
32.862	Key into NUM OF NUMERICS SLOTS	<u>EG</u> : 48 (Multiples of 16 only)
32.864	Key into NUM OF STRINGS SLOTS*	<u>EG</u> : 16 (Multiples of 16 only)
32.866	Key into NUM OF TIMES SLOTS*	<u>EG</u> : 32 (Multiples of 32 only)
32.868	Key into NUM OF ARRAY SLOTS*	<u>EG</u> : 10
32.870	Select an SI DATA... SCAN Period*	<u>EG</u> : 1.0
32.874	Select a SCAN RATE	<u>EG</u> : [REG1LOG1]
32.880	Select a PACKAGING OPTION (Select REDUN if redundant nodes)	<u>EG</u> : [REDUN]
32.883	Select a Simulation Indicator option*	<u>EG</u> : ON
32.885	Press <PAGE FWD> to next PED page, key data, select options, and press <ENTER> after each page	I/O MODULE CONFIGURATION PED
32.900	Choose an I/O Module number, press <PAGE FWD> if necessary	<u>EG</u> : I/O MODULE NUMBER 1
32.910	Key into FILE NUMBER	<u>EG</u> : 1
32.920	Key into CARD NUMBER (Example is 1st card Number if REDUN is selected)	<u>EG</u> : 11 for APM ; (9 for PM, 03 for HPM)
32.930	Select a MODULE TYPE	<u>EG</u> : [HLAI]
32.932	If the system has redundant IOPs, select REDUN. If not, skip to 32.940	<u>EG</u> : IOMB File and Card number ports
32.934	Enter redundant IOP File and Card No.	<u>EG</u> : File 2, Card No. 11
32.940	Choose another I/O Module number, press <PAGE FWD> if necessary	<u>EG</u> : MODULE NUMBER 2
32.950	Key into IOM-A FILE NUMBER	<u>EG</u> : 1
32.960	Key into IOM-A CARD NUMBER	<u>EG</u> : 12 for APM
32.970	Select MODULE TYPE	<u>EG</u> : [AO]
32.975	Select a FAILURE OPTION	<u>EG</u> : [HOLD]
32.980	Key in any other data/select options, press <ENTER>	
32.990	Press <PAGE FWD> to next PED page (if any) key in data, select options, and press <ENTER> after each page	I/O MODULE CONFIGURATION PED

* Not applicable for PMs

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS
(continued)

Keyboard: Engr's

Lock: ENGR

Ref: PM09-640/AP09-640/HP09-640, & SW09-605

Pers. Loaded: Universal

Step	Action	Response
32.995	Repeat steps 32.940 through 32.990 to configure other I/O Modules	
	WRITE TO IDF	
32.1000	Press <COMND>	COMMAND DISPLAY
32.1020	If this is the <u>first</u> write to this IDF, go to step 32.1030; if not, go to step 32.1060	IDF for this example is APMCONFIG
32.1030	Select [WRITE TO IDF]	Pathname ports and target appear.
32.1040	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
32.1050	Key into pathname for IDF port	<u>EG</u> : APMCONFIG or HPMCONFIG
32.1060	Press <ENTER>	I/O Module Config data written to IDF. IO MODULE CONFIG. display
32.1070	Repeat steps 32.800 – 32.1060 for other APMs on this UCN (if any)	
32.1080	Repeat steps 32.800 – 32.1070 for each other PM, APM, or HPM primary (if any)	All configured PMs, APMs, HPMs on IDFs (use different IDF names for PMs and HPMs)

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS
(continued)

Keyboard: Engr's

Lock: ENGR

Ref: PM09-640/AP09-640/HP09-640, & SW09-605

Pers. Loaded: Universal

Step	Action	Response
32.1100	Press <COMND>	COMMAND DISPLAY
32.1110	Select [LOAD MULTIPLE]	Additional pathname ports appear
32.1115	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>
32.1120	Key into pathname for IDF port	<u>EG</u> : APMCONFIG
32.1125	Press <ENTER>	"WAIT" in yellow;.."OPERATION COMPLETE"
32.1150	Repeat steps 32.1100 – 32.1150 for PMs or HPMs (if any)	PMs, APMs, HPMs loaded with Node Specific Configuration Data
	PM/APM/HPM POINT BUILDING	
32.1200	Hold <CTL> down and press <9>	NIM BUILD TYPE SELECT MENU
32.1220	Select [PROCESS POINT BUILDING]	NIM PROC. POINT BUILD TYPE display
32.1230	Select a Point Build Type	<u>EG</u> : [REGULATORY CONTROL] NIM-POINT ASSIGNMENT display
32.1240	Key into TAG NAME port	<u>EG</u> : FC2401
32.1244	Select a NODE TYPE	<u>EG</u> : APM
32.1250	Select a POINT FORM option	<u>EG</u> : [FULL]
32.1260	Key into the POINT DESCRIPTOR port	<u>EG</u> : TOWER FLOW CONTROL
32.1265	Key into the E. U. DESCRIPTOR port	<u>EG</u> : KB/D
32.1275	Key into the POINT KEYWORD port	<u>EG</u> : FLOW
32.1285	Key into the UNIT ID port	<u>EG</u> : 01
32.1290	Key into the NETWORK NUMBER port	<u>EG</u> : 03
32.1295	Key into the NODE NUMBER port	<u>EG</u> : 03
32.1300	Key into the SLOT NUMBER port	<u>EG</u> : 01
32.1310	Key in other data/select options press <ENTER>	
32.1315	Press <PAGE FWD>	Next PED page
32.1320	Select a Control Algorithm; press <ENTER>	<u>EG</u> : [PID]

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS
(continued)

Keyboard: Engr's

Lock: ENGR

Ref: PM09-640/AP09-640/HP09-640, & SW09-605

Pers. Loaded: Universal

Step	Action	Response
32.1325	Press <PAGE FWD>	Next PED page
32.1330	Key into PV RANGE HIGH port	<u>EG</u> : 25.00000
32.1335	Select a PV SOURCE OPTION	<u>EG</u> : [ALL]
32.1340	Press <ENTER> after each page and after any entry boxed in yellow	
32.1345	Press <PAGE FWD>	Next PED page
32.1350	Select a NORMAL MODE	<u>EG</u> : [AUTO]
32.1355	Select a NORMAL MODE ATTRIBUTE	<u>EG</u> : [OPERATOR]
32.1360	Key in any other data/select options; press <ENTER> after each page	
32.1365	Press <PAGE FWD>	Next PED page
32.1370	Key into SET POINT HIGH LIMIT port	<u>EG</u> : 25.00000
32.1380	Key into SET POINT LOW LIMIT port	<u>EG</u> : 0.000000
32.1390	Key into SET POINT port	<u>EG</u> : 15.00000
32.1395	Key in any other data/select options; press <ENTER> after each page	
32.1400	Press <PAGE FWD>	Next PED page
32.1420	Select an ALGORITHM FORM option	<u>EG</u> : [IDEAL]
32.1430	Key into OVERALL GAIN VALUE port	<u>EG</u> : 0.699200
32.1440	Key into INTEG. TIME IN MINUTES port	<u>EG</u> : 0.100000
32.1445	Key in any other data/select options; press <ENTER> after each page	
32.1450	Press <PAGE FWD>	Next PED page
32.1460	Key into INPUT CONN. SOURCE port	<u>EG</u> : - - - -
32.1462	Key into OUTPUT DEST. SOURCE port	<u>EG</u> : !AO02S02.OP
32.1465	Key in any other data/select options; press <ENTER> after each page	
32.1470	Press <PAGE FWD>	Next PED page

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS
(continued)

Keyboard: Engr's

Lock: ENGR

Ref: PM09-640/AP09-640/HP09-640, & SW09-605

Pers. Loaded: Universal

Step	Action	Response
32.1485	Key in any other data/select options; press <ENTER> after each page	
32.1490	Press <PAGE FWD>	Next PED page
32.1500	Key into PV HIGH TRIP POINT port	<u>EG</u> : 25.00000
32.1502	Key into PV LOW TRIP POINT port	<u>EG</u> : 0.000000
32.1504	Key into POSITIVE PV ROC TRIP port	<u>EG</u> : 32.00000
32.1508	Press <ENTER> after each page and after each entry boxed in yellow	
32.1510	Press <PAGE FWD>	Next PED page
32.1512	Key into NEG. PVROC TRIP POINT port	<u>EG</u> : 20.00000
32.1514	Key into DEV HIGH TRIP POINT port	<u>EG</u> : 5.000000
32.1516	Key into DEV LOW TRIP POINT port	<u>EG</u> : 5.000000
32.1518	Press <ENTER>	
32.1520	Press <PAGE FWD> to next PED page. Enter any other data/select options and press <ENTER> after each page until all pages are configured.	
32.1522	Hold down <CTL> and press <8>	PED STATUS display
32.1524	Check for proper status	"Display Set Entered?" Yes Errors? No
32.1526	If status is OK, go to step 32.1530; else select faulty item; enter corrections.	
WRITE TO IDF		
32.1530	Press <COMND>	COMMAND DISPLAY
32.1532	If this is the <u>first</u> write to this IDF, go to step 32.1534; if not, go to step 32.1560	IDF for this example is APMPNTS
32.1534	Select [WRITE TO IDF]	Pathname ports and target appear.
32.1540	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
32.1550	Key into pathname for IDF port	<u>EG</u> : APMPNTS

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS
(continued)

Keyboard: Engr's

Lock: ENGR

Ref: PM09-640/AP09-640/HP09-640, & SW09-605

Pers. Loaded: Universal

Step	Action	Response
32.1560	Press <ENTER>	NIM Point Config. data written to IDF. NIM Point PED display
32.1570	Repeat steps 32.1240 – 32.1560 to build other points of this type, or hold <CTL> down and press <9>. Then repeat steps 32.1230 – 32.1570 to build other types of NIM Points until all NIM points are built If you want to clone points from a master point, go to Appendix D	
32.1600	Press <COMND>	COMMAND DISPLAY
32.1610	Select [LOAD MULTIPLE]	Additional pathname ports appear
32.1620	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>
32.1630	Key into pathname for IDF port	<u>EG</u> : APMPNTS
32.1640	Press <ENTER> Build/load points for PMs and HPMS (if any) by repeating steps 32.1200 - 32.1640	"WAIT" in yellow; "PROCESSING ..." "OPERATION COMPLETE" Use different IDF names for PM or HPM points
NIM LIBRARY CONFIGURATION		
\$NhhLIBn Where hh is the UCN number and n is the library number		
Note that the NIM Library is usually automatically loaded while compiling an APM Sequence program. The following steps illustrate how to access, manually load, or modify it.		
32.1710	If the COMND or Point CONF. display is on screen, hold <CTL> down and press <9> <u>twice</u> . Then go to step 32.1720. Else hold <CTL> down and press <HELP>	POINT TYPE BUILD MENU, then NIM BUILD TYPE SELECT MENU ENGINEERING MAIN MENU
32.1715	Select [NETWORK INTF. MODULE]	NIM BUILD TYPE SELECT MENU
32.1720	Select [LIBRARY CONFIGURATION]	NIM LIBRARY CONFIGURATION display
32.1722	Key into the NETWORK NUMBER port	<u>EG</u> : 03
32.1724	Key into the Library Number port	<u>EG</u> : 1
32.1728	Press <ENTER>	

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS Keyboard: Engr's Lock: ENGR
(continued)

Ref: PM09-640/AP09-640/HP09-640, & SW09-605 Pers. Loaded: Universal

Step	Action	Response
32.1730	<PAGE FWD>	<u>EG</u> : TRUE for 002, FALSE for 003, etc.
32.1732	Key into the INDEX ENTRY ports	
32.1734	Press <ENTER>	
32.1736	<PAGE FWD>. Enter any other data, press <ENTER> after each page.	
WRITE TO IDF		
32.1740	Press <COMND>	COMMAND DISPLAY
32.1745	If this is the <u>first</u> write to this IDF, go to step 32.1755; if not, go to step 32.1765	IDF for this example is LIBCNFG
32.1750	Select [WRITE TO IDF]	Pathname ports and target appear.
32.1755	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
32.1760	Key into pathname for IDF port	<u>EG</u> : LIBCNFG
32.1765	Press <ENTER>	NIM Library Config data written to IDF. NIM LIBRARY CONFIGURATION display
32.1767	Repeat steps 32.1724 – 32.1765 to build Libraries 2/3 (if desired) for this UCN	
32.1770	Repeat steps 32.1722 – 32.1767 to build Libraries for other UCNs.	
32.1800	Press <COMND>	COMMAND DISPLAY
32.1810	Select [LOAD MULTIPLE]	Additional pathname ports appear
32.1820	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>
32.1830	Key into pathname for IDF port	<u>EG</u> : LIBCNFG
32.1840	Press <ENTER>	"WAIT" in yellow; "PROCESSING ..." "OPERATION COMPLETE"

TASK 32 CONFIGURE UCN NODES & SUBSYSTEMS
(continued)

Keyboard: Engr.

Lock: ENGR

Ref: PM09-640/AP09-640/HP09-640, & SW09-605

Pers. Loaded: Universal

NOTE

In the following steps you will not be able to save data for any IO Processor for which you have not previously built and loaded at least one data point. When saving data, all IOPs that are configured for a given PM, APM, or HPM will assume a "DB Invalid" status (and should not be validated until a point has been built and loaded).

Step	Action	Response
	SAVE CHECKPOINT DATA FOR A NIM AND PM, APM or HPM	
32.1910	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
32.1920	CKPT cartridge into FD1	
32.1930	Select [SYSTEM STATUS]	SYSTEM STATUS display
32.1934	Select the NIM Node	Node Box is selected
32.1938	Select [NTWK/HWY STATUS]	UCN nn STATUS display
32.1945	Select the appropriate APM target	nn APM nn target fills with color
32.1950	Select [LOAD/SAVE RESTORE]	Additional targets appear.
32.1955	Select [SAVE DATA]	Target changes color; additional targets
32.1960	Select [ENTER]	SAVE UCN NODE display
32.1965	If you wish to save checkpoints on cartridge disk, go to step 32.1970. If you wish to save checkpoints on an HM, select [DEFAULT SOURCE] and then go to step 32.1980.	Note: Both NIM data and the associated APM data is saved by this procedure.
32.1970	Select [ALTERNATE SOURCE]	Targets [1] and [2] etc. appear
32.1975	Select [1]	<u>EG</u> : Volume ID = CKPT
32.1980	Select [EXECUTE COMMAND]	"BUSY UCN nn display and " * "
32.1999	Wait for status to change	"SAVED"
32.2000	Repeat steps 32.1945 – 32.1999 for other PMs, APMs, and HPMS.	
If you entered Task 32 from Task 22, you should go to Task 23 next.		

ERROR RECOVERY PROCEDURES

Section 4

Task #	Error Conditions	Reason and Recovery Procedure
Many	US Node fails with an Internal File Manager error.	1. Copy to existing, protected file. 2. Installing to a protected NCF.CF. 3. HG checkpoint to a protected file. 4. Picture Editor/Button Configurator read or write to protected file. 5. Using DEB commands to write to a protected IDF or Entity List File.
	HG Fails	Manually reset the HG and then repeat Task 18.
2	Universal Personality will not load.	One of the nodes is active. Check that <u>all</u> on the NET are in the RESET state. Then repeat Task 2. You may have to press RESET on the HM an extra time because it may have autobooted.
12	1. A change to &ASY was made to a copy of &ASY on the HM. 2. The changed &ASY on the HM <u>was not</u> copied to a cartridge. 3. A load of the Universal Personality is being attempted, using cartridges that include the CNCF cartridge that existed before changing the &ASY on HM. 4. The CNCF cartridge will not read in and the Station dies.	The time-stamp on the CNCF cartridge disk is not the same as the time-stamp for &ASY in one of the nodes. To recover, do the following: 1. Load the Universal Personality, using the &C1 or &C2 cartridge disk and answering the NCF? prompt with N (NET) for HM as the source of the NCF. 2. If recovery step 1 works, copy the &ASY from HM to the CNCF cartridge you were trying to load from. 3. Try the new CNCF cartridge to load the Universal Personality from cartridges. If successful, mark the same date on it as exists on the &ASY on the HM.

Task #	Error Conditions	Reason and Recovery Procedure
15	Got a message of "HM INIT DID RESPOND FOR 1 MINUTE(REVIEW)"	Power to the Winchester disks is off, even though power to the HM node is on. Turn the power to Winchester disks to the "on" position. Repeat Tasks 14 and 15.
18	On-Process Personality isn't successfully loaded into a module or gateway.	Select the red failure message at the top of the display, to call up the module or gateway status display. On that display, select the node number for the node you are trying to load. This causes an error message to appear. If the message is TARGET NODE FAILED DURING LOAD, check the module or gateway to see if power is on, all boards have a green light on, and the node number appears in the red digits on the front of the unit. If, instead, the red digits show -189 or -190, reset the module or gateway and try to load it again. If another error code appears, call a hardware technician. If you were attempting to load a node or gateway using an HM as the program and data source, and the error message is FILES NOT FOUND, check the HM's status as described in the preceding paragraph. If it has failed, turn its power (power to both the module and disk drive) off, wait five seconds, and turn power back on. In about two minutes, its status on the module status display should become OK. If so, reset the node you were trying to load and try again. If you get the same error message again: a) Use the Universal Personality's Command Processor functions to make sure that the volumes mentioned in Task 18 are all on the source HM. b) Try to load the node using cartridge disks. If this works, you may have to delete and recopy all the volumes on the HM. Get a hardware technician's assistance.

NOTE

If you do not find a description of your problem in these error recovery procedures, see the *TPS Messages Directory* (see References).

TASK 13

Appendix A

TASK 13 LOAD THE OPERATOR OR UNIVERSAL PERSONALITY INTO ANOTHER UNIVERSAL STATION

Keyboard: Both Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Step	Action	Response
13.10	&C1 or &C2 into FD1, CNCF into FD2	
13.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
13.30	Select [CONSOLE STATUS]	Console Status display <u>Other US Status seen on This US</u> PWR_ON or QUALIF
13.40	Select the number of the other Universal Station.	<u>EG: 2</u> (for STN #2, node no. 2) LOAD or DUMP NODE targets appear
13.50	Select [LOAD/DUMP]	LOAD/DUMP targets on screen
13.55	Select [MANUAL LOAD].	Node Load targets appear
13.60	Select [OPERATOR PROGRAM] or [UNIVERSAL PROGRAM]	PGM SOURCE FOR NODE NN
13.70	Select [ALTERNATE SOURCE]	Boxes with 1 and 2 appear
13.90	Select box [1]	VOL ID: "&C1" or "&C2" and targets appear

TASK 13 LOAD THE OPERATOR OR UNIVERSAL PERSONALITY TO THE OTHER US (continued)

Keyboard: Engr's **Lock:** ENGR

Reference: This Document

Personality Loaded: Universal

Step	Action	Response
13.100	Select [EXECUTE COMMAND]	DATA SOURCE FOR NODE NN display
13.110	Select [ALTERNATE SOURCE]	Boxes with 1 and 2 appear
13.120	Select box [2]	VOL ID: "CNCF" and targets appear
13.130	Select [EXECUTE COMMAND]	CONSOLE STATUS AND ASSGN display PRESS ENTER TO EXECUTE
13.140	Press <ENTER>	Status of other Station = NET LOAD "SINGLE LOAD BUSY: US nn" "MOUNT STD ABS for US nn"
13.150	Select [MOUNT STD ABS for US nn]	SOURCE targets appear
13.160	Select [ALTERNATE SOURCE]	Boxes with 1 and 2 appear
13.165	Select box [1]	VOL ID: "&C1" or "&C2" and targets appear
13.170	Select [EXECUTE COMMAND]	CONSOLE STATUS AND ASSGN. display "BUSY: US NN" in upper middle "MOUNT \$Dnn> -USnn"
13.200	Insert into FD2 the AREA cartridge disk you made back in Task 4.	
13.210	Select [MOUNT \$Dnn> -USnn]	SOURCE targets appear
13.220	Select [DEFAULT SOURCE]	VOL ID and targets appear
13.230	Select [EXECUTE COMMAND] It may be necessary to repeat the two previous steps	CONSOLE STATUS AND ASSGN display "BUSY: US NN" "NTWRK VIEW <>" "MOUNT STD ABSTS FOR US NN"
13.250	Select [MOUNT STD ABSTS FOR US]	SOURCE targets appear
13.260	Select [ALTERNATE SOURCE]	Boxes with 1 and 2 appear
13.265	Select box [1]	VOL ID: "&C1" or "&C2" and targets appear
13.270	Select [EXECUTE COMMAND] System Status Display appears on Other US.	CONSOLE STATUS AND ASSGN. display "BUSY: US NN" in upper middle Status for Other US is "WARNING" because minimum data was available. US will be loaded with full database in a later task.

SAVE HG OR PLCG CHECKPOINTS

Appendix B

Perform the task described below if you wish to SAVE checkpoint data for an HG or PLCG and its Boxes. You must have created a Checkpoint cartridge disk (CKPT) in Task 4 before performing this task. You should save the HG and Box checkpoint data on a disk first, and then save it on the HM. No HM volume has to be created as the HG checkpoint volume is preconfigured.

Appendix B SAVE CHECKPOINT DATA FOR A HIWAY GATEWAY AND ITS BOXES

Keyboard: Engr. Lock: ENGR

Reference: SW11-601 & SW09-605

Personality Loaded: Universal

Step	Action	Response
<u>START</u>	SAVE HG/BOX CHECKPOINT	
10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
20	Your HG checkpoint disk into FD1	<u>EG</u> : Volume ID = CKPT
30	Select [SYSTEM STATUS]	SYSTEM STATUS display
34	Select HG Node	Node Box is selected
40	Select [NTWK/HWY STATUS] Hiway/box status must be running	HIWAY STATUS display
44	Select [BOX CMND]	
48	Enter Box Number	<u>EG</u> : 05
52	Press <ENTER>	BOX STATUS DISPLAY
56	Select [ENABLE PROC]	Target changes color
60	Select [EXECUTE COMMAND]	Wait...Box Status: OK
62	Repeat steps 44 - 60 for other boxes	All Box Status = OK
64	Select [SAVE DATA]	[ALL BOXES] target appears.
68	Select [ALL BOXES]	SELECT SAVE BOX PATH display
70	If you wish to save checkpoints on cartridge disk, go to step 80. If you wish to save checkpoints on an HM, select [DEFAULT SOURCE] and then go to step 100.	

**Appendix B SAVE CHECKPOINT DATA FOR A
HIWAY GATEWAY AND ITS BOXES
Continued.**

Keyboard: Engr. **Lock:** ENGR

Reference: SW11-601 & SW09-605

Personality Loaded: Universal

Step	Action	Response
80	Select [ALTERNATE SOURCE]	Boxes [1] and [2] appear
90	Select [1]	<u>EG</u> : Volume ID = CKPT
100	Select [EXECUTE COMMAND]	HIWAY STATUS display
110	Select [START FUNCTION]	Asterisk appears by box number as data is saved. "Busy HWY nn"
120	Wait for status of last applicable box to change to "SAVED."	

SAVE AM CHECKPOINT Appendix C

Perform parts of the task described below to SAVE checkpoint data for an Application Module, A^XM, or a Computing Module. You must have created a checkpoint cartridge disk (CKPT) in Task 4 before performing this task. You should always save the AM checkpoint data on a cartridge disk first, and then save it on the HM. No HM volume has to be created as the AM and CM checkpoint volumes are preconfigured.

Appendix C

SAVE CHECKPOINT DATA FOR AM OR CM MODULES

Keyboard: Engr. Lock: ENGR

Reference: SW11-601

Personality Loaded: Universal

Step	Action	Response
10	Checkpoint cartridge disk into FD1	<u>EG</u> : Volume ID = CKPT
20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
30	Select [SYSTEM STATUS]	SYSTEM STATUS display
34	Select AM Node	Node Box is selected (Similar for Computing Module)
40	Select [NODE STATUS]	Application Modules display
50	Select NODE number	<u>EG</u> : 40
60	Select [SAVE DATA]	CHKPT DEST FOR NODE NN display
70	If you are saving checkpoint data on a cartridge disk, go to step 90.	
80	If you have saved checkpoint data on a cartridge disk and now wish to save it on HM, select [DEFAULT SOURCE] and go to step 110.	
90	Select [ALTERNATE SOURCE]	Boxes [1] and [2] appear.
100	Select [1]	
110	Select [EXECUTE COMMAND] "OK" for the selected node in step 50.	STATUS changes from "OK" to "SAVE"
120	Repeat steps 60 – 110 to save checkpoint data to the HM.	

EXCEPTION BUILDING

Appendix D

The steps listed describe the basic exception build procedure of cloning a point from a master point that you have previously built using Parameter Entry Displays. Refer to Section 7 in the *Data Entity Builder* manual for variations.

Appendix D BUILD POINTS BY EXCEPTION **Keyboard:** Engr. **Lock:** ENGR
FOR HG, AM, A^XM, CM, PLCG, AND PM/APM/HPM POINTS

Reference: SW11-506 & SW11-511 **Personality Loaded:** Universal

Step	Action	Response
	CREATE THE EXCEPTION BUILD SOURCE FILE	
05	If the appropriate Data Entity Builder Menu function is already selected, go to step 10; otherwise go to step 15.	<u>EG:</u> [HIWAY GATEWAY], [APPLICATION MODULE], [NETWORK INTERFACE MODULE], and so on.
10	Press <COMND>, then go to step 30	COMMAND Display
15	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
20	Select [BUILDER COMMANDS]	COMMAND Display
30	Select [PRINT ENTITIES]	PATHNAMES display
40	Key into REFERENCE PATHNAME port	<u>EG:</u> NET>H MV1>
50	Select [PRINT IDF ENTITIES]	Additional pathnames ports
60	Key into PATHNAME FOR IDF port	<u>EG:</u> HGPNTS (or AMPNTS, and so on.)
70	Key master point ID into PATHNAME FOR SELECTION LIST	<u>EG:</u> HG1001 (or AM101, and so on.)
80	Key a file name into PRINT DEVICE ID or DESTINATION PATHNAME	<u>EG:</u> HGEXCBLD.EB (or AMEXCBLD.EB, or PMEXCBLD.EB, and so on.)
100	Press <ENTER>	"BUSY...OPERATION COMPLETE"
110	Press <ESC>	"ESCAPED FROM DEB"
120	Key in an Edit command and the pathname of the source file	<u>EG:</u> ED NET>H MV1>HGEXCBLD.EB or ED NET>H MV1>AMEXCBLD.EB
130	Press <ENTER>	Point parameters displayed

**Appendix D BUILD POINTS BY EXCEPTION
BUILDING (continued)**

Keyboard: Engr.

Lock: ENGR

Reference: SW11-506 & SW11-511

Personality Loaded: Universal

Step	Action	Response
140	Tab* down to &N item; tab over and type new point name over old point name.	<u>EG</u> : &N HG1002
150	Tab to other parameters that you want to change; key in changes	<u>EG</u> : SLOTNUM = 02 <u>EG</u> : EUDESC = BBL5/HR
160	Hold down <CTL> and press <3>	JUMP Menu
170	Hold down <CTL> and press <2>	Jump to End-Of-File
180	Press <RETURN> three times	Inserts three blank lines
190	Key in an end of source file indicator	<u>EG</u> : &E
BUILD ADDITIONAL POINTS		
200	If you want to clone another similar point, copy the parameters as follows	If no other points need to be cloned, go to step 310.
210	Hold down <CTL> and press <3>	JUMP Menu
220	Hold down <CTL> and press <1>	Jump to Beginning-Of-File
230	Tab down to &T item	
240	Hold down <CTL> and press <2>	BLOCK Menu
250	Hold down <CTL> and press <1>	"MARK AT LINE: N"
260	<PAGE FWD>, <TAB> to end of parameters	<u>EG</u> : All parameters highlighted

*Press <HELP> to display the Text Editor Help functions; <Page FWD> to see more. Press <HELP> to return to the editor.

**Appendix D BUILD POINTS BY EXCEPTION
BUILDING (continued)**

Keyboard: Engr. **Lock:** ENGR

Reference: SW11-506 & SW11-511

Personality Loaded: Universal

Step	Action	Response
270	Hold down <CTL> and press <2>	"BUSY"... "LINES WRITTEN: NN"
280	Place cursor on blank line just above the &E	
285	Hold down <CTL> and press <4>	"LINES READ: NN" (parameters copied)
290	Tab down/over to &N item; Type new point name over old point name.	<u>EG:</u> &N HG1003
300	Tab to other parameters that you want to change; key in changes	<u>EG:</u> SLOTNUM = 03
305	If there are other similar points to clone, repeat Steps 210 - 300	
	WRITE SOURCE TO SOURCE FILE	
310	Hold down <CTL> and press <1>	QUIT Menu
320	Hold down <CTL> and press <2>	Message : "FILE UPDATED NN LINES WRITTEN"
	PROCESS THE EB SOURCE FILE	
330	Hold <CTL> down and press <HELP>	COMMAND DISPLAY
340	Select [EXCEPTION BUILD]	Ports/targets appear
350	Key into REFERENCE PATHNAME port	<u>EG:</u> NET>H MV1>
360	Key into pathname for SOURCE file port	<u>EG:</u> HGEXCBLD.EB
370	Key into pathname for IDF port	<u>EG:</u> HGPNTS
375	Key into pathname for SELECTION LIST port	<u>EG:</u> blank to select all points in Exception Build File
380	Key into OPT path for TEMPLATE port	<u>EG:</u> blank to select all points in Exception Build File
390	Press <ENTER>	"PROCESSING ENTITY: nnnnnn"... "OPERATION COMPLETE"
395	Wait until the COMMAND DISPLAY reappears.	Points built by exception are added to the named IDF

**Appendix D BUILD POINTS BY EXCEPTION
BUILDING (continued)**

Keyboard: Engr. **Lock:** ENGR

Reference: SW11-506 & SW11-511

Personality Loaded: Universal

Step	Action	Response
	SAVE/LOAD CLONED POINTS	
460	Select [LOAD MULTIPLE]	Additional ports/target appear
470	Key into REFERENCE PATH NAME port	<u>EG:</u> NET>HNV1>
475	Select [with OVERWRITE]	
480	Key into pathname for IDF port	<u>EG:</u> HGPNTS, AMPNTS, APMPTNS, and so on.
485	Key into pathname for SELECTION LIST port	<u>EG:</u> blank for all points in IDF
487	Press <ENTER>	"WAIT" in yellow; "ENTITY NAMES". Pt. Config data written to IDF. "OPERATION COMPLETE"
495	If errors occur, select the point on the error display. Make corrections; then repeat Steps 460 – 495.	

CREATE EXCEPTION BUILD FILES FROM IDFS

Appendix E

The steps listed describe the basic procedure to generate exception build files from Intermediate Data Files (IDFs). Refer to Section 7 in the *Data Entity Builder* manual for basic information on the Command Menu functions. This procedure does not destroy the IDFs. Intermediate Data Files have the suffix: .db. Exception Build Files have the suffix .eb.

Appendix E CREATE EXCEPTION BUILD FILES Keyboard: Engr. Lock: ENGR
FROM IDFS

Reference: SW11-506 & SW11-511 **Personality Loaded:** Universal

Step	Action	Response
05	If any Data Entity Builder Menu function is already selected, go to step 10; otherwise go to step 15	<u>EG</u> : Any ped or the Command Display.
10	Press <COMND>, then go to step 30	COMMAND Display
15	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
20	Select [BUILDER COMMANDS]	COMMAND Display
30	Select [PRINT ENTITIES]	PATHNAMES display
40	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HMV1>
50	Select [PRINT IDF ENTITIES]	Additional pathnames ports
60	Key into pathname for IDF port	<u>EG</u> : HGPNTS (or AMPNTS, and so on.)
65	If you do not remember all the IDF names, this procedure will list them—press <ESC>. Key in a Data Out command to your printer. Press <ENTER>.	<u>EG</u> : DO \$P1 "DATA OUT COMPLETE"
	Key in a List command with a .db suffix, press <ENTER>.	<u>EG</u> : LS NET>HMV1>*.db DB files are listed and printed.
	Key in another DO command to stop printing. Press <ENTER>.	<u>EG</u> : DO "DATA OUT COMPLETE"
	To return to the Command Display, Hold <CTL> down and press <HELP>	
70	Leave Pathname for SELECTION LIST blank	
80	Key a file name into PRINT device ID or DESTINATION pathname	<u>EG</u> : HGPNTS.EB (or AMPNTS.EB, or PMPNTS.EB, HPMPNTS, and so on.)

Appendix E **CREATE EXCEPTION BUILD FILES FROM IDFS (continued)**

Keyboard: Engr. **Lock:** ENGR

Reference: SW11-506 & SW11-511

Personality Loaded: Universal

Step	Action	Response
100	Press <ENTER>	"WAIT...PROCESSING ENTITY: nnnn OPERATION COMPLETE"
110	Repeat steps 60 - 100 for other IDFs that you want to convert to .EB files. If you want to see which exception build files have been generated, press <ESC>. Key in a List command with an .eb suffix (EG: LS NET>H MV1>*eb).	To review the contents of an .eb file, escape as before, then key in an edit command EG: ED NET>H MV1>HGPNTS.EB
114	To exit the Text Editor, press F1 twice	"Do you really wish to abort this edit?"
118	Key in the letter y	Command Processor Menu
122	To return to the Command Menu, Hold <CTL> down and press <HELP>	

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